

• D) DNA polymerases only have 3'-OH → 5'-OH polarity—false; no known DNA polymerase synthesizes 3' → 5'. If they could, Okazaki fragments would not be necessary, because DNA synthesis could proceed continuously on both strands. Thus, Okazaki fragments exist precisely because DNA polymerase is strictly a 5' → 3' enzyme, making discontinuous synthesis unavoidable on the lagging strand.	
4) The role of a DNA helicase during replication is to:	
A) Maintain the DNA helix as a duplex after replication B) Neutralize DNA supercoils during replication C) Synthesize short helical stretches of RNA prior to replication D) Separate DNA strands of a chromosome prior to replication The correct answer is D) Separate DNA strands of a chromosome prior to replication. DNA helicase is the enzyme responsible for unwinding the parental DNA duplex, breaking the hydrogen bonds between complementary bases and generating the single-stranded templates required by DNA polymerases. Because polymerases can only copy single-stranded DNA, helicase activity is indispensable at the replication fork; without it, the double helix would remain closed and inaccessible to the replication machinery. The other options are incorrect because they describe functions carried out by different enzymes—or functions that do not occur at all: • A) Maintain the DNA helix as a duplex after replication — This is not a biological role of any replication enzyme. Once replication is complete, daughter molecules naturally reform stable duplexes through base pairing; helicase does <i>not</i> maintain or promote duplex structure. • B) Neutralize DNA supercoils during replication — This function belongs to topoisomerases, not helicases. As helicase unwinds DNA, it actually <i>introduces</i> positive supercoils ahead of the fork, which topoisomerase must relieve. • C) Synthesize short helical stretches of RNA prior to replication — The enzyme that synthesizes short RNA primers is primase, not helicase. Primase generates the RNA primers needed for DNA polymerase to initiate synthesis, but this activity is unrelated to strand separation. Thus, helicase's essential contribution is the active unwinding of the DNA double helix, creating the single-stranded templates on which replication can proceed.	
5) DNA polymerases have high fidelity due to:	
A) Perfect base-pairing is always required before a new nucleotide is added to a growing chain B) DNA polymerase core complexes contain an exonucleotide activity.	

C) DNA polymerase core complexes contain an endonucleotide activity. D) DNA polymerases use their 3-OH to 5-OH synthesis activity The correct answer is B) DNA polymerase core complexes contain an exonucleolytic activity. DNA polymerases achieve high fidelity largely because many of them possess a 3'→5' exonuclease proofreading function. When an incorrect nucleotide is incorporated, the polymerase senses the mispairing, pauses, and uses this exonuclease activity to remove the misincorporated base. Only after excision does the polymerase resume synthesis in the correct 5'→3' direction. This proofreading dramatically reduces the error rate, ensuring faithful replication of the genome. The other options are incorrect for mechanistic reasons: • A) Perfect base-pairing is always required before a new nucleotide is added — This is not strictly true. DNA polymerases strongly <i>favor</i> correct base-pairing, but mistakes do occur; the enzyme does not enforce perfect pairing before incorporation. Fidelity therefore cannot rely solely on base-pair selection—it depends on the proofreading described in option B. • C) DNA polymerase core complexes contain an endonuclease activity — Endonucleases cut nucleic acids within a strand; DNA polymerases do not use endonucleolytic cleavage for proofreading. Their corrective function specifically involves exonuclease activity, which removes nucleotides from the end of the newly synthesized strand. • D) DNA polymerases use their 3'-OH to 5'-OH synthesis activity — No naturally occurring DNA polymerase synthesizes DNA in the 3'→5' direction. If they did, proofreading would be incompatible with maintaining proper nucleotide triphosphate chemistry. DNA synthesis is exclusively 5'→3', and fidelity arises from the exonuclease, not a nonexistent 3'→5' synthetic mode. Thus, the high fidelity of DNA polymerases is fundamentally a consequence of their 3'→5' exonucleolytic proofreading ability, making option B the correct answer.	
6) Hemi-Methylated DNA at the <i>E. coli</i> origin of replication, <i>oriC</i>, means that: A) One strand of DNA duplex contains methylated bases at <i>oriC</i> B) Both strands of DNA duplex contains methylated bases at <i>oriC</i> C) One strand of DNA duplex contains methylated deoxyriboses at <i>oriC</i> D) Both strands of DNA duplex contains methylated deoxyriboses at <i>oriC</i> The correct answer is A) One strand of the DNA duplex contains methylated bases at <i>oriC</i>. In <i>E. coli</i>, the origin of replication (<i>oriC</i>) contains specific GATC sequences that are normally methylated on the adenine residue by the enzyme Dam	

methylase. Immediately after DNA replication, each daughter duplex contains one parental strand (already methylated) and one newly synthesized strand (initially unmethylated). This asymmetric methylation state is called hemi-methylation. It is biologically important because the replication initiation protein DnaA cannot immediately re-initiate replication at a hemi-methylated origin; <i>oriC</i> must first return to the fully methylated state, ensuring that replication occurs only once per cell cycle. The other answers are incorrect for clear mechanistic reasons: • B) Both strands of DNA duplex contain methylated bases at <i>oriC</i> — This describes fully methylated DNA, not hemi-methylated DNA. Only after Dam methylase acts on the new strand does full methylation occur. • C) One strand of DNA duplex contains methylated deoxyriboses at <i>oriC</i> — Methylation in <i>E. coli</i> does not occur on the sugar (deoxyribose) backbone. It occurs specifically on bases (adenines in GATC motifs). Thus, methylated deoxyriboses is chemically incorrect. • D) Both strands of DNA duplex contain methylated deoxyriboses at <i>oriC</i> — Incorrect for two reasons: (1) true hemi-methylation does not involve both strands, and (2) methylation is on bases, not on the sugar. Therefore, hemi-methylated DNA at <i>oriC</i> refers precisely to one methylated strand and one unmethylated strand, with methylation occurring on the bases, not the sugar, making A the only correct choice.	
7) Which of the following nucleotides binds replication licensing factor DnaA?	
A) ATP B) CTP C) GTP D) TTP The correct answer is A) ATP. DnaA, the key replication-licensing factor in <i>E. coli</i>, must bind ATP to adopt its active conformation. Only ATP-bound DnaA can oligomerize at <i>oriC</i>, wrap the DNA, and induce the local strand separation necessary for replication initiation. In contrast, ADP-bound DnaA is inactive. Thus, ATP binding is a central regulatory mechanism ensuring that replication begins only when the cell is ready and that <i>oriC</i> is “licensed” properly. The other options are incorrect because they involve nucleotides that DnaA does <i>not</i> use as regulatory ligands: • B) CTP — DnaA does not bind CTP for activation. Cytidine nucleotides play no role in DnaA-driven initiation or its conformational switching. • C) GTP — Many GTP-binding proteins (like G-proteins) use GTP for activation, but DnaA is not a GTPase and does not use GTP as a regulatory ligand. • D) TTP — Thymidine triphosphate is simply one of the four deoxynucleotides used during DNA synthesis; it has no regulatory role in replication initiation and does not interact with DnaA.	

Therefore, ATP is the only nucleotide whose binding activates DnaA and thereby licenses oriC for replication initiation.	
8) Polymerase I in <i>E. coli</i> is	
A) responsible for replicating bulk DNA B) responsible for repairing DNA lesions including RNA primers C) responsible for replicating tRNA genes D) responsible for repairing telomeres The correct answer is B) responsible for repairing DNA lesions including RNA primers. In <i>E. coli</i>, DNA polymerase I is not the main replicative polymerase; instead, its essential role is in DNA repair and in processing Okazaki fragments. After primase lays down RNA primers during replication, Pol I removes these RNA primers using its 5'→3' exonuclease activity and replaces them with DNA using its 5'→3' polymerase activity. It also participates in various repair pathways, where its combined exonuclease and polymerase activities allow it to excise damaged nucleotides and fill in the resulting gaps. Thus, Pol I functions as a versatile repair and primer-removal enzyme, not a genome-replicating enzyme. The other options are incorrect for mechanistically clear reasons: • A) responsible for replicating bulk DNA — This is incorrect because bulk DNA synthesis in <i>E. coli</i> is performed primarily by DNA polymerase III holoenzyme, a highly processive complex capable of copying the entire genome. Pol I is far too slow and lacks the processivity required for genome replication. • C) responsible for replicating tRNA genes — tRNA genes are transcribed, not replicated, by RNA polymerase, specifically RNA polymerase encoded by rpo genes in bacteria. DNA replication enzymes do not selectively “replicate” specific genes; all genes are replicated as part of the global duplication of the chromosome by Pol III. • D) responsible for repairing telomeres — <i>E. coli</i> has no telomeres, as it possesses a circular chromosome. Telomere maintenance is a feature of eukaryotic linear chromosomes and involves telomerase, not any bacterial DNA polymerase. Thus, Polymerase I's central and defining role in <i>E. coli</i> lies in repair functions and removal/replacement of RNA primers, making B the correct answer.	
9) In eukaryotes, the RNA primer used for DNA synthesis is removed by:	
A) DNA polymerase I B) A DNA Topoisomerase C) The flap endonuclease Fen1 D) DNA polymerase III The correct answer is C) The flap endonuclease Fen1. In eukaryotic DNA replication, RNA primers laid down by primase–Pol α must be removed before the lagging strand can be fully ligated. As DNA polymerase	

δ extends the Okazaki fragment, it displaces the RNA primer into a single-stranded flap. The endonuclease Fen1 (Flap Endonuclease 1) specifically recognizes and cleaves these RNA–DNA flaps, thereby removing the primer. After Fen1 cleaves the flap, DNA ligase seals the remaining nick, producing a continuous DNA strand. Thus, Fen1 is the principal enzyme responsible for primer removal in eukaryotes. The other options are incorrect because they either refer to enzymes that do not exist in eukaryotes or perform unrelated functions: • A) DNA polymerase I — This enzyme exists only in prokaryotes (e.g., <i>E. coli</i>) and is responsible for removing RNA primers there via its 5'→3' exonuclease activity. Eukaryotes have no Pol I; therefore, it cannot remove eukaryotic RNA primers. • B) A DNA topoisomerase — Topoisomerases relieve torsional stress and resolve supercoils during replication and transcription. They do not remove RNA primers, nor do they cleave RNA–DNA hybrids as Fen1 does. • D) DNA polymerase III — This is the main replicative polymerase of bacteria, not eukaryotes. It has no role in RNA primer removal and does not exist in eukaryotic replication machinery. Thus, in eukaryotic cells, the dedicated enzyme for removing RNA primers is Fen1, making C the correct answer.	
10) A main role of the sliding clamp during DNA synthesis is to: A) link nucleotides together B) separate DNA strands C) remove supercoils at the replication fork D) ensure the processivity of DNA polymerase The correct answer is D) ensure the processivity of DNA polymerase. The sliding clamp—β-clamp in bacteria and PCNA in eukaryotes—is a ring-shaped protein complex that encircles DNA and binds tightly to DNA polymerase. Its principal role is to prevent the polymerase from dissociating from the DNA template, allowing it to synthesize thousands of nucleotides in a single binding event. This property, known as processivity, is essential for the rapid and efficient replication of an entire genome. Without the clamp, polymerase would fall off frequently, drastically slowing replication and reducing accuracy. The other options describe activities belonging to entirely different components of the replication machinery: • A) link nucleotides together — This is the catalytic job of DNA polymerase, which forms the phosphodiester bonds that extend the DNA chain. The sliding clamp does not perform any chemical catalysis. • B) separate DNA strands — This is the function of DNA helicase, which unwinds the double helix to create single-stranded templates. The clamp cannot disrupt hydrogen bonds or unwind DNA.	

- **C) remove supercoils at the replication fork** — This is performed by **topoisomerases**, which cut and reseal DNA to relieve torsional stress generated by helicase. The sliding clamp has no role in managing DNA topology.

Thus, the sliding clamp is essential not for catalysis, unwinding, or supercoil removal, but for mechanically stabilizing the DNA polymerase on the DNA, ensuring long, uninterrupted synthesis—making D the correct answer.

Task set B

Task set B contains 3 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

- 1) Why do we need a gyrase in front of a moving *E. coli* replication fork.

A gyrase (a type II topoisomerase) is essential in front of a moving *E. coli* replication fork because the unwinding of the parental DNA helix by helicase generates intense positive supercoiling ahead of the fork.

As helicase separates the two DNA strands, the remaining double helix becomes overwound. This overwinding rapidly increases torsional stress, which—if not relieved—would physically stall the replication fork, prevent further unwinding, and potentially cause DNA breakage.

DNA gyrase solves this mechanical problem. It introduces negative supercoils by cutting both strands of the DNA, passing another segment through the break, and resealing it. This reaction counteracts the positive supercoils produced by replication, maintaining the DNA in a topologically workable state. Only with continuous gyrase activity can the helicase continue unwinding DNA and the polymerases continue elongating efficiently.

In short, we need gyrase to relieve the torsional stress and positive supercoiling that arise ahead of the replication fork, ensuring smooth, continuous progression of DNA replication.

Esperanza's answer:

A gyrase is a topoisomerase, When a replication fork moves forward to two strands of the parental double helix are separated. As the two strands are copied and the fork moves forward positive DNA supercoils are generated in front of the fork as it cannot rotate freely in the cell due to the size of the chromosome (and chromosomal attachment to structures in the cell). The supercoils will make it progressively harder and harder to separate the two parental strands for further copying. If the problem is not solved the fork will stop and eventually collapse. Fortunately, the topoisomerase activity of gyrase resolves the positive supercoils to allow for progressive replication.

- 2) Why does a replication fork contain a leading and a lagging strand during replication?

A replication fork contains a leading and a lagging strand because of two fundamental constraints of DNA replication:

1. The two template strands of DNA run antiparallel (one 5'→3', the other 3'→5').
2. DNA polymerases can synthesize DNA only in the 5'→3' direction, by adding nucleotides to a free 3'-OH group.

When the replication fork opens, helicase exposes both template strands at the same time.

However, because the strands run in opposite directions, only one of them—the one oriented

3'→5' relative to the direction of the fork—can be copied continuously in the required 5'→3' direction. This strand becomes the leading strand.

The other template strand is oriented 5'→3' in the direction of fork movement, meaning the polymerase would have to synthesize 3'→5' to copy it continuously—something no known DNA polymerase can do. As new template DNA becomes available, the cell solves this problem by synthesizing the complementary strand discontinuously, in short 5'→3' segments called Okazaki fragments, each initiated by an RNA primer. This discontinuous strand is the lagging strand.

Thus, a replication fork necessarily contains a leading and a lagging strand because:

- DNA strands are antiparallel, and
- DNA polymerase is directionally constrained to 5'→3' synthesis.

The combination of these structural and biochemical rules makes continuous synthesis possible on one strand only, while the other must proceed in fragments.

Esperanza's answer:

DNA polymerases can only synthesize DNA in the 5'-OH to 3'-OH direction. For that reason one of the two strands in a parental DNA helix (the 3'-OH to 5'-OH strand) can in principle be replicated from one end to the other. This strand is called leading strand as the DNA polymerase synthesizing this strand is physically ahead of the DNA polymerase that synthesizes the other strand. The other strand to be copied (the 5'-OH to 3'-OH strand) cannot be replicated in a continuous manner since the polymerase has the wrong polarity for this job.. However, as a short single stranded section of the lagging strand is generated as the leading strand is synthesized; these sections 1-2000 bases can be replicated by DNA polymerase. The resulting small double stranded section is called an Okazaki fragment. When a new Okazaki fragment is formed, the lagging strand DNA polymerase will be dismantled from the template and reform just behind the DNA polymerase that synthesizes the leading strand to generate yet another Okazaki fragment. In this way, the lagging strand will be composed by a large number of Okazaki fragments that eventually will be joined by DNA ligase. Remembering that synthesis of DNA begins with an RNA primer also means that many small RNAs need to be excised from the lagging strand before the Okazaki fragments can be joined.

- 3) Speculate why DNA Polymerase I is the major DNA Polymerase Activity in E. coli despite that DNA Polymerase III is the Polymerase, which is responsible for synthesizing the chromosome!

DNA polymerase I is the major DNA polymerase activity in assays because it is abundant, multifunctional, constantly active in repair and primer removal, and biochemically easy to detect.

DNA polymerase III synthesizes the chromosome but is present in very low amounts and acts only within a complex machine at the replication fork.

Dumbing it down:

Hundreds of molecules of Pol I exist, but only a few copies of Pol III. So when you measure total activity, Pol I dominates simply because there's so much more of it. Pol I removes every RNA primer and fixes all the little gaps left behind during replication. There are thousands of these per chromosome, so Pol I is constantly active. It also helps repair damaged DNA, which happens continuously. Pol III copies the chromosome, but only at the two replication forks, and only when the cell is actually replicating its DNA. It's powerful but doesn't act very often, and it requires a big protein machine to function. Pol I is everywhere, working on many small tasks all the time — so when scientists measure polymerase activity, Pol I shows up as the major player, even though Pol III is the real replicator.

Esperanza's answer:

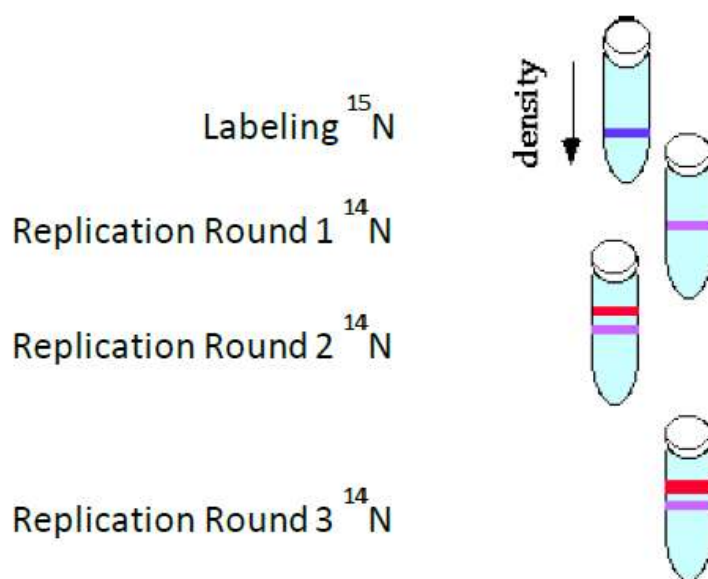
In *E. coli*, Polymerase III replicates the entire chromosome by entering at the single bi-directional Origin of Replication Site, *oriC*. Hence, only two polymerases are actually required to complete the job. In contrast, numerous RNA primers between Okazaki (and other DNA lesions) are removed from DNA by DNA Polymerase I. Moreover, these lesions will appear at random location in the bacteria. To find and repair them take high concentrations of DNA Pol I.

Task set C-1

Task set C contains a case problem with two sub-questions (four in the exam). The answers to each sub-question should be written using the space allocated below each question.

In a famous experiment Matthew Meselson and Franklin Stahl demonstrated that DNA replication is semiconservative. The trick was to grow cells in a medium containing heavy ^{15}N as nitrogen source. In this way “heavy” DNA could be made as nitrogen is a major constituent of DNA. Next, they switched the cells to normal medium with regular light ^{14}N nitrogen and allowed the cells to replicate. After each round of replication, a sample of DNA was sampled and subjected to ultracentrifugation in Cesium chloride solution. After ultracentrifugation Cesium chloride forms a gradient and within this gradient DNA will be positioned according to mass. Specifically, high masses will be closer to the bottom of the tube than lighter species.

A Meselson-Stahl experiment could produce a result looking like:



Question 1. Does the pattern fit what you expect from semiconservative replication? Draw a pedigree of DNA strands containing heavy and light strands as they would form during the experiment to prove your answer,

Yes—the pattern shown is exactly what we expect from semiconservative DNA replication. In semiconservative replication, each daughter DNA molecule contains one original (parental) strand and one newly synthesized strand. Meselson and Stahl demonstrated this by first growing cells in heavy nitrogen (^{15}N), producing DNA in which *both* strands were heavy. When the cells were then transferred to light nitrogen (^{14}N), the composition of the DNA strands changed in a predictable way with each replication round.

Before the medium switch, all DNA molecules consisted entirely of heavy strands (HH), which formed a single heavy band after ultracentrifugation. After one generation in ^{14}N medium, each heavy strand served as template for a newly synthesized light strand, producing only hybrid molecules (HL). This resulted in a single band at an intermediate density.

After the second round of replication in ^{14}N , each hybrid molecule produced one new hybrid (HL) and one completely light molecule (LL). Thus, the DNA population consisted of 50% HL and 50% LL molecules, giving rise to two bands in the density gradient—one intermediate and one light.

In the third round, the same logic continued: hybrid molecules again produced one HL and one LL, while fully light molecules produced only LL. As a result, the proportion of LL molecules increased to 75%, while HL molecules decreased to 25%. The density gradient still showed two bands, but the light band became much more prominent.

This pattern—heavy DNA shifting to a single hybrid band, then to a mixture of hybrid and light bands—matches exactly what semiconservative replication predicts, confirming that each new DNA double helix contains one old strand and one newly synthesized strand.

Question 2. How would the bands distribute in the centrifugation tubes in case replication was conservative during the same number of replication rounds?

If DNA replication had been conservative, the banding pattern in the centrifugation tubes would have looked very different from what Meselson and Stahl actually observed. In the conservative model, the parental DNA duplex remains completely intact during replication, and an entirely new daughter duplex is made from fresh nucleotides. This means that the two strands of the original heavy (^{15}N -labeled) DNA would *never* separate, and no hybrid molecules containing one heavy and one light strand would ever be produced.

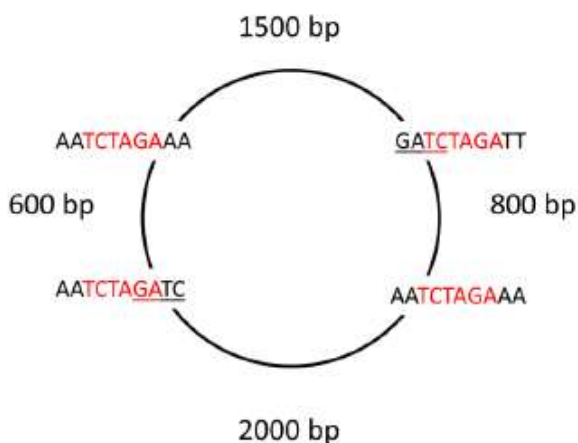
Before the switch to ^{14}N medium, all DNA molecules would appear as a single heavy band. After the first round of replication in ^{14}N , the original heavy duplex would still be present, but the newly synthesized DNA would be completely light, producing a second, distinctly lighter band. Thus, instead of forming a single intermediate-density band—as expected from semiconservative replication—the conservative model predicts two separate bands immediately after the first replication round: one heavy and one light.

With each additional replication round in ^{14}N medium, the same pattern would continue. The original heavy DNA would remain unchanged and would continue to appear as a heavy band, while all newly produced DNA would be light. The proportion of light DNA would increase with each generation, making the light band progressively stronger, but the heavy band would never disappear. Importantly, the conservative model would never produce an intermediate band, because hybrid molecules do not exist in this mechanism.

In summary, conservative replication would yield a clear separation between heavy and light DNA in every replication round, with no hybrid DNA at any stage. This predicted pattern is fundamentally different from the intermediate bands observed experimentally, which is why the conservative model could be ruled out.

Task set C-2

The Dam Methylase Recognizes the sequence motif GATC and then it methylates the A. The restriction enzyme XbaI recognizes and cleaves the sequence TCTAGA, however it is inhibited by overlapping Dam methylation. I.e if one of the A's in its recognition sequence is methylated (remember DNA is a double-stranded molecule); then, that sequence will not be cleaved. A researcher uses XbaI to cleave a plasmid (see below), which has been purified from a wild-type *E. coli* strain and from a Dam methylase mutant strain *E. coli* strain that does not produce the methylase. The plasmid is composed by 4900 bp and the XbaI recognition sites are indicated on the figure below. The distance in base pairs between the four different sites are indicated.



Question 1.

- How many fragments do you obtain when the plasmid from the mutant *E. coli* strain is digested with XbaI? And how many do you get when the plasmid from the wild-type *E. coli* strain is digested with XbaI?

When the plasmid from the mutant *E. coli* strain is digested with XbaI we get 4 fragments, as no sites are methylated and thus all sites are cut.

When the plasmid from the wild-type *E. coli* strain is digested with XbaI we get one fragment, which is the intact 4900 bp plasmid, because all four sites get methylated and thus not cut.

For understanding:

To answer this question, you must understand several core concepts in molecular biology that relate to DNA modification and restriction enzyme behavior. First, bacteria often methylate their own DNA as part of restriction–modification systems, which protect the genome from cleavage by the organism's own restriction endonucleases. In *E. coli*, the Dam methylase specifically methylates adenine within the GATC sequence. This methylation is

important because many restriction enzymes are sensitive to chemical modification of their recognition sites: if a base within the target sequence is methylated, the enzyme may fail to bind or cleave.

A second essential concept is the specificity of restriction enzymes such as XbaI, which recognize short palindromic DNA sequences—in this case TCTAGA—and cut both DNA strands at defined positions. When a recognition sequence overlaps a methylation motif, as XbaI's site overlaps the Dam site (CTAG/GATC), methylation can prevent enzyme binding and thereby block cleavage.

Finally, you need a basic understanding of how circular plasmid DNA behaves when digested with restriction enzymes. Each cleavage site on a circular plasmid produces an additional fragment; cutting at four sites yields four linear fragments whose sizes reflect the distances between the sites, while cutting at no sites leaves the plasmid intact. Together, these principles allow you to predict how methylation status alters restriction digestion patterns and how those patterns translate into discrete DNA fragments on a gel.

When the researcher transformed the methylated plasmid into wild-type and Dam methylase mutant *E. coli* strains they obtained 106 and 2 colonies on the plates selecting for the plasmids, respectively, even that the same amount of plasmid was used in the two transformation experiments. When the researcher purified the plasmid from the Dam methylase mutant strain and re-transformed the un-methylated plasmid into the Dam mutant strain 106 colonies were obtained. The plasmid concentration was identical to what was used in the first experiment.

Question 2. Explain the results produced by the two transformation experiments.

Bacteria use DNA methylation as a way to mark their own DNA so their restriction enzymes do not destroy it. Wild-type *E. coli* has Dam methylase, so a plasmid purified from these cells carries the correct methylation pattern and is easily accepted, giving many colonies. In contrast, Dam-mutant cells have no methylation on their own DNA. When they receive a methylated plasmid, they see it as foreign and their restriction enzymes cut it up, so almost no colonies appear. However, once the plasmid has been grown in the Dam-mutant strain, it becomes unmethylated. When this unmethylated plasmid is transformed back into the Dam-mutant strain, it is no longer recognized as foreign, allowing normal transformation and a high number of colonies.

1. First transformation into the Dam⁻ strain:

The methylated plasmid is mostly destroyed. Only 2 colonies survive because only 2 cells managed to receive a plasmid that escaped restriction.

2. Those 2 surviving colonies now contain the plasmid.

Inside a Dam^- strain, the plasmid is replicated without methylation (because the strain cannot methylate DNA).

3. The researcher purifies the plasmid from those surviving Dam^- colonies.

This newly purified plasmid is now fully unmethylated.

4. Second transformation into the Dam^- strain:

The unmethylated plasmid looks “normal” to a Dam^- cell, so it is not destroyed, and transformation efficiency returns to 10⁶ colonies.

So the researcher used the plasmid from the surviving 2 colonies to perform the next transformation, and that is why the colony number was restored.

- *NOTE: I was a bit confused whether the plasmid is essential for growth or if the colonies were grown on selective plates – chat says that the plasmid gives the cells antibiotic resistance so the professors has used selective plates where all the cells with no plasmid would be killed by the antibiotic.*

Longer explanation:

The key to understanding the transformation results is that bacteria use methylation marks on their DNA as a kind of “tag” to show what belongs to them. In normal, wild-type *E. coli*, an enzyme called Dam methylase places methyl groups on specific DNA sequences. These marks tell the cell’s restriction enzymes not to destroy that DNA. When the researcher transformed a plasmid that came from a wild-type strain, the plasmid already carried the correct methylation pattern. Because the wild-type bacteria recognized these marks as normal and safe, they did not attack the plasmid, and many cells were able to take it up successfully, resulting in the high number of colonies.

When the same methylated plasmid was transformed into a Dam^- mutant strain, the situation changed. Dam^- mutant bacteria do not add methyl groups to their own DNA, so their genome is completely unmethylated. When these cells encounter DNA that *does* contain methylation, they interpret it as foreign. Their restriction enzymes then cut and destroy most of the incoming plasmid DNA. Only a tiny fraction survives this attack, which explains why only two colonies appeared.

Finally, when the researcher purified the plasmid from the Dam^- mutant strain, the plasmid was now unmethylated, just like the host’s own DNA. When this unmethylated plasmid was transformed back into the Dam^- mutant strain, the cells no longer viewed it as foreign, so they did not destroy it. As a result, transformation efficiency returned to normal, and many colonies formed. This behavior shows how important DNA methylation patterns are for bacteria in distinguishing self from non-self DNA.

Question 3. *oriC* is hemimethylated for around 13 minutes after replication while most other chromosomal DNA is hemimethylated after replication only for a few minutes before getting fully methylated due to the Dam methylase activity. One reason for this could be that hemimethylated DNA at *oriC* is sequestered to the membrane preventing access of the Dam methylase. Why is it an advantage of the cell to delay methylation of *oriC* compared to other sequences?

- *NOTE: Hemi is Greek and means “half” and/or “one sided” – so hemimethylated DNA means that only one of the DNA strands are methylated.*

Delaying methylation at *oriC* temporarily blocks re-initiation of DNA replication. This protects the cell from starting replication too early or too often, allowing replication to occur once per cell cycle, in a controlled and coordinated way.

After DNA replication, every new DNA region is briefly hemimethylated, but *oriC* stays hemimethylated much longer than the rest of the chromosome. This delay is *not an accident*—it is an important regulatory strategy.

Why delaying methylation at *oriC* is useful

oriC is the start site for DNA replication.

For the cell to initiate a new round of replication, *oriC* must be fully methylated on both strands. Only then can the initiation protein DnaA bind efficiently and start replication again.

If *oriC* were methylated immediately after replication—like the rest of the chromosome—the cell could accidentally start another round of replication too soon, before the cell is ready. This would create too many copies of the chromosome or cause replication forks to collide, which is dangerous for the cell.

By keeping *oriC* hemimethylated for ~13 minutes, the cell creates a built-in “waiting period.” During this time:

- *oriC* cannot start another round of replication
- DnaA is prevented from re-initiating
- the cell has time to grow and prepare for the next proper initiation

This ensures that each cell cycle begins replication only once, at the correct time, and avoids uncontrolled or over-initiation.

Question 4. *In vitro*, DnaA binds non-methylated, hemi methylated and fully methylated *oriC* fragments quite similarly. However, *in vivo* it does not bind to hemimethylated *oriC*. Interestingly, another protein, SeqA, has been shown *in vitro* to interact with hemi methylated *oriC* regions. Moreover, if *oriC* is incubated with SeqA *in vitro*, then DnaA does not bind. Based on this information propose a simple model to explain how replication re-initiation is delayed.

In short: seqA binds to the same site as DnaA binds to, GATC, so seqA binds to *oriC* as long as it is hemimethylated and slowly disassociates from it when it gets fully methylated – it works as a blocker of the DnaA so the initiation does not happen too soon.

Same thing but explained better:

Right after replication passes through *oriC*, the new *oriC* copies are hemimethylated. *In vitro*, DnaA could in principle still bind these sequences, but in the cell something else happens first: SeqA has a strong preference for hemimethylated GATC sites, which are abundant in *oriC*. As soon as *oriC* becomes hemimethylated, SeqA binds to it and, together with other factors, helps sequester this region at the membrane.

Because SeqA is already bound, DnaA cannot access its binding sites at *oriC*, even though the DNA sequence itself would allow it. In other words, SeqA acts as a physical and functional blocker of DnaA binding on newly replicated *oriC*. While SeqA is bound, Dam methylase cannot easily reach the *oriC* GATC sites either, so *oriC* remains hemimethylated for a longer time than the rest of the chromosome.

Only after some time, SeqA gradually dissociates and Dam methylase can fully methylate *oriC*. Once *oriC* is fully methylated, SeqA no longer has high affinity for it, DnaA can again bind, and a new round of replication can be initiated—but only then. In this simple model, SeqA binding to hemimethylated *oriC* creates a protected “no-reinitiation” window, ensuring that replication cannot restart immediately and thus helping to enforce just one initiation per cell cycle.

Assignment 2. DNA, genomes and chromosomes

Task set A

Task set A contains 10 multiple choice questions. The correct answer should be indicated by a letter in the cell next to the possible answers.

	correct answer
1) What class of enzymes changes the linkage number of DNA?	
A) Exonucleases B) Endonucleases C) Restriction Endonucleases D) Topoisomerases Topoisomerases are a specialized class of enzymes that regulate the topology of DNA by altering its linking number (Lk), which is the number of times one strand of DNA winds around the other in a closed circular molecule. DNA supercoiling and entanglement occur frequently during replication, transcription, and recombination, and topoisomerases resolve these issues by introducing transient breaks in the DNA backbone. There are two major types of topoisomerases. Type I topoisomerases cut a single DNA strand, change the linking number in steps of one, and then reseal the break. Type II topoisomerases cut both DNA strands, change the linking number in steps of two, and are essential for processes like chromosome segregation. Because they are the only enzymes capable of directly changing the linking number, topoisomerases are the correct answer to this question. The other enzyme classes listed do not alter DNA topology in this way: • Exonucleases degrade DNA by removing nucleotides from the ends of DNA molecules. They are involved in processes like DNA repair, proofreading, and degradation, but they do not change the linking number. • Endonucleases cleave DNA internally by breaking phosphodiester bonds within the strand. While they cut DNA, they do not reseal it in a controlled way that modifies DNA topology. • Restriction endonucleases are a specific type of endonuclease that recognize short, specific DNA sequences and cleave at or near them. They are important tools in molecular biology but do not change the overall linking number of DNA molecules. In summary, only topoisomerases have the unique ability to increase or decrease the linking number of DNA, relieving or introducing supercoils as needed, which is why option D is the correct choice.	
2) The core nucleosome does not include:	
A) H1 B) H2A C) H2B D) H3 E) H4	

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The nucleosome is the fundamental unit of chromatin, consisting of ~146 base pairs of DNA wrapped around a histone protein core. This core is an octamer made up of two molecules each of histones H2A, H2B, H3, and H4. These histones are small, positively charged proteins that interact strongly with the negatively charged DNA backbone. Histone H1 is different. It is not part of the nucleosome core particle. Instead, H1 is often called the linker histone because it binds to the DNA between nucleosomes (the "linker DNA") and helps stabilize the entry and exit points of DNA on the nucleosome. H1 plays a key role in folding nucleosomes into higher-order structures such as the 30 nm fiber, but it is not considered part of the core nucleosome. This makes H1 the correct answer. • H2A, H2B, H3, H4 → These four histones form the nucleosome core octamer. Each pair (H2A–H2B dimers and H3–H4 tetramers) is essential for the wrapping of DNA in the nucleosome. • H1 → Only binds to the outside, with linker DNA, and organizes nucleosomes into compact chromatin fibers. Since it is not a component of the core particle, it is excluded.	
3) In a polynucleotide, a phosphate group is linked to the _____ carbons of two pentoses.	
A) 1' and 2' B) 1' and 3' C) 3' and 5' D) 1' and 5' E) 2' and 3' DNA and RNA are polymers of nucleotides, and each nucleotide is made up of three parts: a nitrogenous base, a sugar (pentose), and a phosphate group. The sugar in DNA is deoxyribose and in RNA is ribose. Both sugars are numbered from 1' to 5': • 1' carbon → attached to the nitrogenous base. • 2' carbon → distinguishes DNA (H) from RNA (OH). • 3' carbon → has a hydroxyl (–OH) group essential for forming the phosphodiester bond. • 5' carbon → attached to the phosphate group. In a polynucleotide chain, the phosphate group forms a phosphodiester bond between the 3' hydroxyl group of one nucleotide's sugar and the 5' phosphate group of the next nucleotide's sugar. This linkage gives DNA and RNA their polarity and directionality (we always read sequences from the 5' end to the 3' end). That is why the correct answer is 3' and 5'.	
Why the other options are incorrect: • A) 1' and 2' → The 1' carbon binds to the base, and the 2' carbon determines whether the sugar is deoxyribose or ribose. Neither is directly involved in the phosphodiester bond.	

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• B) 1' and 3' → The 1' carbon is connected to the base, not to the phosphate. The 3' carbon does participate, but the pairing with 1' is wrong. • D) 1' and 5' → Again, the 1' is for the base attachment. While the 5' carbon is correct (it links to phosphate), the 1' is not. • E) 2' and 3' → The 2' carbon is only important for distinguishing RNA (OH) from DNA (H). Only the 3' hydroxyl participates in the bond, not the 2' carbon.	
In summary: The 3'–5' phosphodiester bond is what connects nucleotides into a stable chain, creating the backbone of DNA and RNA molecules.	
4) Negative supercoiling is important for DNA structure, because	
A) it prevents RNA from pairing with DNA in the double helix. B) it holds together the antiparallel strands of DNA in the double helix. C) it facilitates for transcription. D) it prevents the replication fork to move forward. DNA supercoiling refers to the overwinding or underwinding of the DNA double helix. In cells, DNA is usually found in a negatively supercoiled state, meaning it is slightly underwound. This underwinding introduces torsional strain that makes it easier for the two strands of DNA to separate. This property is critical for biological processes that require strand separation, especially transcription, where RNA polymerase must unwind DNA to expose the template strand for RNA synthesis. Negative supercoiling reduces the energy barrier for strand separation, thereby facilitating the initiation and progression of transcription. It also helps during replication, recombination, and repair. That is why C) it facilitates for transcription is the correct answer.	
Why the other options are incorrect: • A) it prevents RNA from pairing with DNA in the double helix. This is not true. RNA can transiently base-pair with DNA (e.g., during transcription, when the RNA transcript pairs with the template DNA strand). Negative supercoiling does not prevent this—it actually promotes it by opening the helix. • B) it holds together the antiparallel strands of DNA in the double helix. The stability of the double helix is provided by hydrogen bonds between bases and base stacking interactions, not by negative supercoiling. Supercoiling changes torsional stress, not the direct bonding forces. • D) it prevents the replication fork to move forward. The opposite is true. Negative supercoiling helps replication forks progress by lowering the energy needed to separate strands. If DNA were positively supercoiled, the replication fork would indeed be hindered.	

In summary: Negative supercoiling makes DNA easier to unwind, which is essential for transcription (and replication). The other options either confuse supercoiling with base-pairing forces or claim the opposite effect.	
5) An example of correct nucleotide pairing in DNA is	
A) C and U. B) T and U. C) G and U. D) A and T. DNA follows Watson–Crick base pairing rules, where bases form specific hydrogen-bonded pairs: • Adenine (A) always pairs with Thymine (T) through two hydrogen bonds. • Guanine (G) always pairs with Cytosine (C) through three hydrogen bonds. These rules are essential for maintaining the uniform width of the DNA double helix and ensuring accurate replication and transcription. Therefore, A and T is the correct example of complementary base pairing in DNA.	
Why the other options are incorrect: • A) C and U → Uracil (U) is not found in DNA; it is a base in RNA where it replaces thymine. In DNA, cytosine (C) pairs with guanine (G), not uracil. • B) T and U → Both thymine and uracil are pyrimidines, so they cannot pair with each other. Also, uracil does not normally occur in DNA. • C) G and U → Guanine (a purine) can form a “wobble” pair with uracil in RNA contexts (like in tRNA decoding), but in DNA this is not part of the standard Watson–Crick base pairing.	
6) The two strands of the DNA double helix are held together by	
A) peptide bonds between nucleotide bases. B) hydrogen bonds between nucleotide bases. C) 5' to 3' attraction. D) ionic bonds What this means: In DNA, the two complementary strands of the double helix are stabilized by hydrogen bonds between the nitrogenous bases. Adenine (A) pairs with Thymine (T) via two hydrogen bonds, while Guanine (G) pairs with Cytosine (C) via three hydrogen bonds. These bonds are weak individually, but collectively they make the DNA molecule stable while still allowing the strands to separate during replication and transcription. Why the other options are incorrect: • A) Peptide bonds between nucleotide bases → Peptide bonds link amino acids in proteins, not DNA bases. • C) 5' to 3' attraction → DNA polarity (5' → 3' direction) describes strand orientation but is not a physical force that holds strands together.	

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• D) Ionic bonds → DNA is negatively charged due to phosphate groups, but ionic bonds do not hold the double helix together. Hydrogen bonds between bases are the true force stabilizing strand pairing.	
7) Which feature is common between prokaryotic and eukaryotic chromosomes?	
A) They both contain telomeres B) They both contain domains with different degrees of superhelicity C) They both contain centromeres D) They both contain poly-cistronic genes What this means: Both prokaryotic and eukaryotic DNA molecules are organized into domains of supercoiling (superhelicity). Supercoiling refers to the overwinding or underwinding of DNA, which helps compact the molecule and regulate accessibility for processes like transcription and replication. Even though eukaryotic DNA is linear and packaged with histones while prokaryotic DNA is typically circular, both rely on regions of differing supercoiling to organize and manage their genetic material. Why the other options are incorrect: • A) They both contain telomeres → Only eukaryotic chromosomes have telomeres at the ends of linear DNA; prokaryotic chromosomes are usually circular and don't need them. • C) They both contain centromeres → Only eukaryotes have centromeres for spindle attachment during mitosis/meiosis. • D) They both contain polycistronic genes → Common in prokaryotes (operons), but eukaryotic genes are generally monocistronic. Superhelicity is the one shared feature between both chromosome types.	
8) The presence of uracil in DNA can be accurately removed because ...	
A) It cannot form Watson-Crick base-pairs to any other of the DNA bases in DNA B) It does not need repair – it spontaneously hydrolyses into the desired nucleotide C) It can use the other strand in the DNA duplex as a template for accurate repair D) It can use a complementary RNA molecule for repair What this means: Uracil (U) sometimes appears in DNA when cytosine (C) undergoes spontaneous deamination. This is a mutagenic event, because U normally pairs with A instead of G. DNA repair systems (e.g., base excision repair) can detect uracil as an error, remove it, and then use the complementary undamaged DNA strand as a template to restore the correct cytosine. This is possible because DNA is double-stranded and contains redundant information. Why the other options are incorrect: • A) It cannot form Watson–Crick base pairs → Wrong, uracil can base-pair with adenine. • B) It does not need repair – it spontaneously hydrolyses into the desired nucleotide → False; uracil does not revert back into cytosine on its own, it must be repaired.	

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D) It can use a complementary RNA molecule for repair → Incorrect; repair uses the DNA duplex, not RNA. The correct mechanism is template-based repair using the opposite DNA strand.	
9) Telomeres are -----:	
A) attachment sites of the mitotic spindle B) used to code for telophase proteins C) protecting the ends of chromosomes. D) tRNA coding sequences What telomeres are: Telomeres are repetitive DNA sequences (in humans, typically the repeat TTAGGG) located at the ends of linear eukaryotic chromosomes. They do not code for proteins, but instead act as protective caps. Their main roles are: Protect chromosome ends from being mistaken as DNA breaks that need repair. Prevent loss of essential genetic information during replication, since DNA polymerases cannot fully replicate the ends of linear DNA (the “end replication problem”). Maintain genomic stability, because without telomeres, chromosomes could fuse together or degrade. Special enzymes called telomerases maintain telomere length in certain cells (e.g., stem cells, germline cells, cancer cells). This is why the correct answer is C) protecting the ends of chromosomes.	
Why the other options are incorrect: A) attachment sites of the mitotic spindle → That function belongs to the centromere and its associated kinetochore, not telomeres. B) used to code for telophase proteins → Telomeres are noncoding, repetitive DNA; they do not produce proteins. D) tRNA coding sequences → tRNAs are encoded by specific genes scattered throughout the genome, not by telomeres.	
In summary: Telomeres are protective, noncoding DNA sequences at chromosome ends that prevent degradation and fusion, making “protecting the ends of chromosomes” the correct answer.	
10) What is the basic structural subunit of chromatin?	
A) 30 nm fiber B) histone C) nucleosome D) ribosome What chromatin is: Chromatin is the complex of DNA and proteins (mainly histones) that makes up the chromosomes of eukaryotic cells. Its primary role is to package long DNA	

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molecules into a compact, organized structure while still allowing regulated access for processes like replication, transcription, and repair. What a subunit is in this context: When we say “the basic structural subunit” of chromatin, we mean the smallest repeating organizational unit that builds up the higher-order chromatin fiber. What nucleosomes are: The nucleosome is the fundamental repeating unit of chromatin. Each nucleosome consists of ~146 base pairs of DNA wrapped around a histone octamer, which is composed of two copies each of histones H2A, H2B, H3, and H4. Linker histone H1 binds outside the core to help compact nucleosomes into higher-order fibers. This structure explains how meters of DNA can be tightly packed inside the nucleus while still being accessible when needed. That is why nucleosomes are the correct answer: they are the basic building blocks of chromatin.	
Why the other answers are incorrect: • A) 30 nm fiber → This is a higher-order structure of chromatin formed by the folding and coiling of nucleosomes, not the fundamental unit itself. • B) histone → Histones are essential protein components of nucleosomes, but by themselves they are not the repeating structural unit. • D) ribosome → Ribosomes are molecular machines for protein synthesis, completely unrelated to chromatin structure.	
In summary: Chromatin is organized DNA-protein material, its smallest repeating subunit is the nucleosome, which allows efficient DNA packaging. The other options refer either to larger structures (30 nm fiber), components of nucleosomes (histones), or unrelated organelles (ribosomes).	

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Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

1)

a) Explain why GC-rich DNA requires a higher temperature to denature or melt than AT-rich DNA.

DNA denaturation, or “melting,” is the process of separating the two strands of the DNA double helix. The stability of the helix depends largely on the hydrogen bonding between base pairs and base-stacking interactions. Adenine (A) pairs with Thymine (T) through two hydrogen bonds, while Guanine (G) pairs with Cytosine (C) through three hydrogen bonds. Because GC pairs have one more hydrogen bond than AT pairs, regions of DNA that are GC-rich form stronger, more stable interactions. In addition to hydrogen bonding, stacking interactions between G–C pairs are stronger than those between A–T pairs, further stabilizing GC-rich DNA.

As a result, more thermal energy (higher temperature) is required to disrupt these interactions and separate GC-rich DNA strands, leading to a higher melting temperature (T_m) compared with AT-rich DNA.

Solution:

The three hydrogen bonds formed in GC basepairs are stronger than the two hydrogen bonds formed in AT basepairs

b) The genomes of bacteria that live in hot springs generally have a higher GC content as compared to bacteria that live at 20 degrees Celcius. Explain why.

Thermophilic bacteria (those living in hot environments) face constant stress from elevated temperatures, which tend to destabilize DNA by breaking hydrogen bonds. To withstand this, their genomes have evolved a higher GC content, which provides additional hydrogen bonds and stronger stacking interactions, making the DNA helix more resistant to denaturation.

This adaptation ensures that their genetic material remains intact and functional under extreme conditions. Without such stability, DNA in hot environments would denature too easily, disrupting replication and transcription. Therefore, high GC content is a molecular adaptation for thermal stability.

Solution:

The high temperature in hot springs will break weak hydrogen bonds. This is compensated by a higher GC content in the genome as they form more hydrogen bonds than AT pair

c) Moreover, living at the really hot places even produces an enzyme that introduces positive supercoils. How could that enzyme facilitate survival?

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In mesophilic organisms, DNA is usually maintained in a negatively supercoiled state. Negative supercoiling helps open the helix, facilitating transcription and replication at moderate temperatures. However, in thermophiles, where heat already promotes spontaneous strand separation, too much negative supercoiling would cause excessive unwinding and unwanted single-stranded DNA regions, making the genome fragile and prone to damage.

To counteract this, some thermophiles use specialized reverse gyrase enzymes, which introduce positive supercoils. Positive supercoiling overwinds the DNA helix, making it harder to separate the strands. This stabilizes the DNA at high temperatures, prevents excessive denaturation, and ensures genomic integrity. Thus, introducing positive supercoils is a crucial survival mechanism in thermophiles.

Solution:

The high temperature in hot springs will break weak hydrogen bonds to produce single stranded patches. Some single stranded patches are necessary for many biological function including transcription initiation, however too much is not desirable as the genome will be too fragile. At low temperature single stranded patches are generated by introducing negative supercoiled DNA. However, at high temperature where DNA denaturation of relaxed DNA is spontaneous; the task is rather to prevent forming too much single stranded DNA and this is solved by introducing positive supercoils.

- 2) **Model organisms of eukaryotes are typically haploid or diploid. However, if possible, researchers will prefer to work with haploid organisms. Considering that researchers often introduce gene deletions to understand the function of a gene; and that it is often only possible to delete one gene at a time – explain why a haploid organism is preferred for these experiments.**

In genetics research, haploid organisms (with one set of chromosomes) are often preferred over diploid organisms (with two sets of chromosomes). The key reason is that in haploids, each gene exists in only one copy, so **the phenotype resulting from a mutation or deletion is immediately visible**. If a gene is deleted, the cell has no backup copy, and the resulting phenotype reflects the direct consequence of losing that gene.

In contrast, in diploids, deleting a gene on one chromosome may not produce a phenotype because the second copy (allele) can often compensate, masking the effect. To see the phenotype in diploids, both copies of the gene generally need to be deleted, which is technically more challenging.

Additionally, when studying gene function, researchers often perform **targeted deletions or knockouts one gene at a time**. Haploid systems are much more efficient for this approach because the genotype-phenotype relationship is **simple and unambiguous**. This is why haploid organisms such as *Saccharomyces cerevisiae* (baker's yeast) are widely used as genetic model systems.

Solution:

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In a haploid organism, the phenotype is directly visible after deleting the gene of interest. In a diploid, both homologs of the gene (in most cases) need to be deleted to produce the phenotypic.

Task set C

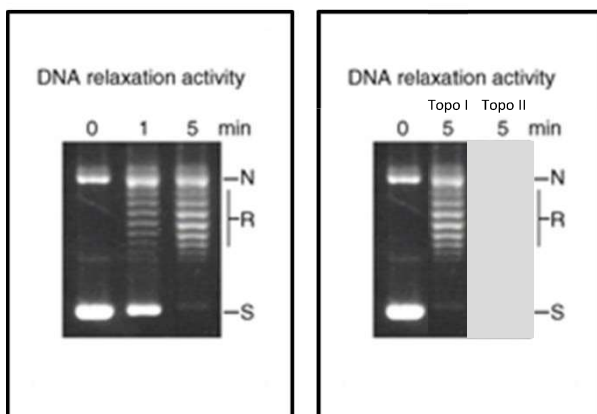
Task set C contains a case problem with two sub-questions. The answers to each sub-question should be written using the space allocated below each question.

Task C1

A scientist has isolated a new topoisomerase. The scientist was familiar with type I topoisomerases and has previously made a plasmid based assay where a population of negatively supercoiled plasmids was partially relaxed by type I topoisomerase before analysis on agarose gels. See Figure below. The scientist now did the same experiment with the new topoisomerase.

Question 1.A: What would the result look like if the enzyme was a type II topoisomerase? Draw the expected band pattern characteristic of a type II topoisomerase in the drawing below.

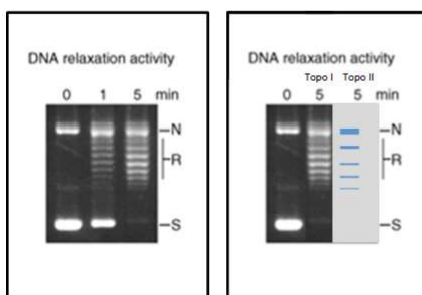
Topo 1 Topo 1 & Topo 2



Fill in bands in the gray lane

Answer:

Topo 1 Topo 1 & Topo 2



Type I topo:
Changes linkage number
In steps of 1

Fill in bands in the gray lane

Type II topo:
Changes linkage number
In steps of 2

Chat:

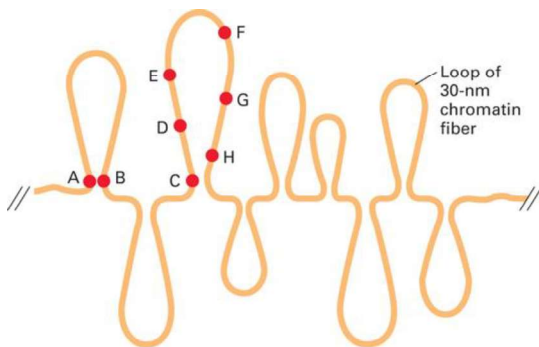
If the enzyme is a type II topoisomerase, the banding pattern on the agarose gel would differ from that produced by a type I enzyme. Whereas type I topoisomerases relax supercoiled DNA by introducing single-strand breaks and removing one supercoil at a time, type II topoisomerases act by introducing transient double-strand breaks and passing another double helix through this break. This changes the linking number of the plasmid in steps of two ($\Delta Lk = \pm 2$). On an agarose gel, this

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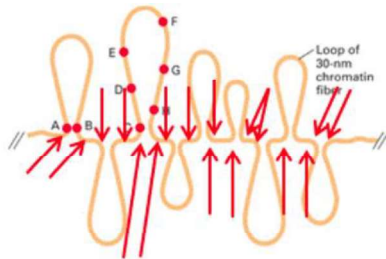
would be seen as plasmid topoisomers separated by larger intervals, corresponding only to even-numbered relaxation states, rather than the stepwise consecutive bands typical of type I topoisomerases. Thus, the expected pattern for type II would be fewer, more widely spaced bands, reflecting the mechanism by which the enzyme alters DNA topology.

The scientist developed an antibody against the new Topo and used it to stain a sample of fixed chromatin using a procedure where proteins in close proximity to DNA are crosslinked to DNA. The DNA was visualized by another stain to produce the figure below. In this experiment the new Topo was shown to bind at specific locations on the chromatin sample.

Question 1.B: In the figure below, indicate by arrows where you could expect to find topo II using the new antibody.



Answer:



Chat:

Topo II does not bind uniformly across chromatin but instead localizes to specific genomic regions where its enzymatic activity is required. These include centromeres and pericentromeric heterochromatin, where Topo II is essential for resolving DNA catenanes during chromosome segregation, as well as scaffold or matrix attachment regions that help organize chromatin into higher-order structures. In addition, Topo II is often enriched in transcriptionally active regions, where it relieves torsional stress created by RNA polymerase during elongation. Therefore, when visualized with an antibody in the chromatin staining experiment, Topo II would be expected to appear as punctate signals at centromeres, at transcriptionally active domains, and at chromatin loop boundaries, rather than being evenly distributed throughout the chromatin.

Task C2

A team of researchers were interested in understanding gene expression of the Globin gene in mice. Accordingly, they extracted mouse chromatin from young red blood cells, which has still not lost their nuclei. Next, the extracted chromatin was fragmented by DNaseI. The researchers envisioned that DNaseI would cleave DNA in chromatin according to the illustration in Figure C2-A. They also added the restriction enzyme BamHI that cleaves at positions flanking the Globin gene. After the digestion, the designed a PCR reaction that should be able to amplify a 4.6 kb fragment of the

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Globin gene using primers that close at the BamHI sites (but on the side facing the Globin gene). The results of the PCR reaction is shown in in Figure C2-B.

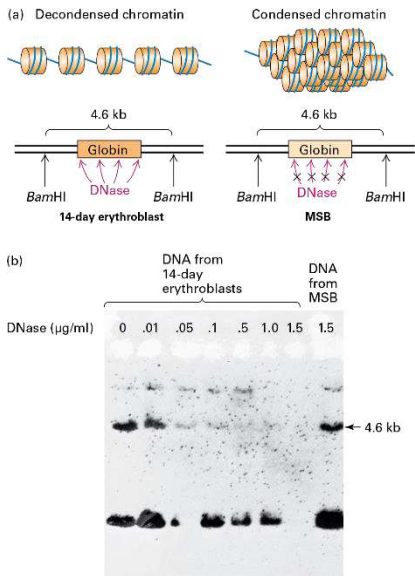


Figure C2_1

Question C2_A. Explain why the PCR reaction produced a band in chromatin that was not treated with DNaseI

Answer: This is a control experiment showing that mice has a globin gene in their genome (not a big surprise). The intact Globin gene serves as a template for a productive PCR reaction

Chat:

The presence of a PCR band in chromatin not treated with DNaseI reflects the intact nature of the globin locus in the mouse genome. After BamHI digestion, the DNA fragment containing the 4.6 kb globin gene remains undamaged and can therefore serve as a proper template for PCR amplification with the designed primers. This result serves as a control, demonstrating that the globin gene exists and that the PCR setup works as intended.

Question C2_B. Explain why the PCR reactions produced fainter and fainter bands as the DNaseI concentrations were increased.

Answer: The experiment indicates that the Globin gene is digested by DNaseI. If so, it cannot serve as template for a PCR reaction with the added primers.

Chat:

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When DNaseI is added to erythroblast chromatin, the globin locus becomes progressively fragmented, since DNaseI preferentially cleaves DNA in nucleosome-free or open chromatin regions. Because PCR requires an intact DNA template between the primer sites, fragmentation of the globin gene reduces the availability of intact templates. As DNaseI concentration increases, more globin loci are cut, leading to weaker amplification and fainter PCR bands. This indicates that the globin gene is accessible to DNaseI digestion, consistent with an open chromatin configuration.

Question C2_C. In a parallel experiment, the researchers isolated chromatin from a lymphoblastoid cell line (MSB). What does the experiment tell you about production of Globin in these cells?

Answer: The experiment indicates that the Globin gene is not digested by DNaseI in MSB cells. Hence, it can serve as a template for a PCR reaction with the added primer

Chat:

In lymphoblastoid (MSB) cells, the globin gene remains intact even after DNaseI treatment, as shown by the persistence of a strong PCR band. This suggests that the globin locus is not accessible to DNaseI in these cells, meaning it is maintained in a closed chromatin state. The locus therefore remains resistant to nuclease digestion and continues to serve as an intact template for PCR amplification. This indicates that globin is not being expressed in MSB cells.

Question C2_D. Does the results of these experiments make biological sense?

Answer: The experiment indicates that the Globin gene is situated in euchromatin in erythroblasts, but in heterochromatin in MSB cells.

Chat:

Together, these experiments demonstrate that the chromatin status of the globin locus differs between cell types. In erythroblasts, the locus is DNaseI-sensitive, indicating that it is found in euchromatin, an open chromatin configuration compatible with active transcription. In contrast, in MSB cells the locus is DNaseI-resistant, showing that it is packaged into heterochromatin, a compact form of chromatin that prevents transcription. Thus, the experiments clearly distinguish transcriptionally active from silent chromatin states at the globin gene.

Question C2_E. Does the results of these experiments make biological sense?

Answer: Yes, erythroblasts are supposed to produce haemoglobin; white blood cells are not.

Chat:

These results make biological sense when considering the function of the two cell types. Erythroblasts are precursors of red blood cells, which require high levels of haemoglobin for oxygen transport, and therefore must actively transcribe the globin gene. By contrast, lymphoblastoid cells are precursors of white blood cells, which play roles in immunity and have no need for haemoglobin. In these cells, the globin locus is transcriptionally silent and embedded in

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heterochromatin. The observed differences in chromatin structure at the globin locus therefore reflect normal cell-type-specific regulation of gene expression.

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Exercise

Task set A

Multiple choice

1. RNA generally functions as a regulator by:

A) base-pairing with a single-stranded region.

B) influencing ribosomal activity.

C) recruiting proteins to gene promoters.

D) catalysing biochemical reactions.

E) participating in chromatin remodelling.

RNA generally functions as a regulator through **complementary base-pairing with single-stranded regions of target nucleic acids**. This mechanism is the basis for the activity of many regulatory RNAs, such as microRNAs, small interfering RNAs, and antisense RNAs. By forming short double-stranded regions with their targets, these RNAs can alter mRNA stability, block or promote translation, and influence gene expression post-transcriptionally. This base-pairing mechanism represents the most fundamental and widespread way in which RNA exerts regulatory control.

B) Influencing ribosomal activity – this is an indirect consequence of base-pairing, not the primary mechanism.

C) Recruiting proteins to gene promoters – this is mainly the role of transcription factors, not small RNAs.

D) Catalysing biochemical reactions – only specific RNAs (ribozymes) do this; it is not the general regulatory role of RNA.

E) Participating in chromatin remodelling – some long noncoding RNAs are involved, but this is a specialized function rather than the main regulatory mechanism.

2. Which of the following is generally **NOT** a way that antisense RNA might inhibit gene expression?

A) It might prevent translation.

B) It might affect RNA processing.

C) It might affect stability of the protein product.

D) It might prevent transcription.

E) All of these are ways that antisense RNA might inhibit gene expression.

Antisense RNA regulates gene expression primarily by forming complementary base pairs with target mRNA, which can block ribosome binding sites and thereby prevent translation, interfere

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with normal RNA processing such as splicing or polyadenylation, or even hinder transcription by forming RNA–DNA hybrids at the template strand. What antisense RNA does **not** do is alter the **stability of the protein product**, since its action occurs exclusively at the level of nucleic acids rather than proteins.

Why the other options are incorrect:

- **A) Prevent translation** – this *is* a well-established function of antisense RNA, so it is not incorrect.
- **B) Affect RNA processing** – antisense RNA can interfere with processing events, so this is also correct.
- **D) Prevent transcription** – antisense RNA can block transcription initiation or elongation through hybrid formation, so this is valid.
- **E) All of these are ways...** – incorrect, because antisense RNA cannot influence the stability of the protein product (C).

Generally, about antisense RNA:

General function of antisense RNA

Antisense RNAs are RNA molecules that are complementary to messenger RNAs (mRNAs). Because of this complementarity, they can bind to the mRNA and change how it behaves. One common effect is that they **block the ribosome from attaching** to the mRNA; when this happens, the protein cannot be produced, so the gene is effectively silenced. Antisense RNA can also **interfere with RNA processing steps** such as splicing or polyadenylation, which means the mRNA is not shaped correctly and cannot be used to make a proper protein. In other cases, the binding of antisense RNA makes the mRNA **unstable and more likely to be degraded**, so the message is destroyed before any protein can be produced. Finally, antisense RNA can sometimes **hybridize with the DNA template itself**, which prevents transcription from continuing, meaning the mRNA is never even made in the first place.

Why they are needed

This form of regulation is important because it allows the cell to quickly and reversibly adjust gene expression without changing the DNA. By turning down or turning off specific mRNAs, the cell saves resources, avoids producing harmful amounts of proteins, and responds more flexibly to changing conditions. In short, antisense RNAs are needed because they provide the cell with an **efficient, fine-tuned way of controlling which proteins are made and when**.

3. In mammals and other animals, the expression of a specific gene can be turned off by the introduction in the cells of:

- A) long dsDNA.**
- B) short dsRNA.
- C) short dsDNA.
- D) short ssDNA.

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E) long ssDNA.

In mammals and other animals, gene expression can be specifically silenced through a process called **RNA interference (RNAi)**. This mechanism is triggered by the introduction of **short double-stranded RNAs (dsRNAs)**, typically 20–25 nucleotides long, known as **small interfering RNAs (siRNAs)**.

These siRNAs are incorporated into the **RNA-induced silencing complex (RISC)**. One strand of the siRNA guides RISC to a complementary mRNA, where it base-pairs and causes cleavage or prevents translation. The consequence is that the target gene's mRNA is destroyed or silenced, and the protein is not produced.

☞ Other options (long dsDNA, short/long ssDNA, short dsDNA) do not trigger this RNAi pathway in animals. DNA forms are involved in replication or gene editing, but not in post-transcriptional silencing.

4. siRNAs, unlike miRNAs...

A) Have multiple targets

B) Have a single target

C) are processed by Dicer and loaded into Argonaute proteins forming RISC complexes

D) Promotes mRNA degradation

Both **siRNAs (small interfering RNAs)** and **miRNAs (microRNAs)** are short double-stranded RNAs processed by **Dicer** and loaded into **Argonaute proteins** within the **RISC complex**. Both can promote degradation or translational repression of target mRNAs.

The key difference lies in their **targeting specificity**:

- **siRNAs** are usually perfectly complementary to their target mRNA, meaning they are highly specific and typically silence **one gene**. They exist on the same RNA
- **miRNAs**, on the other hand, usually bind with imperfect complementarity, especially in animals, so a single miRNA can regulate **many different mRNAs**. They exist on different positions in the genome.

☞ Therefore, the unique feature of siRNAs compared to miRNAs is that they generally **have a single, specific target**

siRNAs and miRNAs are small regulatory RNAs that silence genes after transcription.

- **siRNAs** usually pair perfectly with one specific mRNA, leading to its cleavage and degradation. This stops the production of that single protein, giving very precise control.
- **miRNAs** usually pair imperfectly with many mRNAs, often just blocking translation or slightly destabilizing the transcript. This allows them to fine-tune the expression of whole networks of genes.

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Why they are important: Both are essential for controlling gene expression, protecting the cell from viruses and transposons, and ensuring proteins are made in the right amounts at the right time. The consequence of their action is reduced or prevented protein production from target genes, which keeps cellular processes balanced and responsive.

5. The nucleoprotein complex Argonaute/RISC:miRNA has **NOT** the following potential function

- A) Inhibition of translation elongation
- B) Co-translation protein degradation
- C) Inhibition of ribosomal subunit joining
- D) Inhibition of gene activation**

The **Argonaute/RISC:miRNA complex** is the active unit of RNA interference. It uses the miRNA as a “guide” to recognize complementary sequences in target mRNAs and then regulates them mainly at the post-transcriptional level.

- **Inhibition of translation elongation** – when RISC binds to the mRNA, it physically interferes with ribosome movement along the transcript, so the ribosome stalls and cannot finish making the protein.
- **Co-translation protein degradation** – RISC can recruit additional cellular machinery (like ubiquitin–proteasome pathways) that mark the newly synthesized protein for destruction even as it is being translated.
- **Inhibition of ribosomal subunit joining** – RISC binding near the start codon can block the assembly of the small and large ribosomal subunits, preventing translation from starting.

However, the complex **cannot inhibit gene activation**, since this occurs at the **DNA level** (through transcription factors and chromatin state), while Argonaute/RISC:miRNA only acts after mRNA is already made.

6. The enzyme that processes pre-miRNAs into small miRNAs is called _____.

- A) Restriction enzymes.
- B) CRISPRs.
- C) miRNA maturation enzymes.
- D) Dicer.**
- E) Catalytic siRNAs.

After a primary miRNA transcript (pri-miRNA) is first cropped in the nucleus by **Drosha**, it is exported to the cytoplasm as a pre-miRNA hairpin. The enzyme **Dicer**, an RNase III endonuclease, then cuts this hairpin into a short double-stranded RNA of about 21–23 nucleotides. One strand becomes the mature **miRNA**, which is loaded into the Argonaute/RISC complex to regulate gene expression.

☞ The other options are incorrect because:

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- **Restriction enzymes (A)** cut DNA, not RNA.
- **CRISPRs (B)** are a bacterial defense system involving RNA but not miRNA processing.
- **miRNA maturation enzymes (C)** is not a real enzyme name — it's just a description.
- **Catalytic siRNAs (E)** are products, not enzymes.

7. Which of the following is **NOT** a ncRNA?

A) mRNA.

B) rRNA.

C) tRNA.

D) snRNA.

E) antisense RNA.

The only RNA in the list that is **not a non-coding RNA (ncRNA)** is **mRNA (messenger RNA)**. mRNA is a **coding RNA** because it carries the genetic instructions from DNA and serves as a direct template for protein synthesis during translation. This makes it fundamentally different from ncRNAs, whose roles are structural or regulatory rather than coding.

Why the other options are incorrect:

- **rRNA (ribosomal RNA)** – an ncRNA that forms the structural and catalytic core of ribosomes, essential for protein synthesis but not translated itself.
- **tRNA (transfer RNA)** – an ncRNA that carries amino acids to the ribosome and matches them to the codons on the mRNA.
- **snRNA (small nuclear RNA)** – an ncRNA involved in splicing pre-mRNA within the spliceosome.
- **antisense RNA** – an ncRNA that regulates gene expression by base-pairing with complementary mRNAs to block translation or promote degradation.

☞ Thus, all of these are non-coding RNAs, leaving **mRNA** as the only coding RNA among the options.

9. Which mutation will result in decreased expression of the *Escherichia coli trp* operon?

A) Leader peptide start codon

B) A deletion of region 2 of the *trp* leader

C) A deletion of region 3 of the *trp* leader

D) A deletion of region 4 of the *trp* leader

E) TrpR operator.

The correct answer is **B) A deletion of region 2 of the *trp* leader**. The *E. coli trp* operon is regulated by attenuation in the leader sequence, which contains four regions capable of base-

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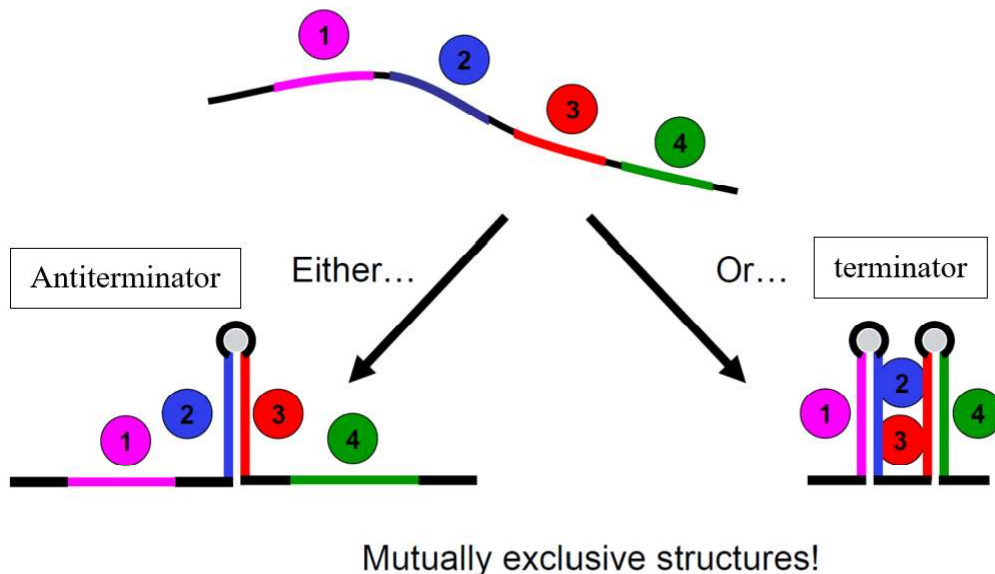
pairing. Normally, regions 2 and 3 can pair to form an anti-terminator structure that allows transcription to continue when tryptophan levels are low. If region 2 is deleted, this anti-terminator can no longer form. As a result, region 3 will always pair with region 4, creating the terminator hairpin and causing premature termination of transcription. This leads to consistently reduced expression of the operon, even under conditions when the cell needs more tryptophan.

Why the other options are wrong:

- **A) Leader peptide start codon** – If this codon is lost, the leader peptide cannot be translated, which causes the ribosome to stall. This favors the anti-terminator (2–3 pairing), so transcription continues and expression increases, not decreases.
- **C) Deletion of region 3** – Region 3 is required to form the terminator hairpin with region 4. Without region 3, attenuation cannot occur, so transcription continues unchecked, leading to increased expression.
- **D) Deletion of region 4** – Region 4 is also essential for forming the terminator hairpin. If it is deleted, attenuation cannot occur, so the operon is expressed at higher levels.
- **E) TrpR operator** – The operator is the binding site for the trp repressor. If it is deleted, the repressor cannot bind, so repression fails and the operon is expressed more strongly, not less.

Attenuation – Transcription Termination/Antitermination

Example: The Tryptophane attenuator in *Escherichia coli*



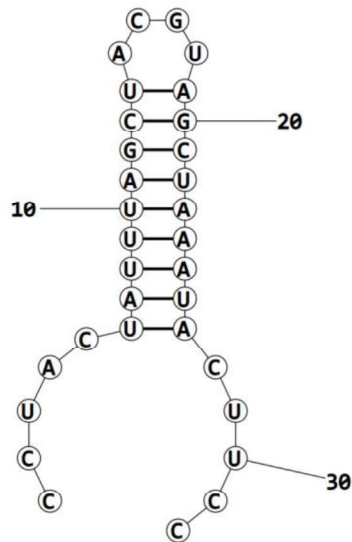
10. Which RNA molecule will form the most stable hairpin? (use the online server <https://rna.urmc.rochester.edu/RNAstructureWeb/Servers/Fold/Fold.html>)

- A) CCUACUAUUUAGCUACGUAGCUAAAUACUUCC
- B) CCUACUACGUAGCUACGUAGCUAUUUACUUCC
- C) CCUACUAGCUAGUUACGUAAUUAGCUACUUCC

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D) CCUACACCUACGUAACGUACCUACCUACUUCC

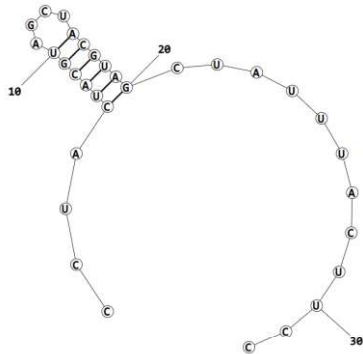
E) CCUACUAGCUAGCUACGUAGCUAGCUACUUCC



a)

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$\Delta G = -8.0$ (Ikke det laveste så ikke korrekte svar)

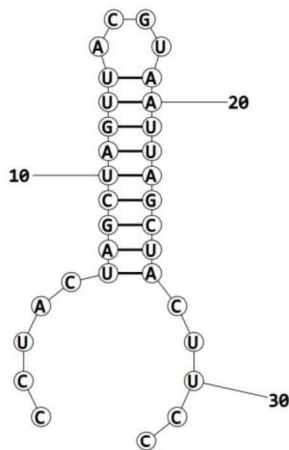


b)

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$\Delta G = -6.0$ (Ikke det laveste så ikke korrekte svar)

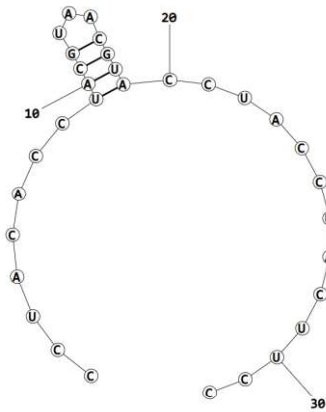
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c)

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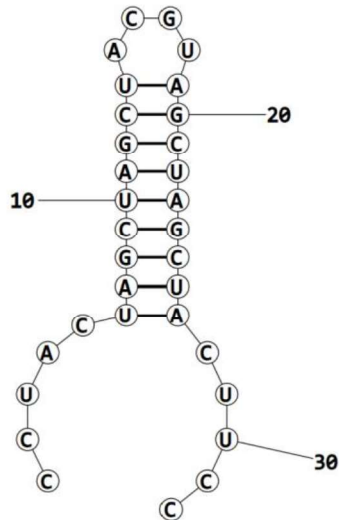
$\Delta G = -8.0$ (Ikke det laveste så ikke korrekte svar)



d)

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$\Delta G = -0.7$ (Ikke det laveste så ikke korrekte svar)



e)

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$\Delta G = -12.7$ (det laveste så det rigtige svar!!)

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Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

1. What is a riboswitch, and how does a riboswitch work?

A **riboswitch** is a regulatory segment of an mRNA molecule that can directly bind a small metabolite and, in response, change the expression of the gene encoded by that mRNA. Unlike protein-based regulators, riboswitches are made entirely of RNA and function without needing additional proteins.

How a riboswitch works

- **Structure:** A riboswitch is typically located in the **5' untranslated region (5' UTR)** of the mRNA. It has two main parts:
 - An **aptamer domain**, which specifically binds to a small molecule (e.g., a metabolite such as thiamine pyrophosphate, flavin mononucleotide, or glycine).
 - An **expression platform**, which changes its RNA secondary structure in response to metabolite binding.
- **Mechanism:** When the metabolite binds the aptamer, the RNA folds into an alternative structure that alters gene expression. This can happen in different ways:
 - **Transcriptional control:** The riboswitch forms a terminator hairpin that halts RNA polymerase, stopping transcription.
 - **Translational control:** The riboswitch hides or exposes the ribosome binding site (Shine–Dalgarno sequence), thereby preventing or allowing translation.
 - **RNA stability:** Some riboswitches affect how quickly the mRNA is degraded.

Consequence

The riboswitch provides a direct link between metabolite levels and gene expression. When the metabolite is abundant, the riboswitch turns off synthesis of enzymes that would make more of it. When the metabolite is scarce, the riboswitch allows expression of those enzymes.

☞ In short: a riboswitch is an RNA “sensor–switch” built into the mRNA that binds a metabolite and changes RNA structure to control whether the gene is expressed.

Solution: A riboswitch is a regulatory RNA located in the leader peptides which may assume alternate base-pairing configurations (with the presence of a ligand or an environmental change) that affect attenuation (termination/antitermination in prokaryotes) or translation

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2. Describe how microRNAs are produced in eukaryotes and three potential regulatory functions

Production of microRNAs (miRNAs) in eukaryotes

1. **Transcription:** miRNAs are first transcribed by RNA polymerase II (or sometimes III) as long primary transcripts called **pri-miRNAs**, which may contain one or several hairpin structures.
2. **Nuclear processing:** The enzyme **Drosha**, together with its cofactor DGCR8 (also called Pasha), cleaves the pri-miRNA into a shorter hairpin precursor of about 70 nucleotides, called a **pre-miRNA**.
3. **Export to cytoplasm:** The pre-miRNA is transported out of the nucleus by **Exportin-5**.
4. **Cytoplasmic processing:** The enzyme **Dicer** cuts the pre-miRNA hairpin into a ~21–23 nucleotide double-stranded RNA fragment.
5. **RISC loading:** One strand (the “guide strand”) is incorporated into the **RNA-induced silencing complex (RISC)**, centered on an Argonaute protein. The other strand (the “passenger strand”) is degraded.
6. **Target recognition:** The mature miRNA guides RISC to partially complementary sequences in target mRNAs, usually in their 3' untranslated region (3' UTR).

Three potential regulatory functions of miRNAs

1. **Inhibition of translation** – Binding of miRNA–RISC to the mRNA can block ribosome access, reducing or preventing translation into protein.
2. **mRNA degradation** – When complementarity is strong, the Argonaute protein in RISC cleaves the mRNA, leading to rapid decay.
3. **mRNA destabilization and storage** – miRNAs can recruit enzymes that shorten poly(A) tails, destabilizing transcripts, or sequester mRNAs in processing bodies (P-bodies) for storage and later degradation.

🔗 **In summary:** miRNAs are produced through a carefully orchestrated processing pathway involving Drosha and Dicer, and once loaded into RISC, they regulate gene expression mainly by blocking protein production, promoting mRNA degradation, or controlling mRNA stability.

Solution: The microRNA gene is transcribed, and a double-stranded region is formed in the RNA. This region is recognized by the Dicer nuclease, which cleaves the RNA, forming the active miRNA. miRNA affect mRNA translation by Inhibiting translation elongation, Co-translation protein degradation or Inhibition of ribosomal subunit joining.

Task set C

1. The Lysine biosynthetic operon of *Staphylococcus aureus*

Most microorganisms synthesize lysine aspartic acid through eight enzymatic steps, catalyzed by enzymes encoded on the chromosome. In *Staphylococcus aureus*, six of the eight genes are expressed as an operon *lys*, with eight open reading frames, two of which encode proteins with unknown function (Figure 1). The two remaining genes in lysine biosynthesis are found at other locations on the chromosome. The *lys* operon is expressed from a single unregulated promoter (P), as shown below. The sequence in front of *lysC* is shown including the first four amino acids of LysC is shown in Figure 2. The leader can fold into two alternative structures, as shown in Figure 3. Structure 1 is only formed when lysine is present. Without lysine, the energetically most favorable Structure 2 will form.

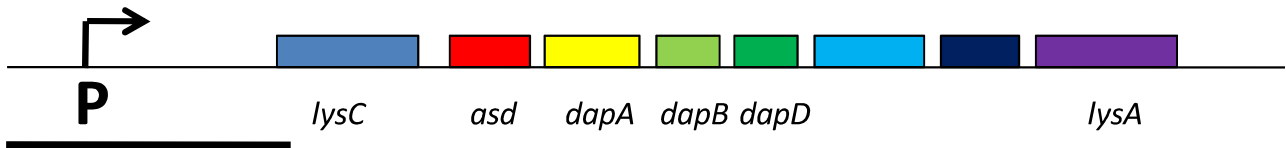


Figure 1. Genetic map of the *lys* operon in *Staphylococcus aureus*.

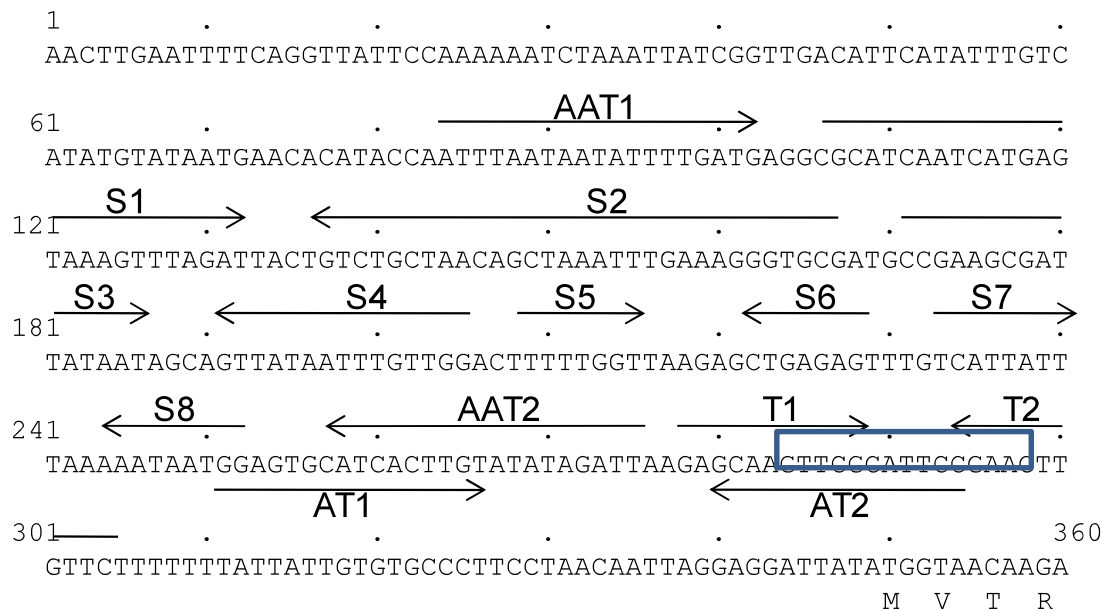


Figure 2. The sequence of the *lys* operon leader

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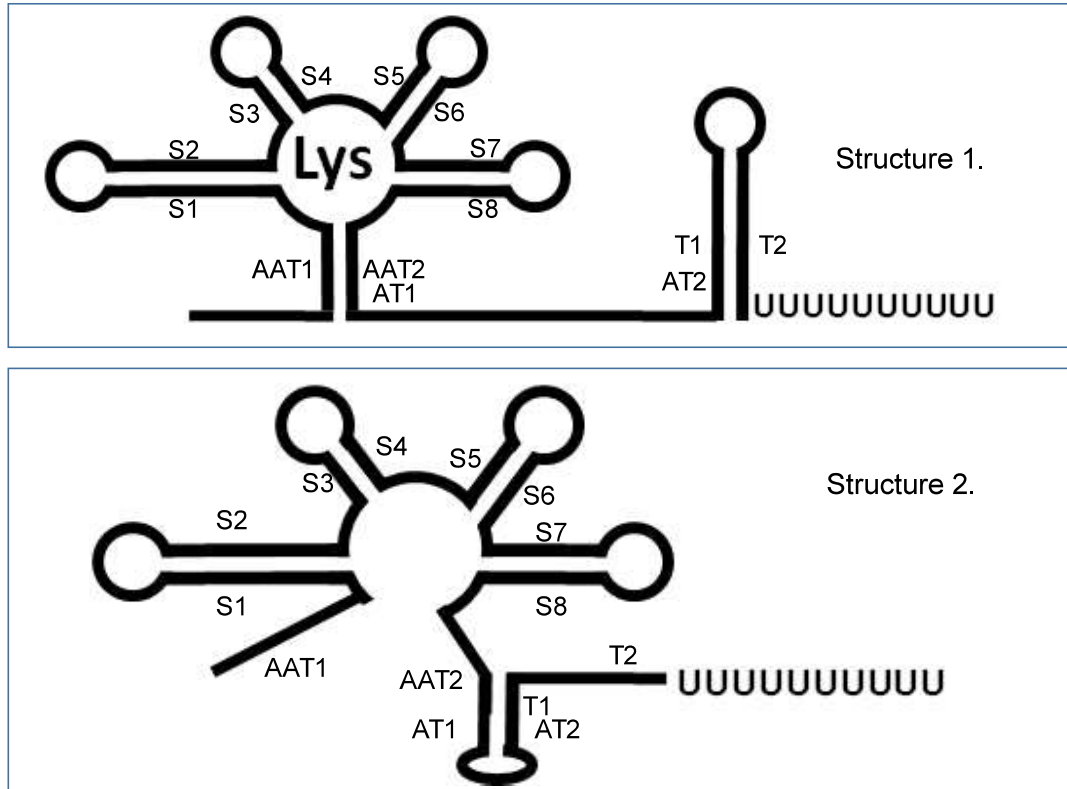


Figure 3. The secondary structures of the *lys* operon leader in the absence and presence of Lysine

1. T1-T2 in structure 1 adopts a structure with consequences for the gene transcription. How is it called? What is the consequence of the formation of such a structure in the transcription process?

T1–T2 i struktur 1 danner en **intrinsisk transkriptionsterminator (attenuator)**: en stabil hårnål efterfulgt af en U-rig sekvens. Når denne struktur dannes (ved tilstedeværelse af lysin), **standser RNA-polymerasen tidligt**, så transkriptionen af det nedstrøms *lys*-operon **afbrydes/påbegyndes ikke**, og genekspressionen falder.

Solution: T1–T2 form a loop followed by a poly-U tail, which functions as a transcription terminator for RNA polymerase. When the leader sequence forms this terminator, transcription of the downstream genes stops, and as a result no mRNA is produced for translation, leading to reduced concentration of the gene products.

Different *lys* operon leader and promoter fragments were fused to a *lacZ* reporter gene which starts with ATG, and the β -galactosidase activity was determined.

2. Predict the β -galactosidase activity for each mutant by filling in the missing data in the table below. (Hint: identify critical sequences in the promoter. You have a additional final in DTU Learn)

	Sequence inserted - coordinates refer to the sequence in figure 2	Relative β -galactosidase activity - units pr mg dry weight	
		With lysine	No lysine

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wt	1 - 360	10	100
1	1 - 320		
2	1 - 295		
3	1 - 240		
4	1 - 80		
5	70 -360		
6	70 - 295		
7	1 - 360 Δ (90 – 105)		

	Sequence inserted - coordinates refer to the sequence in figure 2	Relative β -galactosidase activity - units pr mg dry weight	
		With lysine	No lysine
wt	1 - 360	10	100
1	1 - 320	10	100
2	1 - 295	>100	>100
3	1 - 240	>100	>100
4	1 - 80	>100	>100
5	70 -360	Below detection	Below detection
6	70 - 295	Below detection	Below detection
7	1 - 360 Δ (90 – 105)	>100	>100

1. The first codons encoding for the protein from the *lys* operon are removed. However, the *lacZ* reporter gene has its own initiation codon; thus, it behaves like the wild type (wt).
2. The section T2 is compromised, preventing the leader peptide from forming a terminator. As a result, regardless of the presence or absence of Lys, the operon undergoes uncontrolled transcription.
3. The fragments T1 and T2 are excised. Similar to mutant 2, this prevents the formation of an antiterminator, irrespective of Lys presence. Additionally, the affected section S8 might hinder its ability to bind Lys, rendering the system unresponsive to the metabolite.
4. The RNA structure is missing the sequence essential for Lys binding or terminator formation, resulting in unchecked reporter expression.
5. The -10 sequence is removed, preventing the RNA polymerase from transcribing the gene.
6. Same as 5.
7. The promoter lacks the segment crucial for Lys binding. Consequently, it remains unresponsive to the metabolite and consistently assumes an antiterminator configuration.
3. What would you call the regulatory mechanism of the *lys* operon?

It is a riboswitch

4. As a result of a single mutation where the T residue at position 169 is changed to a C, the expression of the *lys* operon is no longer affected by lysine addition. What could explain that observation?

Lysine makes specific contact to T169 – consequently a T169C mutation would prevent lysine binding, and hence, the favored structure – the antiterminator conformation will be formed

2. miRNA in Cancer Metastasis and Angiogenesis

Recent findings highlight a unique trans-encoding miRNA, named miRNA-CM1, which is notably overexpressed during the metastasis of certain cancer types. It's believed that miRNA-CM1 might boost angiogenesis by suppressing a proapoptotic factor.

You have the capability to measure:

- Concentration of the precursor form (pre-miRNA-CM1).
- Concentration of the mature form (miRNA-CM1).
- Amount of miRNA-CM1 loaded onto the Argonaute protein.
- The concentration of the proapoptotic factor

A cancer cell line has been genetically engineered with various mutations targeting miRNA-CM1. Your task is to analyze how these mutations impact the miRNA's sequence. The goal is to devise a gene therapy that replaces the original miRNA-CM1 sequence with one that diminishes angiogenesis, all while maintaining optimal cell viability.

You also measured all the parameters in a non-cancerous cell line to determine the parameter profile of a healthy cell.

	pre-miRNA-CM1 (pM)	miRNA-CM1 (pM)	Argonaute:miRNA-CM1- (pM)	proapoptotic factor (pM)	Cell proliferation (cell div/hour)	Cell viability (%)
Cancer cell line miRNA-CM1 wt	0.5	1	10	0.1	200	100
Non-cancer cell line miRNA-CM1 wt	0.01	0.02	0.2	20	60	100
Cancer cell line mutant 1	1.5	0.01	0.001	100	20	30
Cancer cell line mutant 2	0.5	15	2	30	100	85
Cancer cell line mutant 3	0.5	1	10	25	70	100
Cancer cell line mutant 4	0.5	1	10	0.1	50	20

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1. What does it mean that the miRNA-CM1 is “trans-encoding”?

When miRNA-CM1 is described as "trans-encoding," it means that it regulates the expression of genes other than its own encoding gene. In other words, miRNA-CM1 affects the expression of genes that are distinct from the gene from which miRNA-CM1 itself is derived.

2. How does the concentration of the Argonaute:miRNA-CM1 correlate with the proapoptotic factor in the cancer cell line compared to the non-cancer cell line? How does it affect cell proliferation and cell viability?

When the concentration of Argonaute:miRNA-CM1 is high (10 pM), it reduces the concentration of the proapoptotic factor. This increases the cell proliferation and the cell viability is 100%. In a non-cancer cell line, the concentration of the precursors and the Argonaute:miRNA-CM1 nucleoprotein complex is low, the cell proliferation is reduced, and the cell viability is 100%

3. Discuss the experiment results for all the mutants, explaining how the processing of the miRNA-CM1 can affect cell proliferation and cell viability

Mutant 1: In mutant 1, the concentration of pre-miRNA-CM1 is higher than in the wild type. This suggests that the mutation impacts the processing by the nuclease Dicer. Consequently, the Argonaute:miRNA-CM1 concentration is low, and the concentration of the apoptotic factor is very high. This significantly reduces cell proliferation and cell viability, with values even surpassing those of healthy non-cancer cells.

Mutant 2: For mutant 2, the concentration of free miRNA-CM1 is higher than in control, and the Argonaute:miRNA-CM1 complex concentration is low. This indicates that the mutation affects the recognition of miRNA-CM1 by Argonaute. As a result, the concentration of the proapoptotic factor rises, leading to decreased cell proliferation. However, this mutation also diminishes cell viability, suggesting that the miRNA might have off-target activity on other genes involved in the process.

Mutant 3: In mutant 3, pre-miRNA-CM1 is processed by Drosha and loaded into Argonaute. However, the concentration of the proapoptotic factor is higher than in the cancer cell control. This suggests that the mutations prevent it from binding to the mRNA that encodes the proapoptotic factor. Consequently, cell proliferation decreases, but cell viability remains at 100%. It appears that this mutation does not impact other signaling pathways.

Mutant 4: In mutant 4, the profiles of pre-miRNA-CM1, free miRNA-CM1, and Argonaute:miRNA-CM1 complex mirror those of the wild type. Yet, cell proliferation decreases and cell viability drops to 20%. These results suggest that the mutant miRNA affects another signaling pathway involved in the process

4. Which mutant would you choose as the lead candidate for gene therapy and why?

Mutant 3 has the closer cell proliferation and cell viability profile compared to the non-cancer cell line.

The main concepts from today

1. RNA as a regulator

- RNA is not just a messenger for protein coding.
 - It can regulate gene expression by **base-pairing with complementary RNA or DNA**, altering transcription, translation, or RNA stability.
 - Key types: **antisense RNAs, siRNAs, miRNAs, riboswitches, ribozymes, and lncRNAs**.
-

2. Antisense RNA

- Antisense RNAs bind complementary mRNAs.
 - **Consequences:**
 - Block ribosome binding → no translation.
 - Cause mRNA degradation → less protein.
 - Affect RNA processing (e.g., splicing).
 - Antisense RNAs work at the **RNA level**; they do not act on proteins directly.
-

3. RNA interference (RNAi) with siRNAs and miRNAs

- **siRNAs:**
 - Derived from double-stranded RNAs (dsRNA).
 - Usually perfectly complementary to a single mRNA.
 - Cause cleavage and degradation of that mRNA.
 - **miRNAs:**
 - Encoded in the genome.
 - Processed by **Drosha (nucleus)** and **Dicer (cytoplasm)** into ~22 nt RNAs.
 - Loaded into Argonaute (RISC complex).
 - Bind imperfectly to many mRNAs, leading to translational repression or destabilization.
 - **Importance:** Fine-tuning gene expression, antiviral defense, control of development, cancer regulation.
-

4. Argonaute/RISC:miRNA complex

- This is the **effector complex**.
- Functions include:
 - Blocking ribosomal subunit joining.
 - Stalling translation elongation.
 - Promoting co-translational degradation of proteins.

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- **It does not regulate transcriptional gene activation**, since it acts post-transcriptionally.
-

5. Riboswitches

- Riboswitch = structured RNA element in the **5' UTR** of certain bacterial mRNAs.
 - Works by binding a metabolite (like thiamine pyrophosphate or lysine).
 - Ligand binding causes the RNA to fold into a different secondary structure that can:
 - Form a **terminator hairpin** → transcription stops.
 - Hide the Shine–Dalgarno sequence → translation blocked.
 - This provides direct metabolite-sensing regulation, without needing proteins.
-

6. Attenuation in operons (like trp or lys operon)

- In bacteria, some operons use **attenuation**, where the RNA leader sequence can form alternative hairpins.
 - Example: **trp operon**.
 - Regions 2–3 form an **anti-terminator** → transcription continues.
 - Regions 3–4 form a **terminator** → transcription stops.
 - Deleting region 2 forces constant attenuation → **decreased expression**.
 - Example: **lys operon riboswitch in Staphylococcus aureus**.
 - In presence of lysine, a terminator forms → no transcription.
 - Without lysine, anti-terminator forms → transcription continues.
-

7. Non-coding RNAs (ncRNAs)

- RNAs that do not encode proteins but have structural or regulatory roles.
 - Examples: **rRNA, tRNA, snRNA, antisense RNA, miRNA, siRNA, lncRNA**.
 - **mRNA** is the exception — it is coding.
-

👉 In summary:

The theory behind these exercises is that **RNA molecules are versatile regulators of gene expression**. They can:

- Directly sense metabolites (riboswitches).
- Control transcription termination (attenuation).
- Pair with mRNAs to block or degrade them (antisense RNA, siRNA, miRNA).
- Act within protein–RNA complexes (RISC/Argonaute).

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These mechanisms allow cells to finely adjust gene expression in response to internal and external signals.

Exercise

Task set A

Multiple choice

1. Which of the following is not generally a way to control the activity of a positive regulatory transcription factor?

- A) Binding of an inhibitor that affects its ability to bind to DNA
- B) Synthesis of protein in a specific tissue type
- C) Covalent modification of a protein
- D) Transport of the protein out of the cell**
- E) Binding of a ligand

Because the transcription happens inside the nucleus, the transportation of the protein out of the cell has more to do with translation than it does transcription

2. Which of the following is not usually the function of an independent domain of an activator transcription factor?

- A) ATPase**
- B) Dimerization (two molecules (often proteins) come together to form a complex.)
- C) DNA binding
- D) Transcription activation
- E) All of these are potential independent domains of an activator.

Though all modelling complexes contain an ATPase catalytic subunit, it does not mean that it is a domain. This means that ATPase subunits is not the same as activator domains as domains are inside a protein and subunits are separate proteins that assemble into a complex.

3. An activator that does not have an activating domain

- A) must directly interact with RNA polymerase.
- B) must also not have a DNA-binding domain.
- C) may require a coactivator with an active domain.**
- D) is mutant and cannot activate a gene.
- E) is called a repressor.

An activator transcription factor normally has two parts: one that binds DNA and another that activates transcription. If the activator does not have its own activation domain, it can still bind to

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DNA but cannot directly stimulate transcription. In this case, the activator recruits a **coactivator** protein that does contain an activation domain or enzymatic activity. The coactivator then interacts with RNA polymerase or with chromatin-modifying enzymes to promote transcription. In other words, the activator provides the DNA-binding specificity, while the coactivator supplies the activation function. This is why the correct answer is that an activator without an activating domain **may require a coactivator with an activation domain**.

4. All chromatin-remodelling complexes have a subunit for _____ that is encoded by one member of a gene superfamily.

- A) histone binding
- B) ATPase**
- C) repressor binding
- D) DNA binding
- E) transcription factor binding

5. Transcription activation via a remodelling complex often involves:

- A) histone deacetylation.
- B) recruiting an activator to a promoter.
- C) DNA methylation.
- D) displacing nucleosomes from a promoter**
- E) breaking apart nucleosomes.

Cells use two main types of mechanisms to control how tightly DNA is packaged and whether genes can be transcribed: histone modification complexes and chromatin remodeling complexes.

Histone modification complexes work by chemically altering the histone proteins around which DNA is wrapped. They add or remove chemical groups, such as acetyl or methyl groups, to the tails of histones. These modifications change how histones interact with DNA and with other proteins. For example, histone acetylation usually loosens the chromatin structure and promotes transcription, whereas histone deacetylation tightens chromatin and represses transcription. Histone methylation is more nuanced: depending on the specific residue, it can either activate or repress gene expression.

Chromatin remodeling complexes, on the other hand, do not modify histones chemically but instead use the energy of ATP hydrolysis to physically reposition nucleosomes. They can slide nucleosomes along DNA, temporarily eject them, or replace standard histones with specialized variants. By doing so, they directly change whether promoter regions and other regulatory DNA elements are accessible to transcription factors and RNA polymerase.

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In summary, histone modification complexes act by writing or erasing chemical marks on histones, while chromatin remodeling complexes act by physically rearranging nucleosomes. Both processes are essential for regulating transcription, but they operate through fundamentally different strategies.

6. Which statement best describes the relationship between methylation and gene activity?

- A) DNA methylation is generally associated with transcriptional inactivity.
- B) DNA methylation is generally associated with translational inactivity.
- C) DNA methylation is generally associated with transcriptional activity.
- D) Histone methylation is generally associated with transcriptional inactivity.
- E) Histone methylation is generally associated with transcriptional activity.

DNA methylation and histone methylation are both epigenetic mechanisms that influence whether genes are turned on or off, but they work in different ways.

DNA methylation usually involves the addition of a methyl group to cytosine bases within CpG islands in promoter regions. When these sites are methylated, transcription factors and the transcriptional machinery cannot easily bind, or repressive proteins are recruited. The result is that the gene becomes transcriptionally inactive. In most cases, therefore, DNA methylation is a stable mark of long-term gene silencing.

Histone methylation, on the other hand, refers to the addition of methyl groups to specific amino acids in the histone proteins around which DNA is wrapped. Unlike DNA methylation, the effect of histone methylation is context-dependent. Methylation of some histone residues, such as H3K9 or H3K27, is associated with chromatin compaction and transcriptional repression. In contrast, methylation of other residues, like H3K4, is linked to open chromatin and active transcription. Thus, histone methylation can either promote or inhibit gene activity depending on which residue is modified and how many methyl groups are added.

7. What circumstances are generally not associated with histone phosphorylation?

- A) with chromatin condensation
- B) during DNA repair
- C) during homologous recombination
- D) cyclically during the cell cycle
- E) Histone phosphorylation is associated with all of the above circumstances.

Histone phosphorylation is a dynamic modification that plays several well-established roles in the cell. One of the best-known examples is during **chromatin condensation** in mitosis, where

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phosphorylation of histone H3 at serine 10 is essential for proper chromosome compaction. Histone phosphorylation is also strongly linked to the **DNA damage response**, most notably through phosphorylation of the histone variant H2AX. This modification, called γ -H2AX, occurs rapidly at sites of double-strand breaks and acts as a signal to recruit DNA repair proteins. In addition, histone phosphorylation occurs **cyclically during the cell cycle**, with levels rising and falling in coordination with cell cycle progression.

What histone phosphorylation is **not generally associated with** is **homologous recombination**. While γ -H2AX signals that DNA damage has occurred, the specific repair process of homologous recombination is carried out by other specialized proteins, such as Rad51 and BRCA1. In other words, histone phosphorylation functions as a broad marker of DNA damage and chromatin state, but it does not directly mediate the homologous recombination pathway.

For this reason, the correct answer is that histone phosphorylation is not generally associated with homologous recombination.

8. How is the yeast GAL gene control system unlike the *E. coli* lac operon?

- A) GAL genes do not have a catabolite repression level of control.
- B) GAL genes are constitutively expressed.
- C) GAL genes are not induced by their substrate.
- D) GAL genes are transcribed when the cell has a sufficient supply of glucose.
- E) GAL genes are not expressed in a single polycistronic mRNA.**

The yeast **GAL gene control system** and the ***E. coli* lac operon** are both classic examples of how cells regulate metabolism based on available nutrients, but they differ in their genetic organization. In *E. coli*, the lac genes are arranged together in a single operon and transcribed into one **polycistronic mRNA**. This allows multiple genes needed for lactose metabolism to be expressed at the same time under the control of a single promoter.

In contrast, the yeast GAL genes are not organized in an operon. Instead, they are scattered across different locations in the genome, and each gene has its own promoter and regulatory sequences. Although they are all regulated by the same control proteins — primarily the activator GAL4, the inhibitor GAL80, and the sensor GAL3 — they are expressed as **separate, monocistronic mRNAs**.

This key difference explains why option E (“**GAL genes are not expressed in a single polycistronic mRNA**”) is correct. The GAL system and the lac operon both respond to sugar availability, but their genetic organization and transcriptional output are fundamentally different.

Polycistronic mRNA

- *Poly-* = many, *-cistronic* = coding regions (genes).
- A **polycistronic mRNA** is a single mRNA molecule that carries the instructions for making **several different proteins**.

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- This is common in **prokaryotes** (like *E. coli*), where multiple genes that function in the same pathway are grouped together in an **operon**.

☞ Example: the **lac operon** in *E. coli*

- One mRNA is made from the operon.
- That mRNA contains coding regions for **LacZ, LacY, and LacA**.
- Each of these regions is translated separately into its own protein.

Monocistronic mRNA

- In contrast, **eukaryotic cells** (like yeast or humans) usually produce **monocistronic mRNAs**, where each mRNA codes for just **one protein**.
- That's why the yeast GAL genes each have their own promoter and are transcribed separately.

9. In the context of eukaryotic gene regulation, a true activator...

A) is a protein that contains a DNA binding domain and an activator domain

B) can be involved in positive and negative control regulation.

C) is always a heterodimer composed by a DNA binding domain and an activator domain

D) make physical contact with the basal apparatus activating gene expression

In eukaryotes, a **true activator** is a transcription factor that both **binds specific DNA sequences** and **directly stimulates transcription** of the associated gene. Unlike regulatory proteins that only provide DNA-binding specificity and rely entirely on co-activators, a true activator carries its own **activation domain** that can recruit or interact with the transcriptional machinery.

A true activator typically has at least two independent functional domains:

1. **DNA-binding domain (DBD):** Recognizes and binds to a specific sequence in enhancers or promoters, ensuring the activator acts on the correct gene.
2. **Activation domain (AD):** Interacts with components of the transcription apparatus — such as the Mediator complex, general transcription factors, or histone-modifying enzymes. Through these interactions, the activator increases the likelihood that RNA polymerase II will initiate transcription.

For example, the yeast protein **GAL4** is considered a true activator. It binds to upstream activating sequences (UAS) in the promoters of GAL genes and, through its activation domain, recruits the transcriptional machinery needed for gene expression.

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☒ **In short:** In eukaryotic gene regulation, a true activator is a transcription factor that not only binds DNA but also contains its own activation domain, allowing it to directly stimulate transcription without relying entirely on other co-activators.

10. Which of the following features is NOT epigenetic

A) DNA methylation affecting gene expression without changing the DNA sequence

B) Inherited histone modifications that influence chromatin structure

C) Mutations in the DNA sequence of a gene that alter its function

D) Protein structures that can perpetuate a specific cellular state across generations

Epigenetics refers to heritable changes in gene expression or cellular state that occur without altering the underlying DNA sequence. Classic examples include DNA methylation, histone modifications, and stable protein conformations that influence chromatin structure or gene activity. These modifications change how the genome is read, not the genetic code itself.

Option C, however, describes mutations in the DNA sequence of a gene. A mutation alters the nucleotide sequence, permanently changing the genetic code. This is a direct genetic change, not an epigenetic one, because the information in the DNA itself is modified. In contrast, epigenetic mechanisms leave the sequence intact but still regulate whether and how genes are expressed.

For this reason, C is the correct answer: mutations are genetic, not epigenetic.

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Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

1. List the six levels at which eukaryotic gene expression can be regulated Which level is the most important?

1) Chromatin remodeling

DNA in eukaryotes is wrapped around histones to form chromatin, which can be either tightly packed (heterochromatin) or open (euchromatin). Chromatin remodeling complexes and histone modifications (e.g., acetylation, methylation) control whether transcription machinery can even access the DNA. This is the **most important checkpoint**, because if chromatin remains closed, none of the downstream steps can occur. It acts as the primary gatekeeper of gene expression.

2) initiating transcription

Once chromatin is accessible, transcription factors and RNA polymerase II assemble at the promoter. Regulation here determines whether a gene is transcribed into RNA. This step allows fine-tuned control of gene activity.

3) mRNA processing

The primary RNA transcript undergoes splicing, capping, and polyadenylation. Alternative splicing in particular allows one gene to produce multiple protein isoforms, expanding regulatory complexity.

4) transporting mRNA from the nucleus to the cytoplasm

Not all processed RNAs are exported. Nuclear pore regulation ensures only correctly processed transcripts reach the cytoplasm for translation.

5) translating the mRNA

Even after export, translation initiation and efficiency can be controlled, for example by regulatory proteins or non-coding RNAs (like miRNAs). This determines how much protein is actually produced from a transcript.

6) mRNA degradation.

The stability of mRNA influences how long it can be translated. Regulatory mechanisms that alter mRNA half-life therefore play a key role in adjusting protein levels.

2. In the context of eukaryotic gene regulation explain the role of the following elements:

Basal transcription factors:

These are the general factors required for RNA polymerase II to initiate transcription at almost all promoters. They assemble at the **core promoter** (e.g., the TATA box) to form the **pre-initiation complex**. Examples include **TFIID**, **TFIIB**, and **TFIIH**. Basal transcription factors do not determine whether a gene is strongly or weakly expressed; instead, they provide the **minimum machinery needed for transcription to start**.

Solution: Basal transcription factors bind to the promoter along with RNA polymerase. This complex is the basal apparatus

Activators:

Activators are transcription factors that bind to **enhancer or upstream activating sequences** near target genes. They contain a **DNA-binding domain** and an **activation domain**. By binding specific DNA sequences, activators increase the rate of transcription by recruiting or stabilizing the transcriptional machinery. For example, the yeast **GAL4** protein binds UAS sites and activates **GAL** gene transcription.

Solution: Activators are transcription factors that bind to the promoter and/or enhancers to facilitate the binding of the basal apparatus

Coactivators:

Coactivators are proteins that do **not directly bind DNA**, but instead act as bridges between activators and the basal transcription machinery. They often function by modifying chromatin (e.g., histone acetyltransferases like **CBP/p300**) or by linking activators to the Mediator complex, which communicates with RNA polymerase II. In this way, coactivators help turn the binding of an activator into an actual increase in transcription.

Solution: Coactivators do not bind to the promoter itself, but instead facilitate connections between activators and the basal apparatus

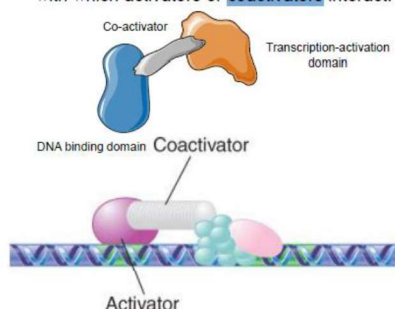
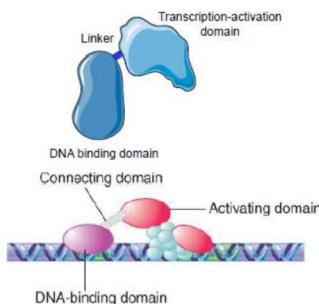


5. Activators and repressors

Activators Interact with the Basal Apparatus



- DNA binding and transcription-activation activities are carried out by independent domains of an activator
- The role of the DNA-binding domain is bringing the transcription-activation domain into the vicinity of the promoter
- An activator that does not have an activating domain may work by binding a coactivator that has an activating domain.
- Several factors in the basal apparatus are targets with which activators or **coactivators** interact.



Task set C

Task set C contains 1 larger problem (2 - 3 in the final exam). The answer should be written using the space allocated below each question.

The proline utilization pathway in *Saccharomyces cerevisiae* proline is converted to glutamate in two enzymatic steps. First proline is converted to 1-pyrroline-5-carboxylate by a proline dehydrogenase encoded by *PUT1*, and subsequently converted to glutamate by the *PUT2* encoded 1-pyrroline-5-carboxylate dehydrogenase.

To analyse the expression of *PUT1* and *PUT2*, a number of mutants unable to grow with proline as sole nitrogen source were isolated, and among the mutants, a strain DB4000 with a deletion of *PUT3* was found.

Q1. What kind of protein could *PUT3* encode? Provide at least two possibilities.

We know that **PUT3 encodes a positive regulator** of the proline utilization pathway because the enzymes responsible for converting proline to glutamate are already identified: **PUT1** encodes the proline dehydrogenase, and **PUT2** encodes the 1-pyrroline-5-carboxylate dehydrogenase. Mutants lacking either of these enzymes would be unable to perform the metabolic steps, explaining their inability to use proline as a nitrogen source. However, in the case of the **DB4000 mutant**, the deletion affects **PUT3**, which is not directly involved in catalysis. The fact that loss of *PUT3* prevents growth on proline suggests that the two enzymatic genes are not expressed at sufficient levels in its absence. Therefore, *PUT3* must function upstream of *PUT1* and *PUT2*, most likely as a transcriptional activator that promotes their expression when proline is the available nitrogen source. If *PUT3* were a negative regulator, its deletion would be expected to increase or deregulate expression of *PUT1* and *PUT2*, not abolish proline utilization. The mutant phenotype therefore provides strong evidence that **PUT3 encodes a positive regulator of the pathway**.

SO:

Likely a **positive regulator** for the *PUT* genes rather than another enzyme (since the enzymes are *PUT1/PUT2*). Two plausible protein types:

1. **Transcriptional activator** that binds upstream of **PUT1/PUT2** and turns them on in the presence of proline (e.g., a DNA-binding regulator whose loss prevents induction, so cells can't use proline).
2. **Proline uptake factor**, such as a **plasma-membrane permease/regulator of permease expression**, needed to import proline before *PUT1/PUT2* can act; without it, proline can't enter the cell, so growth fails even if the enzymes are intact.

(Other close variants: a co-activator/chromatin remodeling factor specifically required for *PUT1/PUT2* induction.)

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Solution: An activator of PUT1 and PUT2; a proline transporter; a gene required for the utilization of glutamate as nitrogen source.

Q2. Which polymerase most likely is responsible for the transcription of *PUT2*?

In *Saccharomyces cerevisiae*, the **PUT2 gene** encodes an enzyme of central metabolism (1-pyrroline-5-carboxylate dehydrogenase). Genes encoding metabolic enzymes are typical **protein-coding genes**, and these are transcribed by **RNA polymerase II**.

So the most likely polymerase responsible for **PUT2 transcription is RNA polymerase II**.

Solution: **RNA polymerase II**

Among the mutants unable to utilize proline as nitrogen source, in strain DB4015 a mutation in BoxA – 161 basepairs upstream of the transcriptional start site was found. The transcription of *PUT2* was quantified in a Reverse Transcriptase (RT) PCR experiment on mRNA isolated from the wildtype yeast and the two mutants DB4000 and DB4015. The amount of *PUT2* mRNA was 100 fold lower in the two mutants compared to the wildtype.

Q3. What role does PUT3 and BoxA play in *PUT2* expression?

PUT3 and BoxA act together as key transcriptional regulatory elements for PUT2 expression.

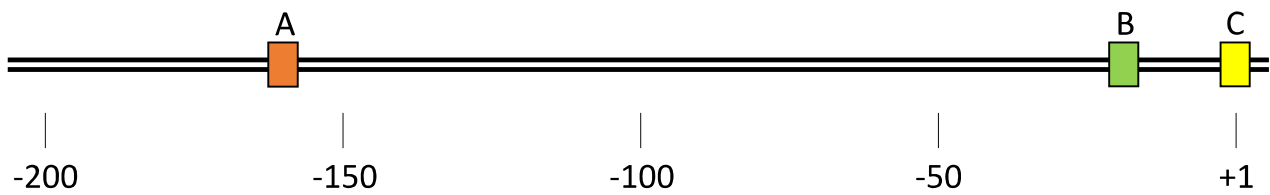
- **PUT3** encodes a transcriptional activator that binds DNA upstream of proline catabolic genes. When proline is available as the nitrogen source, PUT3 activates transcription of both **PUT1** and **PUT2**. Without PUT3, these genes are not expressed at sufficient levels, which explains the inability of the *put3Δ* strain to grow on proline.
- **BoxA** is a specific cis-acting DNA sequence element (a promoter motif) located upstream of **PUT2**. It serves as the **binding site for the PUT3 activator**. The presence of BoxA allows PUT3 to recognize and bind the promoter region, thereby recruiting the transcriptional machinery (RNA polymerase II and associated factors) to initiate transcription of PUT2.

In summary: **PUT3 is the positive transcription factor, and BoxA is the DNA sequence it binds to in order to activate PUT2 expression.**

Solution: PUT3 is an activator that binds in BoxA and facilitates expression

The figure below shows a map of the *PUT2* promoter.

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Q4. What are A, B and C?

- A is the Upstream Activating Sequence (UAS) it is the enhancer element

in yeast genetics, an **Upstream Activating Sequence (UAS)** is the **cis-acting DNA element** that functions as the binding site for transcriptional activators.

It plays a role analogous to enhancers in higher eukaryotes, although UAS elements usually act at shorter distances from the promoter. When an activator protein (like **PUT3** in the proline utilization pathway) binds to a UAS (for example, the **BoxA sequence** upstream of **PUT2**), it recruits or stabilizes the general transcription machinery at the core promoter, thereby increasing transcription by **RNA polymerase II**.

- B is a TATA box

The **TATA box** is a short, conserved DNA sequence found in the **core promoter region** of many eukaryotic genes.

- **Sequence:** It typically has the consensus sequence **TATA(A/T)A(A/T)**, rich in adenine and thymine.
- **Location:** It's usually located about **25–35 base pairs upstream** of the transcription start site (+1).
- **Function:** The TATA box is the binding site for the **TATA-binding protein (TBP)**, which is a component of the **TFIID complex**. Binding of TBP bends the DNA and helps position **RNA polymerase II** at the correct start site for transcription.
- **Role in transcription:** By recruiting general transcription factors, the TATA box provides the platform for assembling the **pre-initiation complex**, which is essential for transcription to begin.

So, while an **Upstream Activating Sequence (UAS)** is where **activator proteins** like **PUT3** bind to regulate gene expression, the **TATA box** is part of the **core promoter** that directly helps **RNA polymerase II** initiate transcription.

- C is the Initiator – Inr

The **initiator (Inr) element** is another type of **core promoter element** in eukaryotic genes.

- **Location:** It spans the **transcription start site (+1)**.
- **Sequence:** The consensus sequence is loosely defined (something like **YYANWYY**, where Y = pyrimidine, W = A/T, and N = any nucleotide).
- **Function:**
 - It **helps define the exact transcription start site**.
 - It can direct the binding of the general transcription machinery, especially **TFIID** (through TBP-associated factors, TAFs).
 - In some promoters that lack a **TATA box**, the initiator can serve as the **main recognition element** for RNA polymerase II initiation.

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- **Role in transcription:** The initiator provides a docking site for transcription factors and helps position RNA polymerase II to start RNA synthesis at the correct nucleotide.

So in short:

The **TATA box** positions RNA pol II by recruiting TBP upstream, while the **initiator** element provides an **anchor at the start site** itself. Together (or individually), they ensure that transcription begins at the right place.

Solution:

- A. Upstream Activating Sequence (UAS) the yeast name for Enhancer element
- B. TATA box
- C. Inr (Initiator)

Q5. Provide the consensus sequences for B and C.

Solution:

- A. TATAA
- B. YYCAYYYYY (Y=pyrimidine)

An experiment proved that PUT3 binds to Box A. The sequence of Box A was as follows:

5'-AACGGGAAGCCAACTCCGTT-3'.

The palindromic sequence is underlined.

Q6. What does that suggest with respect to the quaternary structure of the PUT3 protein?

A palindromic operator strongly suggests **twofold symmetry in the DNA–protein complex**, i.e., **PUT3 binds DNA as a dimer**. Most likely it's a **homodimer**, with each monomer recognizing one half-site of the inverted repeat (one “arm” of Box A). Some yeast activators (e.g., Zn(II)₂-Cys₆ factors) do exactly this: dimerize and contact symmetrical CCG...CCG half-sites. So the quaternary structure inferred is **dimeric (probably a homodimer; possibly a heterodimer in principle)**.

Here's why:

- **Palindromic DNA sequences** have **twofold rotational symmetry** (reading the same 5'→3' on both strands).
- Many transcription factors are **dimers** (two identical or related subunits) that bind to DNA. Each subunit recognizes one half of the palindrome, so the protein–DNA interaction is stabilized by symmetry.
- This is especially common for yeast activators (like zinc cluster proteins) and for prokaryotic repressors (e.g., LacI, TrpR).

☞ However:

- Not every protein that binds a palindrome must be a dimer. Some monomeric proteins have two DNA-binding domains that can contact both halves of a palindrome.

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- Also, not every dimer needs a palindromic site (some bind direct repeats or other arrangements).

So the presence of a **palindromic BoxA sequence** strongly suggests **PUT3 binds DNA as a dimer**, but in principle it could also be a single polypeptide with two DNA-binding motifs.

Solution: **PUT3 is at least a dimer – with a head-to-head configuration**

To pursue the function of *PUT3*, a promoter-less *lacZ* reporter gene was fused right after to the promoter of *PUT2*.

The *lacZ* fusion was transformed into DB4000 (Δ *PUT3*), and subsequently, different variants of *PUT3* were inserted, and the β -galactosidase activity assayed during growth in Ammonium sulfate (an excellent Nitrogen source), Gamma-Amino Butyric Acid (GABA – a very poor Nitrogen source) both in the absence and presence of proline. The results are shown in the following table:

Allele	Mutation	Sp. act. of β -galactosidase ^a with medium containing:			
		Amm	GABA	GABA + Pro	Pro
Vector	None	20	55	44	NG
1. <i>PUT3</i>	Wild type	66	134	385	791
<i>PUT3^c-903</i>	L903R	284	406	624	938
2. <i>PUT3^c-914</i>	N914I	936	855	921	945
<i>PUT3^c-683</i>	S683F	486	518	661	908
3. <i>put3-75</i>	G409D	9	26	17	NG
<i>put3-316</i>	G532R	12	25	41	NG

^a Strain DB1000 (*pep4-3 prb1-1122 prc1-451 put3::LEU2 ura3-52 leu2 trp1*) carried two plasmids in each experiment: the low-copy-number plasmid pMB7 (*TRP1 PUT2-lacZ*) and a high-copy-number (*UR43*) plasmid with the indicated allele of *PUT3*. The strain was grown on minimal media containing glucose (2%) and ammonium sulfate (Amm; 0.2%), GABA (0.1%) without or with proline (Pro; 0.1%), or proline alone (0.1%). Each *PUT3* gene was driven by its natural promoter. Specific activities are the averages of measurements on two independent transformants; variation was <20%. Units are nanomoles of *o*-nitrophenol formed per minute per milligram of protein. The vector is plasmid YEp24, and the enzyme levels measured represent Put3p-independent (background) expression of *PUT2-lacZ*. NG, no growth.

Q7. Discuss the results obtained with the strain 1, 2 and 3.

When the **wild-type PUT3 gene** (strain 1) is introduced, the expected regulation of the *PUT2-lacZ* reporter is observed. In the presence of ammonium, a good nitrogen source, reporter activity is very

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low due to nitrogen catabolite repression. When cells are grown on GABA, a poor nitrogen source, repression is relieved and activity increases moderately. The addition of proline further stimulates expression, and in proline as the sole nitrogen source the strongest induction is seen. These results confirm that wild-type PUT3 functions as an inducible transcriptional activator whose activity is regulated both by the nitrogen source and by the presence of proline.

By contrast, the **PUT3^{c-903} allele (strain 2)** behaves differently. Reporter activity is much higher than wild type under repressive conditions with ammonium, and remains elevated across all conditions tested. This shows that the L903R substitution causes PUT3 to become constitutively active, bypassing the normal regulatory mechanisms. Although some modulation by nitrogen source and proline is still visible, the activator is clearly much less sensitive to repression and induction than the wild type.

Finally, the **PUT3^{c-914} allele (strain 3)** results in a fully constitutive phenotype. Reporter expression is very high under all conditions, including ammonium, where wild-type PUT3 is normally repressed. The N914I mutation thus produces an activator that is permanently active and unresponsive to both nitrogen repression and proline induction.

Together, these results demonstrate that wild-type PUT3 is a regulated activator that requires specific conditions to induce PUT2 expression, while mutations in its C-terminal region (around residues 903–914) abolish regulation and convert it into a constitutively active transcription factor.

Solution:

When supplied with a wild-type PUT3 allele, the defect in DB4000 is complemented, and the strain behaves as the wildtype yeast. In strain 2, PUT3 can activate transcription of PUT2 even in the absence of proline. The protein is trapped in its activating conformation. In strain 3, the PUT3 protein is unable to activate transcription

Task set D

Illustrate two of the following:

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- A. An activator (ActA) binds to its DNA target in heterochromatin. ActA recruits a Histone acetyltransferase (HAT). The unpacking of histones allows a true activator (ActB) to bind to its enhancer. At the same time, an architectural protein brings closer the enhancer region and the TATA box. ActB interacts with the basal apparatus and RNAPolIII initiates the transcription of the target gene.
- B. The different steps where the eukaryotic transcription can be regulated: chromatin remodelling, transcription, splicing and maturation, translocation to the cytoplasm, and translation.
- C. Positive control vs Negative control
- D. A true activator is sequestered in the cytoplasm by an inhibitory protein and the controlled gene is **not** expressed. A ligand binds to the repressor cytoplasmic protein, releasing the true activator, initiation the expression of its target gene.
- E. An activator binds to its DNA target in heterochromatin. ActA recruits a Histone acetyltransferase (HAT). The deacetylation of histones is contained by a boundary element

Exercises

Task set A

Multiple choice

1. A gene that produces a protein that influences the expression of other genes is specifically called a(n):
 - A) activator.
 - B) structural gene.
 - C) repressor.
 - D) regulatory gene.
 - E) constitutive gene.

2. A regulatory circuit in which a repressor requires a metabolite in order to bind to DNA and repress expression, the metabolite is referred to as a(n):
 - A) inducer.
 - B) repressor.
 - C) constitutive effector.
 - D) co-repressor.
 - E) activator.

3. What is an operator site?
 - A) A binding site for the RNA polymerase.
 - B) A binding site for a repressor.
 - C) A constitutive effector binding site.
 - D) A binding site for the ribosome.
 - E) A binding site for the DNA polymerase.

4. In the *lac* operon, in the absence of the inducer:
 - A) the repressor gene is not transcribed.
 - B) the genes encoding enzymes are not translated.
 - C) the repressor gene is not translated.
 - D) the repressor cannot bind to the operator.
 - E) the genes encoding enzymes are only transcribed at a very low rate.

5. A loss-of-function mutation in a repressor gene has the following consequence:
 - A) The repressor gene is not transcribed.
 - B) The genes encoding enzymes are not translated.
 - C) The target genes are constitutively transcribed at a high rate.
 - D) The co-repressor cannot bind to the operator.
 - E) The target genes encoding enzymes are only expressed at a very low rate.

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6. Which statement about regulatory proteins is TRUE?

- A) A repressor binds simultaneously with the RNA polymerase in the promoter.
- B) The lac repressor binds to the operator only in the presence of allolactose.
- C) The target genes are transcribed at a high rate, in the presence of a co-repressor.
- D) The inducer binds to the operator.
- E) Activators facilitates binding of the RNA polymerase to the promoter.

7. Which statement about cis/trans-Acting mutations is TRUE?

- A) A mutation in an operator can be complemented by a plasmid carrying an intact operator site.
- B) A mutation in a repressor gene can be complemented by a plasmid carrying an intact gene site.
- C) A mutation in an activator gene cannot be complemented by a plasmid carrying an intact gene site.
- D) A mutation in an operator can be complemented by a plasmid carrying an intact repressor gene.
- E) A mutation in an operator can be complemented by a plasmid carrying an intact activator gene.

8. Which mutation will result in a decreased expression of the *Escherichia coli lac* operon in the presence of lactose?

- A) *lacI*⁻
- B) *lacO* operator mutation
- C) *crp* (encoding the CRP protein)
- D) *lacA*⁻
- E) *lacZ*⁻

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Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated in the boxes below each question.

Question 1. Define induction, and describe two general types of regulatory circuits involving induction.

In molecular biology, **induction** refers to the process by which the expression of specific genes is activated in response to an external signal, typically the presence of a particular substrate or environmental cue. Induction is a central concept in the theory of gene regulation because it ensures that cells synthesize proteins, especially enzymes, only when they are needed. This regulatory strategy allows organisms to conserve energy and resources by preventing unnecessary gene expression.

Induction is most observed in **catabolic pathways**, where genes encoding enzymes required for the breakdown of specific substrates are normally silent. When the substrate appears in the environment, it acts as an **inducer**, initiating a regulatory cascade that activates transcription. A classic example is the *lac operon* in *Escherichia coli*, which is induced by the presence of lactose.

Two general types of regulation could be: (this is a more overall answer, there are four types listed below this)

1) **Negative control (repressor inactivation):**

In this mechanism, genes are kept in the “off” state by a **repressor protein** that binds to the operator region of DNA, blocking RNA polymerase from initiating transcription. When an inducer molecule is present, it binds to the repressor, changing its conformation and preventing it from binding to the operator. This relief of repression allows transcription to proceed. The *lac operon* provides a well-known example: in the absence of lactose, the LacI repressor inhibits transcription, while in the presence of lactose (converted to allolactose), the repressor is inactivated and the operon is transcribed.

2) **Positive control (activator activation):**

In this case, transcription requires the assistance of an **activator protein** that enhances the binding or activity of RNA polymerase at the promoter. The activator itself must be switched on by an inducer molecule. A prominent example is the regulation of the *lac operon* by the catabolite activator protein (CAP). When glucose levels are low, intracellular cAMP rises, binding to CAP and enabling it to activate transcription of the operon. Thus, the operon is induced only when lactose is present (to inactivate the repressor) and glucose is absent (to activate CAP), illustrating a sophisticated form of combinatorial gene regulation.

Solution answers:

Induction means that the expression of specific structural genes requires the addition of a specific metabolite (the inducer). This may be under either negative or positive control.

- **Inducible expression:** a repressor binds to the operator of a structural gene. When the repressor binds a metabolite (inducer) the repressor adopts an inactive form, it is not capable of binding to the operator and the expression of the structural gene is allowed.
- **Co-repressor dependent repression:** a repressor cannot bind to the operator of a structural gene. When the repressor binds a metabolite (co-repressor) it binds to the operator and the expression of the structural gene is closed
- **Inducible activation:** an activator cannot bind upstream of the promoter of the structural gene. In the presence of a metabolite (inducer/co-activator) the activator can bind and induce the expression of the structural gene
- **Repressible activation:** an activator binds upstream the promoter of the structural gene allowing its expression. In the presence of a metabolite (co-repressor) the activator cannot bind anymore to the activator site and the structural gene is not expressed.

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Question 2. What distinguishes cis-acting elements from trans-acting elements? In a mutant that constitutively expresses a gene typically regulated by a repressor, how can you determine whether the mutation is inactivating the repressor (a trans-acting element) or affecting the operator (a cis-acting element)? Assume that whole genome sequencing is not available as a method.

Cis-acting elements

- Non-coding DNA sequences in or near the genes they regulate. They serve as binding sites for transcription factors or other regulatory proteins
- They are physically linked to the genes they control (i.e., promoter)
- Example: light switch, directly connected to the bulb

Trans-acting elements

- Typically, proteins or RNAs that bind to cis-regulatory elements and influence transcription. They can be produced anywhere in the genome and act at multiple sites, not just the site (or gene) from which they were produced.
- They are diffusible, meaning, they can move around the cell
- Example: person that can switch on or off multiple bulbs

Solution:

Cis-acting sequences are not transcribed to any other form, but function exclusively as a DNA sequence in situ. cis-acting sequences affect only physically linked DNA. Trans-acting sequences encode gene products that are free to diffuse to find their target.

Adding a wild-type copy of the repressor gene to the cells will restore control of the gene if the repressor is mutated (trans-acting element), but not if the operator is mutated (cis-acting element)

Task set C. Task set C contains 2 essay questions (2 to 3 in the final exam). The answer should be written using the space allocated in the boxes below each question.

Problem 1. Regulation of the arginine catabolic genes in *Lactococcus lactis*.

Amino acids may function as an energy source. Arginine is readily degraded through the arginine deiminase pathway (Fig 1A). An analysis of the genome sequence of *L. lactis* KF147, an operon potentially encoding the enzymes in the arginine deiminase pathway was identified (Fig 1B). Deletion mutants of each of the five genes were constructed: DS201 ($\Delta arcA$), DS202 ($\Delta arcB$), DS203 ($\Delta arcC$), DS204 ($\Delta arcD$) and DS205 ($\Delta arcR$).

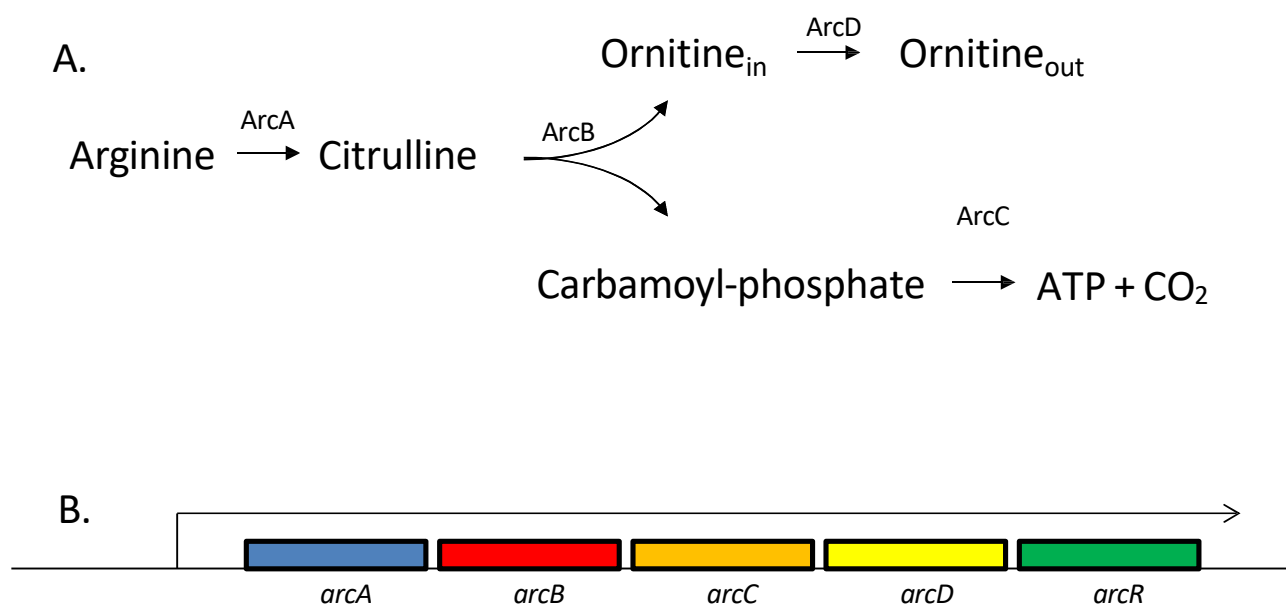


Figure 1A. The arginine deiminase pathway. ArcA: Arginine deiminase. ArgB: Ornithine carbamoyltransferase. ArcC: Carbamate kinase. ArcD: Ornithine transporter that excretes ornithine. ArcR: Regulator of the *arcABCD* operon. B. Genetic organisation of the *arc* operon in *Lactococcus lactis* KF147. The arrow indicates the size of the transcribed mRNA.

Question 1 (10 %). In growth experiments in media with arginine and xylose as carbon sources, the wildtype and mutant DS205 grew significantly faster than the mutants DS201, DS202, and DS203. What explains this observation?

Wild type grows faster than $\Delta arcA-C$ because it can degrade arginine for extra ATP. $\Delta arcR$ grows even faster because the operon is constitutively active without repression. Regulation is by **inducible expression**: ArcR represses the operon in the absence of arginine, and arginine (inducer) relieves repression to allow expression of the ADI pathway.

ArcD is not discussed in the growth comparison because it is not directly required for ATP generation from arginine. While ArcA, ArcB, and ArcC catalyze the key steps of the arginine deiminase pathway and are essential for energy production, ArcD mainly exports ornithine, a by-product of the pathway. Loss of ArcD may reduce efficiency due to metabolite accumulation, but cells can still produce ATP from arginine. Therefore, its phenotype is less dramatic, and the experiment focuses on mutants that either abolish ATP formation ($\Delta arcA-C$) or relieve repression ($\Delta arcR$), which clearly illustrate the regulatory mechanism.

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For understanding we see the system as a machine

- **Wild type** = the complete machine → has all the parts (arcA–C, arcD) **and** the brake (arcR). Normally OFF, but turns ON when arginine is around.
- **DS201 (Δ arcA)** = missing one core part of the engine → machine can't break down arginine, no extra ATP.
- **DS202 (Δ arcB)** = same idea → another broken engine part, can't run.
- **DS203 (Δ arcC)** = same → no engine output.
- **DS204 (Δ arcD, not tested here)** = the machine can still run inside, but it can't export waste efficiently → less efficient, but not totally dead.
- **DS205 (Δ arcR)** = the machine has all its parts, but the brake is gone → it runs all the time as long as arginine is available.

So “**Deletion mutants of each of the five genes were constructed: DS201 (Δ arcA), DS202 (Δ arcB), DS203 (Δ arcC), DS204 (Δ arcD) and DS205 (Δ arcR).**” Means that DS201-205 are five different “machines” (systems) where one of the five “parts” has a mutation. And “wildtype” refers to the whole machine when there is no mutation, so none of the DS201-205 is a wildtype.

Solution

The Mutants DS201, DS202, and DS203 cannot degrade arginine as they have mutations on structural genes involved in the arginine degradation and the formation of additional ATP, explaining the slower growth rate.

DS205's growth is not affected. Therefore arcR is not a gene involved in Arg metabolism and might have a different role

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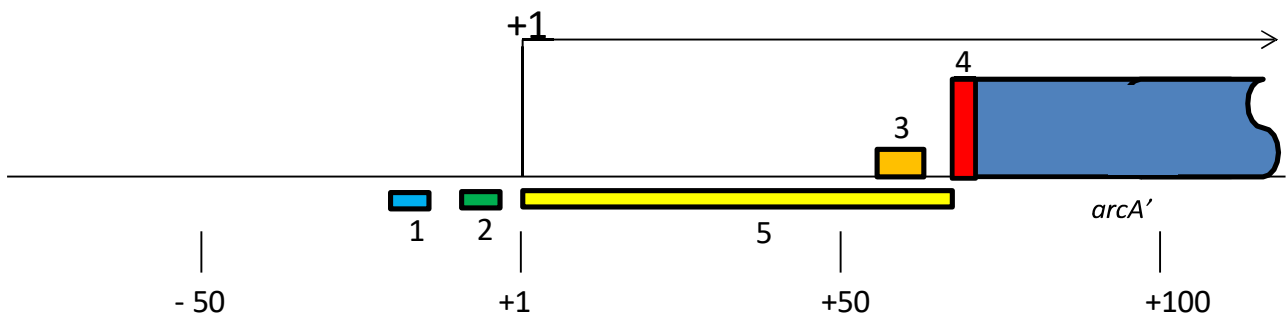


Figure 2. The Promoter area of the *arc*. +1 indicates the transcriptional start site, and the numbers on the DNA are relative to +1. The first part of the *arcA* open reading frame is shown. Five elements (1, 2, 3, 4, and 5) required for gene expression are shown as colored boxes.

Question 2 (20 %). In figure 2 the promoter region of the *arc* operon is shown. What are the five elements 1 to 5?

1. -35 sequence
2. -10 sequence
3. Shine-Delgarno sequence
4. Starts codon
5. Leader

The concentration of *arc* mRNA in the wildtype (DS122) and mutants DS201 and DS205 was determined after growth in a defined medium in the absence and presence of arginine. The relative amounts are shown in table 1.

Strain	DS122 (wt)		DS201 ($\Delta arcA$)		DS205 ($\Delta arcR$)	
		+ arginine		+ arginine		+ arginine
Relative conc <i>arc</i> mRNA	10	100	10	100	100	100

Question 3 (20 %). Describe the effect of the mutation in DS205. Which type of regulation could explain the results, and what would you call ArcR and arginine in such a system?

The mutation in DS205 deletes *arcR*, which normally functions as a repressor of the *arc* operon. In the absence of *arcR*, the operon is expressed constitutively, and high levels of *arc* mRNA are observed both with and without arginine. This shows that the operon is under **negative inducible regulation**, where ArcR acts as the **repressor** and arginine functions as the **inducer** that relieves repression in the wild type.

Solution

A mutation in ArcR leads to constitutive high expression, suggesting a repressor regulation where ArgR is the repressor and arginine the inducer/co-activator.

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Question 4 (10 %). Which method could have been used to determine the mRNA concentration?

- **Northern Blotting**

The concentration of *arc* mRNA could be determined using **Northern blotting**. In this method, total RNA is extracted from the cells and separated by size using gel electrophoresis. The RNA is then transferred to a membrane and hybridized with a labeled DNA or RNA probe that is complementary to the *arc* transcript. The intensity of the resulting signal reflects the abundance of *arc* mRNA, allowing comparison between wild type and mutant strains.

So Northern blotting is like taking all the RNA out of the cells, lining it up by size on a gel, and then looking for just the one you care about (*arc* mRNA). You do this by sticking the RNA onto a membrane and using a “sticky probe” that only binds to *arc* mRNA. The stronger the signal, the more *arc* mRNA there is.

- **Reverse Transcriptase-PCR**

The concentration of *arc* mRNA could be determined using **Reverse Transcriptase-PCR (RT-PCR)**. In this method, the mRNA is first converted into complementary DNA (cDNA) by the enzyme reverse transcriptase. The cDNA is then amplified by PCR using specific primers for the *arc* gene. The amount of PCR product reflects the original concentration of *arc* mRNA, and with quantitative RT-PCR (qRT-PCR), expression levels can be measured very precisely between wild type and mutant strains.

RT-PCR works by first copying the mRNA into DNA, because PCR can only work on DNA. Then that DNA copy gets amplified many times so it's easy to see and measure. The more copies you get, the more mRNA there was to begin with. So it's like making a photocopy of a note before counting it—if you can make lots of copies, you know you had the note to start with.

Question 5 (20 %). After UV mutagenesis, a strain with constitutive high expression of the *arc* genes was isolated. Sequencing revealed that the mutation has acquired a point mutation in position –25 relative to transcription start. What could explain the phenotype?

An ArgR operator site is mutated

The point mutation at +25 lies within the transcribed region, not in the core promoter boxes, so it cannot affect RNA polymerase binding directly. Instead, the constitutive expression phenotype indicates loss of repression. Since repressors typically bind to operator sequences downstream of the transcription start, the most likely explanation is that this mutation disrupts the ArcR operator site. This prevents ArcR from binding and blocking transcription, leading to constant high expression of the *arc* operon.

Question 6 (20 %). In an additional experiment, xylose was substituted with glucose, and the *arc* mRNA determined. In all mutants, the amount of mRNA was below one in both the absence and presence of arginine. What is the explanation for that finding?

If xylose is replaced with glucose, the expression of the *arc* genes is low due to Carbon catabolite repression.

When glucose is used instead of xylose, *arc* mRNA levels drop in all strains due to catabolite repression. Glucose lowers cAMP levels, preventing formation of the cAMP–CRP complex required for transcriptional activation of the *arc* operon. As a result, expression remains low regardless of arginine or the presence of regulatory mutations.

- **Carbon catabolite repression**

Bacteria are “lazy” or smart. If they have **glucose**, they'll always use it first because it's the easiest sugar to break down and gives fast energy.

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To make sure of this, they have a control system called **Carbon Catabolite Repression (CCR)**. CCR = the cell **switches off other metabolic pathways** (like xylose use, lactose use, or arginine degradation) as long as glucose is around.

More scientifically:

Carbon catabolite repression (CCR) is a global regulatory mechanism that ensures bacteria preferentially use glucose over other carbon sources. The system relies on the intracellular concentration of cyclic AMP (cAMP). When glucose is abundant, the cAMP level is low, preventing formation of the cAMP–CRP complex (cAMP receptor protein). Since this complex is required to activate transcription of many catabolic operons, such as those for lactose, xylose, or arginine utilization, their expression is repressed. When glucose is absent, cAMP levels rise, the cAMP–CRP complex forms, and alternative operons are activated, allowing the cell to metabolize other substrates.

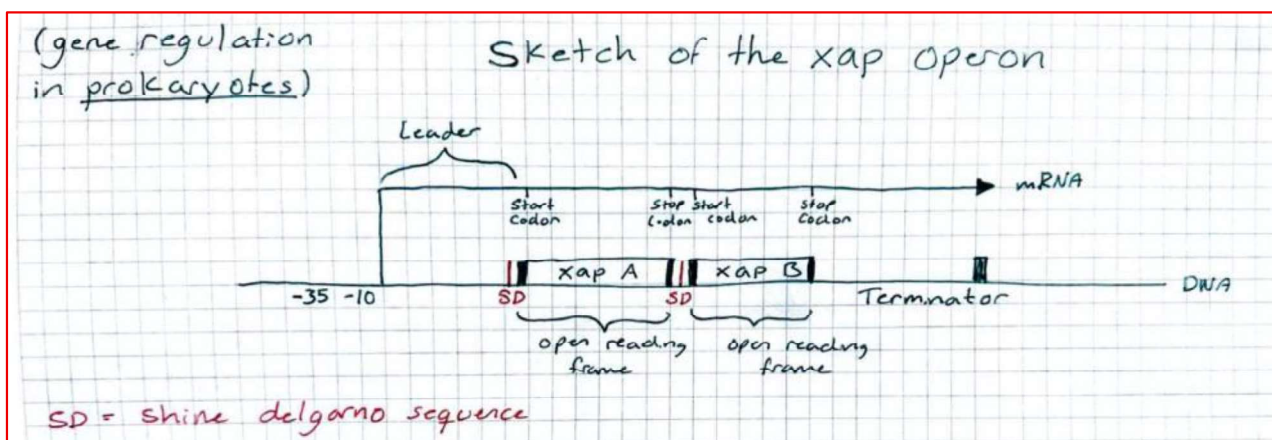
Task 2 set C

Problem 2. Xanthosine utilization in *Salmonella Typhimurium*.

Wild-type *Salmonella Typhimurium* can grow with xanthosine (a purine ribonucleoside) as the sole carbon source. Two genes, *xapA* and *xapB* encoding proteins involved in the catabolism of xanthosine have been identified. The genes *xapA* and *xapB* are in an operon in the order mentioned.

Question 1 (20 %). Draw a sketch of the *xap* operon, where the promoter(s) (-35 and -10 sequence(s)), open reading frames, terminator(s), Shine-Dalgarno sequence(s), startcodon(s) and stopcodon(s) are specified. Also indicate on the drawing the mRNA produced from the operon.

My sketch:



It is important to state that it is in prokaryotes as Shine Delgarno Sequences only exists in prokaryotes and not in eukaryotes.

The Shine–Dalgarno sequence is found only in prokaryotes and serves as the ribosome-binding site in mRNA. It base-pairs with a complementary region of the 16S rRNA in the small ribosomal subunit, thereby positioning the ribosome correctly at the start codon. In contrast, eukaryotes do not have Shine–Dalgarno sequences. Instead, the ribosome binds to the 5' cap structure of the mRNA and scans along the transcript until it reaches the first AUG codon situated in a favorable context known as the Kozak sequence. Thus, translation initiation in prokaryotes and eukaryotes relies on fundamentally different mechanisms.

Short about the kozak sequence: The Kozak sequence is a conserved nucleotide motif in eukaryotic mRNAs that surrounds the start codon and plays a crucial role in translation initiation. During cap-dependent scanning, the small ribosomal subunit binds to the 5' cap and moves along the mRNA until it encounters an AUG codon embedded in a favorable sequence context. The Kozak consensus sequence is 5'-GCC(A/G)CCAUGG-3', with the AUG representing the start codon. The nucleotides at positions -3 (usually A or G) and +4 (usually G) relative to the AUG are particularly important for efficient initiation. When the AUG is in a strong Kozak context, translation begins efficiently; if the context is weak, the ribosome may skip the AUG and continue scanning for a better site. In this way, the Kozak sequence ensures that translation starts at the correct position in eukaryotic mRNAs.

The expression of the XapA protein is measured in a wild-type strain and in the mutant strains 1-3 grown on glycerol respectively in the absence or in the presence of xanthosine. The results are given in Table 1.

Table 1. Enzyme activity of XapA (xanthosine phosphorylase) in units (nmol/min*mg protein)

	XapA level in the different strains - relevant genotype is shown
--	--

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Xanthosine present ^a	Wild type	Mutant 1	Mutant 2	Mutant 3
No	5	0,25	1	1500
Yes	1500	75	1	1500

^{a)} Indicates whether xanthosine was present during growth or not

The regulatory gene of the *xapAB* operon is identified as *xapR*. This gene is expressed constitutively from its own promoter. To investigate which type of mutants had been isolated, a plasmid (*pxapR*⁺) containing wild type gene *xapR*⁺ with its own promoter was transformed into the four strains listed in Table 1. After cultivation, the XapA level was measured in the absence or in the presence of xanthosine. The results are given in Table 2.

Table 2. Enzyme activity of XapA (xanthosine phosphorylase) in strains containing *pxapR*⁺ in units (nmol/min*mg protein)

XapA level in the different strains - relevant genotype is shown				
Xanthosine present ^a	Wildtype/ <i>pxapR</i> ⁺	Mutant 1/ <i>pxapR</i> ⁺	Mutant 2/ <i>pxapR</i> ⁺	Mutant 3/ <i>pxapR</i> ⁺
No	5	0,25	5	1500
Yes	1500	75	1500	1500

^{a)} indicates whether xanthosine was present during growth or not

Question 2 (10 %). How many times is the gene expression induced in the wild-type strain and in mutant 1 upon xanthosine addition?

The gene expression is induced 300 times in the wild-type strain and in mutant 1 upon xanthosine addition as:

For wild type: $\frac{1500 \text{ (XapA level with xanthosine)}}{5 \text{ (XapA level without xanthosine)}} = \text{induced by 300}$

For mutant 1: $\frac{75 \text{ (XapA level with xanthosine)}}{0,25 \text{ (XapA level without xanthosine)}} = \text{induced by 300}$

Question 3 (50 %). Motivate for each of the three mutants 1, 2 and 3 which mutation will provide the measured XapA levels. Take into account the results both from Table 1 and from Table 2.

Mutant 1: there is no difference between the results before and after the constituted gene has been added, thus the mutated gene must be a cis gene, so an operator or a promoter, which cannot be substituted. In this case it is a promoter mutant – the regulation is not affected, but the expression is reduced 20 times.

Mutant 2: When the constituted gene is added the mutant behaves as the wild type, thus the mutated gene must be a trans gene, meaning that it is either a repressor, activator or a sigma factor. As we see a lowered expression of the gene before the addition of the constituted gene both with and without xanthosine, the mutated gene must be an activator (*XapR*).

Mutant 3: we see no difference in expression levels of the gene from before and after the addition of the gene, thus the gene must be cis gene and can therefore not be substituted, so either a promoter or an operator. In this case it is a mutated operator, as the regulation is affected (it is stuck in its activated conformation).

Question 4 (20 %). Which enzyme levels would a *xapR* mutation unable to bind xanthosine give in the absence and presence of xanthosine?

As shown in Question 3, *xapR* encodes an activator of the *xapAB* operon. In the absence of xanthosine, only basal enzyme levels of about 5 units are produced in the wild type. If *XapR* is mutated so that it cannot bind xanthosine, it cannot be activated and induction cannot occur. Consequently, the enzyme level will remain at 5 units in both the absence and the presence of xanthosine.

SO:

With xanthosine: 5 units

Without xanthosine: 5 units

Translation

MC Questions

Choose one correct answer for each of the following:

In bacteria, the function of IF-2 is:

- A) to prevent premature binding of the 50S ribosomal subunit.
- B) to assist the entry of the initiator tRNA into the ribosome.
- C) to facilitate the binding of the 30S ribosomal subunit to mRNA.
- D) to prevent entry of aminoacyl-tRNA into the A site.
- E) to stabilize the 30S ribosomal subunit.

Explanation:

Initiation of translation in bacteria requires three initiation factors: IF-1, IF-2, and IF-3. Each has a distinct role. IF-3 prevents premature joining of the large ribosomal subunit, IF-1 blocks the A site to ensure the initiator tRNA enters the correct position, and IF-2 specifically binds the initiator tRNA (tRNA^{fMet}) together with GTP and helps place it into the P site of the 30S subunit at the start codon. Once correct positioning is achieved, GTP is hydrolyzed, the large 50S subunit associates, and initiation is complete.

🔗 General principle: IF-2 is essential because it ensures that translation begins with the correct tRNA in the correct site, establishing the reading frame.

In bacteria, binding of the 30S ribosomal subunit to mRNA:

- A) involves recognition of the 5' terminus of the mRNA by the 30S subunit.
- B) involves ribosomal proteins binding to the Shine–Dalgarno sequence.
- C) involves base pairing between the Shine–Dalgarno sequence and rRNA.
- D) involves base pairing between the AUG start codon and rRNA.
- E) follows the binding of the 50S subunit.

Explanation:

In bacteria, the small 30S ribosomal subunit does not simply bind to the very beginning of the mRNA. Instead, it recognizes a specific sequence called the Shine–Dalgarno sequence, located in the 5' untranslated region just upstream of the start codon. This sequence is complementary to a region in the 16S rRNA of the 30S subunit, and base-pairing between these sequences positions the ribosome correctly so that the start codon is placed in the P site. This mechanism ensures accurate initiation and prevents errors in reading frame selection.

🔗 General principle: In prokaryotes, translation initiation relies on base-pairing between the Shine–Dalgarno sequence and rRNA, not simply on scanning from the 5' end of the mRNA.

In bacteria, which protein complexes with aminoacyl-tRNAs and GTP to bind to the ribosomal A site?

- A) EF-Tu
- B) eIF2
- C) IF-2
- D) EF-Ts
- E) EF-G

Explanation:

During elongation in bacteria, charged tRNAs must be delivered accurately to the ribosome. This is carried out by **elongation factor EF-Tu**, which forms a complex with GTP and the aminoacyl-tRNA. EF-Tu protects the aminoacyl-tRNA and only releases it into the A site of the ribosome when the codon–anticodon interaction is correct. GTP hydrolysis acts as a checkpoint, ensuring accuracy. After release, EF-Tu is regenerated by EF-Ts, while EF-G is responsible for translocation of the ribosome along the mRNA.

🔗 General principle: EF-Tu ensures fidelity in translation by coupling tRNA delivery with codon recognition and GTP hydrolysis, making elongation both accurate and efficient.

The effects of mutation are minimized by the fact that:

A) there are only three termination codons.

B) DNA is chemically similar to RNA.

C) chemically similar amino acids are represented by similar codons.

D) codons are only three nucleotides in length.

E) there are (usually) only 20 amino acids in polypeptides.

Explanation:

The genetic code is redundant and structured in a way that reduces the impact of mutations.

Multiple codons encode the same amino acid, and codons that differ by a single nucleotide often specify either the same amino acid (silent mutation) or a chemically similar one (conservative substitution). For example, UUU and UUC both code for phenylalanine, while UUU (Phe) and UUA (Leu) specify amino acids with similar hydrophobic properties. This arrangement **buffers the consequences of point mutations**, making proteins less likely to lose function after small changes in DNA.

🔗 General principle: The degeneracy and organization of the genetic code provide a built-in error tolerance, ensuring that mutations do not always result in drastic functional changes.

Short questions

1. Explain the function of the following elements in translation in bacteria and eukaryotes, respectively:

- a. Start codon

The **start codon**, usually AUG, defines the point at which translation begins and establishes the correct reading frame. In bacteria, AUG codes for formyl-methionine (fMet), delivered by a special initiator tRNA, whereas in eukaryotes it codes for methionine (Met), also with a dedicated initiator tRNA. The start codon is therefore essential for ensuring that the polypeptide chain is synthesized in the correct order.

Solutions:

A start codon is where the initiator tRNA binds, and translation is initiated

- b. Stop-codon

The **stop codons** (UAA, UAG, and UGA) do not encode amino acids but instead signal the termination of translation. When a stop codon enters the ribosomal A site, release factors bind and promote the hydrolysis of the bond between the polypeptide chain and the tRNA, releasing the completed protein. This mechanism is conserved in both bacteria and eukaryotes.

Solution:

A stop codon is a tri-nucleotide sequence on the mRNA which leads to termination of translation

- c. Shine-Dalgarno sequence.

In bacteria, the **Shine-Dalgarno sequence** plays a key role in initiation. This purine-rich sequence, located in the 5' untranslated region upstream of the start codon, base-pairs with the 16S rRNA of the small ribosomal subunit. This interaction positions the ribosome so that the start codon is correctly aligned in the P site, ensuring accurate initiation of translation. Eukaryotes do not use a Shine-Dalgarno sequence; instead, they rely on a different mechanism.

Solution:

A Shine-Dalgarno (SD) sequence is the structure on the prokaryotic mRNA where the ribosome binds. If a start-codon exists in the correct distance from the SD, translation will start there. Not found in eukaryotes

- d. 5'-cap

In eukaryotes, the equivalent function is provided by the **5' cap**, a modified guanine nucleotide added to the 5' end of the mRNA. The 5' cap protects the transcript from degradation and also serves as a recognition site for the ribosome. With the help of initiation factors, the ribosome binds to the 5' cap and scans along the mRNA until it encounters the first AUG codon in a favorable Kozak sequence, where translation begins.

Solution:

Only in eukaryotes. Recognized by the small subunit, that subsequently scan the mRNA for the start codon

- List three important differences between translation in prokaryotes and eukaryotes

Solution:

- In prokaryotes translation initiates by the ribosome binding to the SD sequence. In eukaryotes the ribosomal machinery recognizes the 5'-cap and scans for the start-codon.
- In prokaryotes translation is coupled directly to transcription, while in eukaryotes the process is separate (in the nucleus, and in the cytosol)
- In prokaryotes the initiating tRNA is always formyl- methionine tRNA, while in eukaryotes it is methionine-tRNA

- The genetic code is said to be degenerated. Explain what feature of the genetic

Solution:

Several codons code for the same amino acid

5. The genetic code

- The genetic code is universal and degenerated meaning that a single amino acid can be translated from multiple codons.
- 61 of the 64 possible triplets together encode 20 amino acids.
- Three codons (stop codons) do not represent amino acids and cause termination of translation.



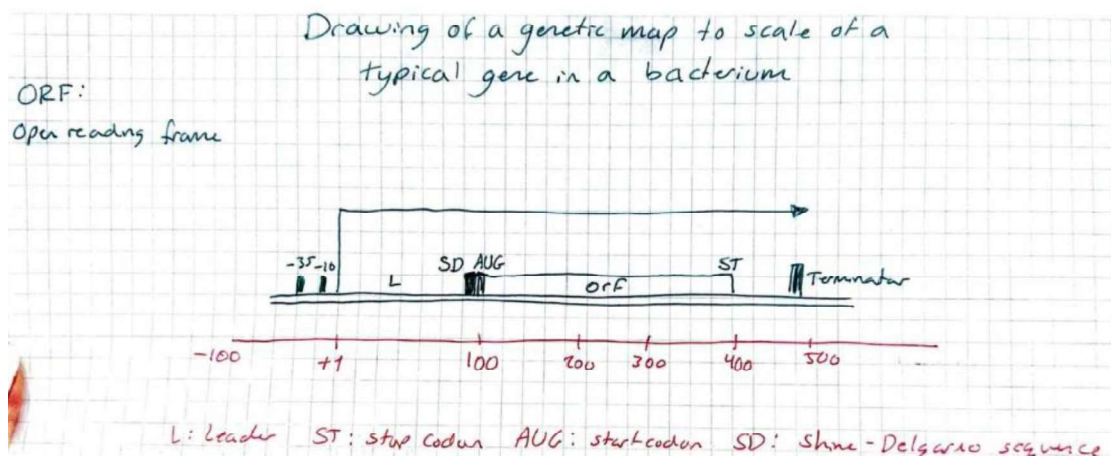
		Second base			
First base		U	C	A	G
	U	UUU Phe UUC F UUA Leu UUG L	UCU Ser UCC S UCA S UCG S	UAU Tyr UAC Y UAA STOP UAG STOP	UGU Cys UGC C UGA STOP UGG Trp
C	G	CUU Leu CUC L CUA L CUG L	CCU Pro CCC P CCA P CCG P	CAU His CAC H CAA Gln CAG Q	CGU Arg CGC R CGA R CGG R
A	A	AUU Ile AUC I AUA Met AUG Met	ACU Thr ACC T ACA T ACG T	AAU Asn AAC N AAA Lys AAG K	AGU Ser AGC S AGA Arg AGG R
G	G	GUU Val GUC V GUA V GUG V	GCU Ala GCC A GCA A GCG A	GAU Asp GAC D GAA Glu GAG E	GGU Gly GGC G GGA G GGG G

- Which kind of molecules are the listed elements? Fill in the table:

Genetic element	DNA	RNA	Protein
DNA polymerase			X
ORF		X	
Origine of replication	X		
Poly-adenylation site		X	
Promoter	X		
Ribosome		X	X
RNA polymerase			X
Transcriptional enhancer	X		
Terminator - Bacteria		X	
Spliceosome		X	X
Stop codon		X	
TATA box	X		
Splice site		X	

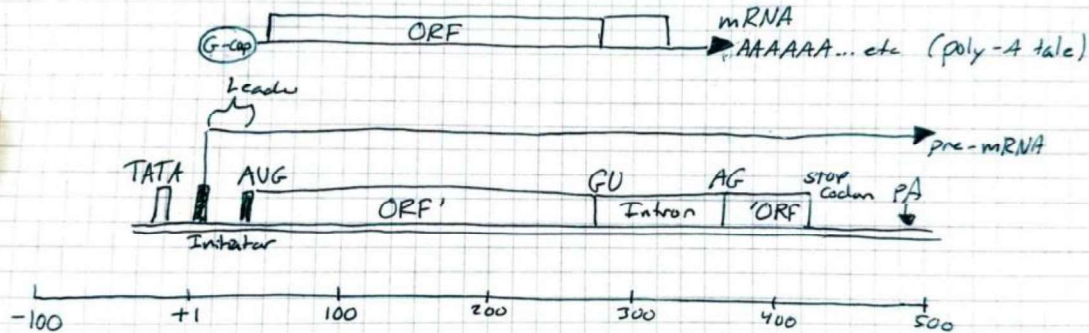
TF _{II} D			X
Shine-Dalgarno sequence		X	
tRNA-synthase			X

5. Draw a genetic map to scale of a typical gene in a bacterium. The gene encodes a protein of 100 amino acids. It has a leader of 100 bps, and is expressed constitutively from a Sigma70/Sigma A promoter. A transcriptional terminator is found 50 bp downstream of the stop codon. On the drawing on the DNA strand clearly indicate the size and position of the following elements: The mRNA, The open reading frame (ORF), the promoter elements (-10 and -35), the leader (L), the Shine-Dalgarno sequence (SD), the start codon (ATG), the stop codon (ST) and the transcription terminator (T). +1 indicate the transcriptional start.



6. Draw a genetic map to scale of a typical gene in a eukaryote. The gene encodes a protein of 100 amino acids. It has a leader of 20 bp, and is expressed constitutively from a Pol II promoter. Transcription is terminated 50 bp downstream of the stop codon. The gene has a single intron of 100 bases located 50 bases from the stop codon. On the drawing clearly indicate on the DNA strand, the size and position of the following elements: The pre-mRNA, the mature mRNA including its modifications, the open reading frame (ORF), the promoter elements (TATA-box and the Initiator (Inr)), the leader (L), the type U2 Splice-sites (GU-AG), the start codon (ATG), the stop codon (ST) and the Polyadenylation site (pA). +1 indicate the transcriptional start.

Drawing of a genetic map to scale of a typical gene in a eukaryote



ORF : open reading frame

TATA-Box: Core promoter element in eukaryotes that positions RNA-polymerase II to start transcription at the correct site.

GU-AG: Type II splice sites

pA: polyadenylation site

+1 indicate the transcriptional start

AUG: start codon

Long problem

(partial, from exam 2003)

Figure 1 shows a fragment of DNA from a temperate phage. The sequence contains a promoter which is recognized by the host RNA polymerase, and the open reading frame (ORF1) for the first gene in the transcript. It is the DNA strand corresponding to the sequence of the mRNA which is shown. ORF1 encodes a protein of 72 amino acids.

```

1      .      .      .      .      .      .      .      .      .      .
      TTTTGTGTTGACAAAGATAAAAAACACATGATATACTAATTTTCATAAAGTTCATGAAACGT
                /\                G  G
                GG                Mut1
                Mut2

61      .      .      .      .      .      .      .      .      .      .
      GAACTGAAAGAAAGGAGAAACATAAATGAGTTATGATTATTCGTCACTACTAGGTAAGAT
                                   G
                                   Mut3

121     .      .      .      .      .      .      .      .      .      .
      CACTGAAAAGTGTGGAAGTCAGTATAACTTCGCAATAGCTATGGGGCTTTCAGAGAGAAC

181     .      .      .      .      .      .      .      .      .      .
      TGTGTCTTTAAAGTTAAATGATAAAGTGACTTGAAAGATGATGAGATTCTAAAGGCAGT

241     .      .      .      .      .      .      .      .      .      .
      TCACGTATTAGAACTTAACCTCAAGATATTCCGAAATATTTTTTAAATGCAAAAGTTCA

301     .      .      .      .      .      .      .      .      .      .
      TTAAACATGAACTTAACTAGAACTAG
  
```

Figure 1. DNA sequence from the temperate bacteriophage (60 basepairs per line). Only coding strand is shown. The numbers refer to the distance from the first basepair of the shown sequence. Sequence

changes for three isolated mutants, Mut1, Mut2 and Mut3 are shown under the wild-type DNA sequence.

The sequence changes in three isolated mutants (Mut1, Mut2 and Mut3) are shown under the wild-type sequence in the figure. Thus, Mut1 is a double mutant containing two basepair substitutions (both a T to G change). Mut2 is an insertion mutation where two additional basepairs have been inserted. Mut3 is a single basepair substitution.

1. Indicate on the wild-type DNA sequence (e.g. by red color) the DNA sequence corresponding to the -35 and -10 regions for the promoter. Determine the distance between the two consensus sequences. Is it in agreement with the sequences could be a promoter?

Tip: If you use Courier New, 10 pitch it is easy to align your text with the sequence

Solution:

Yes, TTGACA is the -35 sequence, and TATACT is the -10 sequence. The distance is 17 base pairs, so it will work as a promoter.

Elaboration:

In bacterial promoters, transcription starts when RNA polymerase holoenzyme binds to two conserved sequences upstream of the transcription start site: the **-35 region** (consensus: TTGACA) and the **-10 region (Pribnow box)** (consensus: TATAAT). The optimal spacing between these two motifs is ~17 bp (acceptable range is 16–18 bp). Correct spacing is critical because it allows the σ -factor of RNA polymerase to contact both sites simultaneously. In this case, the identified sequences TTGACA and TATACT match the consensus well, and their distance (17 bp) falls within the correct range, so they form a functional promoter.

2. Mark the first base in the transcript (the mRNA) when it is known that the first base is an “A”.

Solution:

Solution given: The first base in the mRNA is an A, marked after the -10 sequence.

Elaboration:

In bacteria, the transcription start site is usually located about 6–9 nucleotides downstream of the -10 region. Once RNA polymerase is bound, it unwinds the DNA and begins RNA synthesis at this +1 site. Here, the problem states the transcript begins with an A. This is consistent with typical promoter architecture, where the +1 site is determined relative to the -10 box. Thus, identifying this first base is essential for aligning the coding sequence and correctly defining the mRNA reading frame.

See figure below

Answer shown in red



- Find the reading frame for ORF1. Mark the start codon and underline the Shine-Dalgarno (ribosome binding) sequence.

Solution:

See figure (remember that the distance between SD and ATG must be 3-10 b)

Solution given: See figure (SD and ATG start codon indicated).

Elaboration:

In prokaryotic translation, the ribosome binds to the mRNA by recognizing the **Shine-Dalgarno sequence**, which base-pairs with the 16S rRNA of the 30S ribosomal subunit. This sequence is located a few nucleotides upstream of the start codon (usually 3–10 bp away). The correct start codon (AUG) is then placed into the ribosomal P site, allowing translation to begin. Marking both the SD sequence and the start codon ensures the correct **open reading frame (ORF)** is identified. This guarantees that codons are read in the right triplet order, producing the correct protein.

- Write the first four amino acids encoded by ORF1 and the last amino acid in the protein. Which stop codon is used? Underline the stop codon and the last coding codon.

Solution:

See figure. Stop codon: UAA (TAA på DNA)

The first 4 are methionine, serine, tyrosine, aspartic acid the last is histidine. See figure

Solution given: Start = Methionine, then Serine, Tyrosine, Aspartic acid. Last = Histidine. Stop codon = UAA (TAA on DNA).

Elaboration:

Translation always begins with **methionine (fMet in bacteria)** from the AUG start codon. By

following the codons sequentially in the correct frame, the next three codons are read as serine, tyrosine, and aspartic acid. This shows the reading frame is correct. Translation continues until the ribosome encounters a stop codon (UAA, UAG, or UGA), which is recognized by release factors and not by tRNAs. In this case, the stop codon is UAA, which corresponds to TAA on the DNA coding strand. The last codon before this stop encodes histidine, so histidine is the final amino acid in the protein.

5. Which change(s) do you expect with regard to gene expression of ORF1 and/or the gene product of ORF1 in the three mutants (Mut1, Mut2 and Mut3)? Explain your answers.

Solution:

- Mut 1: There will be low or no promoter activity as the new -10 sequence will have three changes from the consensus sequence.
- Mut 2: The promoter activity will be very low or absent, as the distance between -35 and -10 is increased to 19 bp. (should be 17 bp +/- 1 bp)
- Mut 3: The changed sequence means there will be a UAG stop codon (TAG on the DNA) after amino acid #4. Hence, there is no functional gene product.

Elaboration:

- **Mut1:** The -10 sequence is critical for RNA polymerase binding. Mut1 changes three bases in the -10 region, disrupting the consensus. This would drastically reduce or abolish RNA polymerase recognition, leading to little or no transcription.
- **Mut2:** The distance between the -35 and -10 motifs is critical. Normal promoters require ~17 bp spacing. In Mut2, insertion of two bases increases the spacing to 19 bp. This disrupts the geometry needed for RNA polymerase σ -factor to contact both motifs at once. As a result, promoter activity would drop severely or be lost entirely.
- **Mut3:** This mutation introduces a new stop codon (UAG) early in the ORF. This is a **nonsense mutation**, truncating the protein after just four amino acids. Even if transcription and initiation occur normally, translation terminates prematurely, producing a non-functional peptide.

Amino acids and their codons + egenskaber / chemical properties

Codon	Amino acid	Codon	Amino acid
UUU	Phenylalanine	UCU	Serine
UUC	Phenylalanine	UCC	Serine
UUA	Leucine	UCA	Serine
UUG	Leucine	UCG	Serine
CUU	Leucine	CCU	Proline
CUC	Leucine	CCC	Proline
CUA	Leucine	CCA	Proline
CUG	Leucine	CCG	Proline
AUU	Isoleucine	ACU	Threonine
AUC	Isoleucine	ACC	Threonine
AUA	Isoleucine	ACA	Threonine
AUG (start) ^b	Methionine	ACG	Threonine
GUU	Valine	GCU	Alanine
GUC	Valine	GCC	Alanine
GUA	Valine	GCA	Alanine
GUG	Valine	GCG	Alanine

color coding:

Codon	Amino acid	Codon	Amino acid
UAU	Tyrosine	UGU	Cysteine
UAC	Tyrosine	UGC	Cysteine
UAA	None (stop signal)	UGA	None (stop signal)
UAG	None (stop signal)	UGG	Tryptophan
CAU	Histidine	CGU	Arginine
CAC	Histidine	CGC	Arginine
CAA	Glutamine	CGA	Arginine
CAG	Glutamine	CGG	Arginine
AAU	Asparagine	AGU	Serine
AAC	Asparagine	AGC	Serine
AAA	Lysine	AGA	Arginine
AAG	Lysine	AGG	Arginine
GAU	Aspartic acid	GGU	Glycine
GAC	Aspartic acid	GGC	Glycine
GAA	Glutamic acid	GGA	Glycine
GAG	Glutamic acid	GGG	Glycine

Green → Codons that encode **polar, uncharged amino acids** (e.g., tyrosine, glutamine, asparagine, serine).

Red → Codons that encode **positively charged (basic) or similar group amino acids** (histidine, lysine, arginine).

Yellow → Codons that encode **negatively charged (acidic) amino acids** (aspartic acid, glutamic acid).

Blue → Codons that encode **special cases**: cysteine, tryptophan, glycine, and also the stop codons (UGA, UAA, UAG).

Part A:MC

1. One major function of the 5' cap of a eukaryotic mRNA is:

- A) to prevent the RNA from being translocated to the cytoplasm.
- B) to protect the mRNA from degradation.**
- C) to minimize transcriptional errors.
- D) to provide a binding site for RNA polymerase.
- E) to catalyze intron removal.

2. Pre-mRNA splicing does *not* involve which of the following structural features?

- A) PolyA tail**
- B) Branch site
- C) 3' splice site
- D) 5' splice site
- E) All of these structural features are involved in pre-mRNA splicing.

3. The first stage in the assembly of the spliceosome, the commitment complex, is also known as the:

- A) A complex.
- B) B1 complex.
- C) B2 complex.
- D) C1 complex.
- E) E complex.**

4. The stem-loop structure at the 3' end of a bacterial mRNA functions primarily in:

- A) termination of translation.
- B) regulating the frequency of translation.
- C) termination of transcription.**
- D) mRNA degradation.
- E) mRNA transport.

5. The major advantage of bacteria having unstable mRNAs is:

- A) the ability to rapidly recycle RNA nucleotides.
- B) the ability to rapidly change the proteins produced in the cell.**
- C) to speed up protein synthesis.
- D) the efficiency of repeated transcription of the same mRNAs.
- E) the ability to produce different polypeptides from the same gene.

6. One of the following features is not found in prokaryotic mRNA, which?

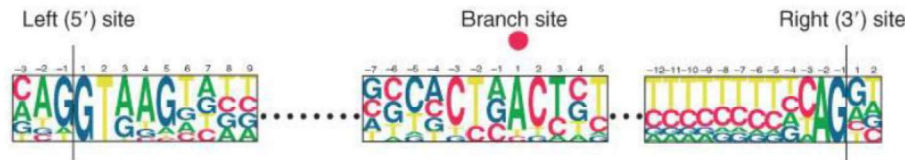
- A) 5'-capping**
- B) 3'-polyadenylation
- C) Group II Intron splicing
- D) Endonuclease cleavage sites
- E) 3' exonuclease degradation

Part B: short questions

1. What is the GU-AG rule?

The GU-AG rule (originally called the GT-AG rule) describes the requirement for these conserved dinucleotides at the first two and last two positions of introns in pre-mRNA's.

Splicing signals for major (U2-type or GU-AG) introns



The **GU-AG rule** is a principle in RNA splicing which states that most introns in eukaryotic pre-mRNA begin with **GU** at the 5' splice site and end with **AG** at the 3' splice site. These conserved sequences are recognized by the spliceosome to ensure accurate removal of introns and proper joining of exons.

Ans: The coding strand sequence of a eukaryotic gene has a 5' splice site consensus sequence of GT and a 3' splice site consensus sequence of AG, so the transcribed pre-mRNA will have splice site consensus

sequences of GU and AG.

2. What are the two stages of pre-mRNA splicing?

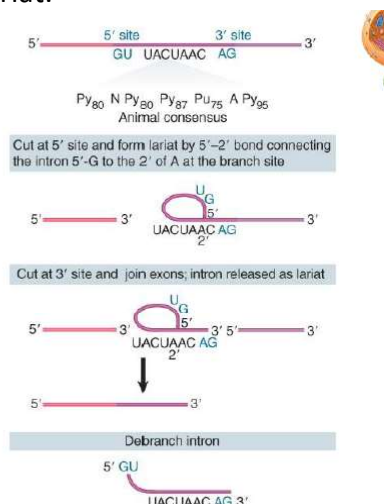
The first stage is lariat formation and then comes exon ligation.

Lariat formation: The 2'-OH of an adenine within the intron (the branch point) attacks the 5' splice site, cutting the RNA and forming a lariat structure.

Exon ligation: The free 3'-OH of the upstream exon attacks the 3' splice site, joining the two exons together and releasing the intron lariat.

3.1 Splicing: splice sites

- An intermediate in intron removal is the formation of a lariat when it is cleaved at the 5' splice site and then ligated to the branch site in a 2'-5-di-esterbond.
- The intron is released as a lariat when it is cleaved at the 3' splice site, and the left and right exons are then ligated together.
- Finally, the lariat is resolved and the intron RNA is then prone for degradation



Ans: In the first stage, the 5' end of the intron is cleaved from the upstream exon and attached to a branch site within the intron, forming a lariat structure. The second stage involves the binding of the upstream exon to the downstream exon

3. What is the role of the U1 snRNP?

Ans: The U1 snRNP initiates the splicing reaction by binding to the 5' splice site of an intron. This binding involves RNA-RNA base pairing between an snRNA of U1 and the 5' splice site. U1 then recruits additional splicing factors

“The **U1 snRNP** plays a key role in starting pre-mRNA splicing. It recognizes and binds to the **5' splice site** of an intron through base-pairing between its snRNA and the pre-mRNA. This early binding helps mark the correct cutting site and recruits other spliceosome components, setting up the splicing process. U1's accuracy is crucial, since mistakes at this step can lead to faulty mRNA and nonfunctional proteins.”

4. Explain how a single gene can result in more than one RNA product due to different ways of processing the pre-mRNA.

Ans: Alternative splicing, in which the intron removal and exon splicing process may vary, can result in two or more different RNA products from the same gene. For example, splicing junctions may be either recognized or skipped to either include or exclude specific exons in the final transcript, or splicing junctions may be skipped to extend the length of an exon to an alternate splice site

“A single gene can produce different RNA products through **alternative splicing**. In this process, introns and exons are joined in different combinations: some exons may be included or skipped, or alternative splice sites may be used. This allows one gene to generate multiple mRNA variants, which can then produce different proteins, greatly increasing the diversity of gene expression.”

5. What are the two types of ribonucleases regarding the kind of substrate used?

Ans: Endoribonucleases cleave RNA molecules at internal phosphodiester bonds and may recognize specific sequences. Exoribonucleases remove nucleotides from the termini of RNA molecules and are unidirectional, either 5' to 3' or 3' to 5'

“There are two main types of ribonucleases based on their substrates. **Endoribonucleases** cut RNA at internal sites within the strand, sometimes recognizing specific sequences. **Exoribonucleases** work from the ends of the RNA, removing nucleotides one by one in either the 5'→3' or 3'→5' direction.”

6. True or False? Generally, deletion of a 3' splice site would result in recognition of the next downstream 3' splice site and the splicing out of all sequences up to that point.

Ans: True

“**True.** If a 3' splice site is deleted, the spliceosome will usually move on to the next available downstream 3' splice site, removing all the intervening sequence as part of the intron.”

7. In the DNA sequence from yeast below (“....” indicates a stretch of bases that is not shown):

- 1) Underline or highlight the following motifs: start-codon, 5'splice sequence, 3'-splice sequence and branch sequence.
- 2) Write out the codons of the (single) resulting exon

AUGGGAUCGAGCUUCGGGUAUUCGGA....UAACGGGAAAUACUACGCGACUUAUGA....UUUCAGGA
AUCCCGAUUGGGCA....

Notice that the branch is given specifically for this yeast in the slides – it can differ ?

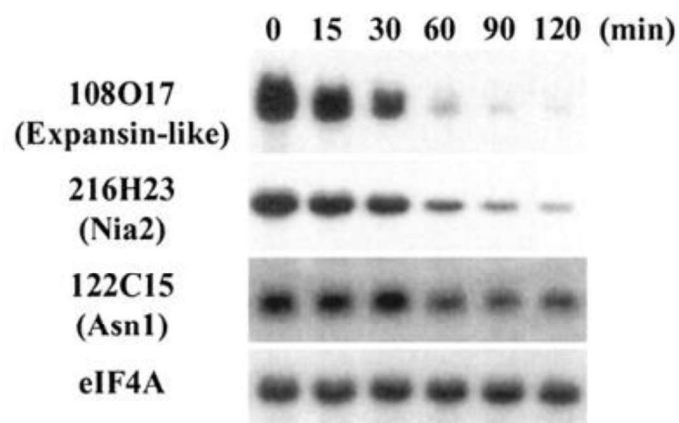
The single resulting exon is the two exons together after the removal of the intron:

AUGGGAUCGAGCUUCGGGAAUCCCGAUUGGGCA

Part C: Problem

Identification of unstable transcripts in *Arabidopsis thaliana*.

Arabidopsis thaliana ecotype Columbia was grown on plates containing 1× Murashige and Skoog salts, 1× Gamborg's vitamins, and 1% sucrose for 2 weeks at 22°C and 16-h/8-h light/dark cycles. The plants were then transferred to a flask with incubation buffer. After a 30-min incubation, 3'-deoxyadenosine (cordycepin) was added to a final concentration of 0.6 mM (time 0). This drug efficiently inhibits transcription in plants. Tissue samples were harvested at regular intervals thereafter and quickly frozen in liquid nitrogen. Total RNA was isolated and analyzed by Northern blot. Northern blot analysis of four mRNAs was performed (108O17, Nia2, Asn1 and eIF4A) with corresponding radioactive probes. Samples consisted of 10 µg of total RNA isolated from the indicated time points. The radioactivity of each band was quantified, and is presented in the table below. eIF4A is used as a stable standard as it is not degraded over time.

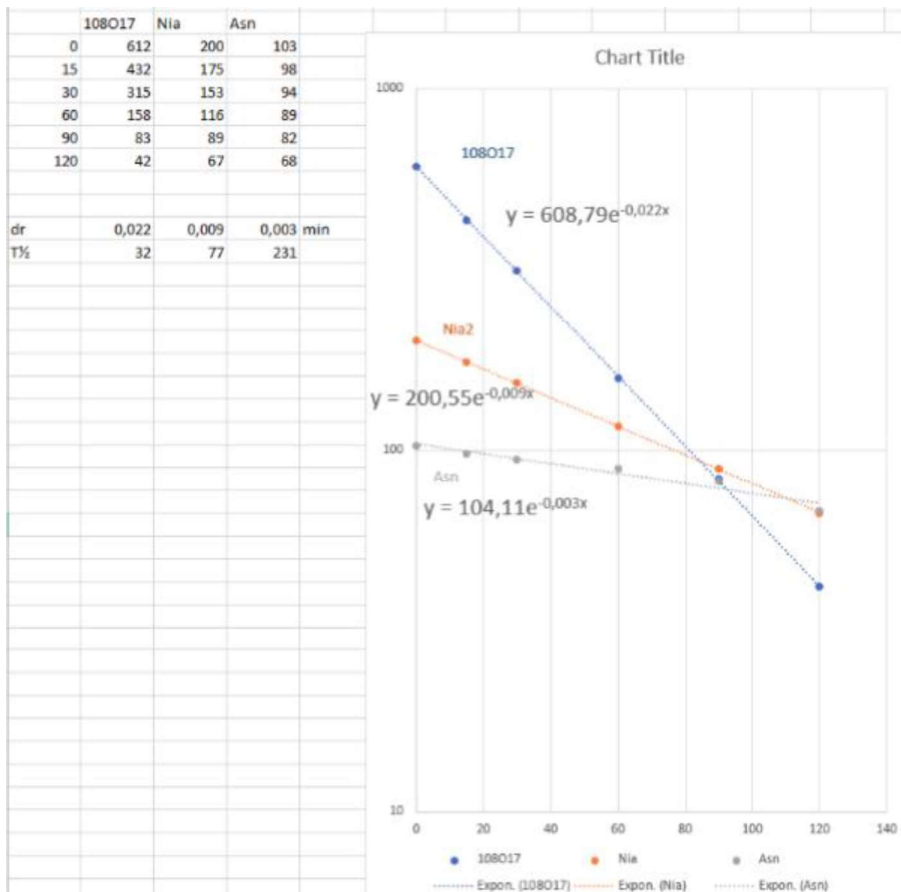


Relative (to eIF4A) radioactivity in the individual bands

Transcript	Time in minutes after cordycepin addition					
	0	15	30	60	90	120
108O17	612	432	315	158	83	42
Nia2	198	175	153	116	89	67
Asn1	103	98	94	89	82	68
eIF4A (standard)	100	100	100	100	100	100

1. Calculate the half-life of the three transcripts.

108O17: 32 min, Nia2: 77 min, Asn1: 231 min



The half-lives calculated (108O17: 32 min, Nia2: 77 min, Asn1: 231 min) reflect differences in transcript stability. mRNA half-life is influenced by sequence motifs in UTRs, secondary structures, and RNA-binding proteins. In plants, unstable transcripts (e.g., 108O17) are often linked to rapid regulation of stress-responsive or metabolic pathways, whereas stable transcripts (e.g., Asn1) usually encode housekeeping or long-term regulatory proteins

- Which reaction initiates mRNA degradation in *Arabidopsis thaliana* and which kind of enzyme catalyses the reaction

“Deadenylation catalyzed by poly(A) nuclease”

The first and rate-limiting step of cytoplasmic mRNA decay is **deadenylation** of the poly(A) tail, catalyzed by **poly(A) nucleases** (e.g., CCR4–NOT complex). Shortening of the poly(A) tail destabilizes the transcript by preventing poly(A)-binding protein (PABP) interaction, which normally protects mRNA from exonucleolytic attack

- In one of the mRNA’s a SE box was identified. Which of the genes could it most likely be?

“Asn1 as it is the most stable (SE = Stabilizing Element)”

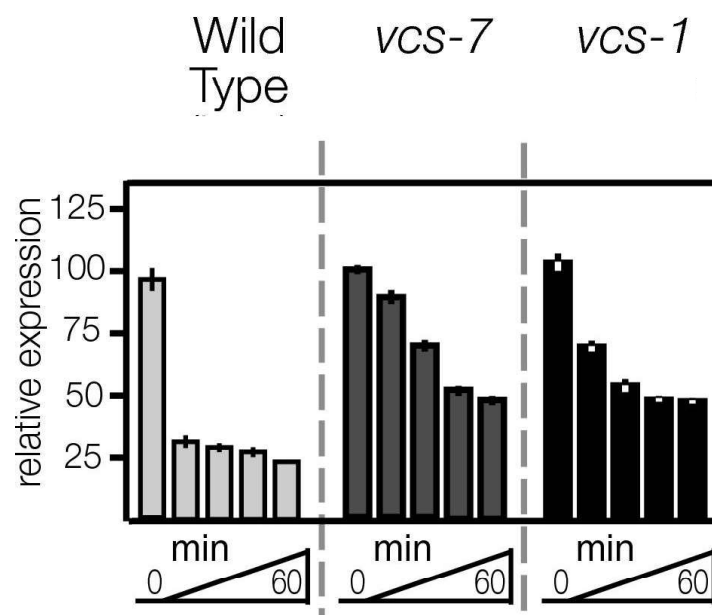
An **SE (Stabilizing Element)** in the 3’ UTR prevents rapid decay by recruiting RNA-binding proteins that block access of nucleases. Since **Asn1** is the most stable transcript, it is most likely to contain an SE box, consistent with literature showing UTR motifs modulate half-life in *Arabidopsis* and other eukaryotes

- mRNA degradation is often associated with the exosome. What is its role?

“It harbors a 3’-5’ exonuclease activity.”

The **exosome complex** is a conserved 3'→5' exonuclease machinery in eukaryotes. It degrades defective, misprocessed, or normal mRNAs after deadenylation, and also participates in nuclear RNA quality control. In *Arabidopsis*, the exosome is essential for RNA turnover, gene silencing, and regulation of developmental mRNAs

In a study of seedling development in *Arabidopsis thaliana* mutants with changed morphology were isolated and studied. It turned out that several proteins involved in seedling development were present in high amounts in one mutant. Common for these proteins is that they are positively regulated by the transcription factor MYP4. To better understand the reason for this observation, two mutants and the wildtype were studied in a Northern blot experiment. Total RNA isolated from samples taken between 0 and 60 minutes after deoxyadenosine (cordycepin) addition were conducted. A *MYP4* specific probe was used. The result was as follows:



5. Explain what has happened to the *MYP4* mRNA in the mutants.

“The stability of the *MYP4* mRNA has been increased.”

Northern blot data show **increased stability of *MYP4* mRNA** in mutants. Normally, *MYP4* is turned over quickly, but in these mutants, degradation is slowed, prolonging transcript availability for translation. This implies a defect in one of the RNA decay pathways

6. Which process in gene expression could have been affected in the mutants. Suggest which protein or protein complex that may have been altered. Several possibilities – just mention one.

“Several possibilities including the exosome, polyA nuclease, de-capping (DCP), 5'-3' exonuclease (XRN1)”

The mutants could be defective in:

- **Deadenylation (poly(A) nucleases)**
- **Decapping (DCP1/DCP2 complex)**
- **5'→3' decay (XRN exonuclease)**

- **3'→5' decay (exosome)**

Even a single mutation in one of these pathways can markedly increase mRNA half-life. For example, loss of **XRN4** in Arabidopsis stabilizes several stress-responsive transcripts.

7. How can the observed effect on *MYP4* mRNA explain the high amount of proteins involved in seedling development?

“As *MYP4* mRNA is more stable, more *MYP4* protein may be translated from the same mRNA. As *MYP4* is a transcriptional activator, it could result in higher expression of the target genes.”

Since *MYP4* is a **transcription factor**, stabilization of its mRNA leads to accumulation of more *MYP4* protein. This increases transcriptional activation of downstream seedling-development genes, explaining the observed overabundance of their protein products in the mutant.

Assignment 6. Transcription

Task set A – MC questions

Task set A contains 10 multiple choice questions (20 in final exam). The correct answer should be indicated by a letter in the cell next to the possible answers.

		correct answer
1) In prokaryotes, transcription starts when RNA polymerase binds to the:		
A) transcription unit. B) coding strand. C) promoter. D) transcription bubble. E) nontemplate strand.	In prokaryotes, transcription begins when RNA polymerase binds to the promoter region of DNA. The promoter is a specific DNA sequence upstream of the gene that signals the starting point for transcription.	C
2) Which statement best describes the role of the eubacterial RNA polymerase sigma factor?		
A) Sigma factor allows assembly of RNA polymerase. B) Sigma factor is responsible for catalyzing the formation of phosphodiester bonds during RNA polymerization. C) Sigma factor shuttles cytoplasmic RNA polymerase to promoter sequences. D) Sigma factor increases the affinity of RNA polymerase for promoter sequences. E) Sigma factor actively removes core enzyme from DNA and attaches it to a promoter.	In eubacteria, the sigma (σ) factor is a subunit of RNA polymerase that helps the enzyme recognize and bind specifically to promoter regions on DNA, ensuring transcription starts at the correct site. It does not catalyze bond formation (that's the role of the RNA polymerase core enzyme).	D
3) Which of the following is unimportant for the recognition of the promoter by the sigma factor?		
A) The –35 element B) The length of the region between the –10 and –35 elements C) The –10 element D) The sequence surrounding the startpoint E) The Shine-Delgarno sequence	The Shine-Dalgarno sequence is involved in translation initiation in prokaryotes (it helps ribosomes align with the start codon), not in transcription. For promoter recognition by the sigma factor, the –10 element (Pribnow box) ,	E

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	the -35 element , and the spacing between them are all critical.	
4) In general, intrinsic termination of transcription in bacteria:		
A) occurs at a specific site that is stable to changes in temperature. B) requires transcription of the sequences signaling termination. C) is unaffected by sequences that are not in proximity to the termination site. D) is triggered by RNA polymerase's interaction with DNA sequences. E) requires accessory proteins.	In bacteria, intrinsic termination (also called Rho-independent termination) occurs when RNA polymerase transcribes a GC-rich inverted repeat followed by a poly-U sequence. This forms a hairpin loop in the RNA followed by a stretch of uracils, destabilizing the RNA-DNA hybrid and causing transcription to stop. It does not require accessory proteins (that's Rho-dependent termination).	B
5) One major difference between bacterial and eukaryotic RNA polymerase is that:		
A) prokaryotes have several RNA polymerases. B) eukaryotic RNA polymerase can polymerize RNA in the 3' to 5' direction. C) prokaryotic RNA polymerase recognizes several different promoters. D) eukaryotic RNA polymerase has only one polypeptide. E) eukaryotic RNA polymerase cannot read DNA to find the promoter sequence.	Eukaryotic RNA polymerase requires transcription factors, such as TBP, to recognize promoters, unlike bacterial RNA polymerase which directly recognizes promoter sequences with the sigma factor.	E
6) Eukaryotic RNA polymerase II synthesizes:		
A) 5S rRNA. B) 28S rRNA. C) mRNA. D) 18S rRNA. E) tRNAs.	Pol II transcribes protein-coding genes into messenger RNA, as well as some snRNAs and miRNAs	C
7) TATA-binding protein (TBP) is required for initiation by:		
A) RNA polymerases I, II, and III. B) RNA polymerases I and III. C) RNA polymerase II. D) RNA polymerase III. E) RNA polymerase I.	TBP is a universal transcription factor used by all three eukaryotic RNA polymerases, although in different complexes: SL1 for Pol I, TFIID for Pol II, and TFIIB for Pol III	A

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8) The TATA box is a common component of promoters for:		
A) RNA polymerases I, II, and III. B) RNA polymerases I and III. C) RNA polymerase II. D) RNA polymerase III. E) RNA polymerase I.	The TATA box is a core promoter element recognized by transcription factor TFIID for genes transcribed by RNA Pol II, mainly protein-coding genes	C
9) Which statement regarding transcriptional termination in eukaryotes is right?		
A) RNA pol II termination is facilitated by an inverted repeat followed by a U's B) RNA polymerase II termination is closely linked to mRNA processing C) RNA polymerase I terminates if the RNA is untranslated D) RNA polymerase III termination is linked to replication process E) None of the statements is right	Termination of Pol II is coupled to cleavage and polyadenylation of the pre-mRNA; it is not like bacterial intrinsic termination	B
10) Generally, a eukaryotic household gene requires ____ for being expressed.		
A) TFIID B) a Shine-Dalgarno sequence C) a Sigma factor. D) an Okazaki fragment. E) RNA polymerase I	TFIID, containing the TBP subunit, is essential for initiating transcription by RNA Pol II, including housekeeping genes	A

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Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

1) Explain the role of sigma factors in RNA synthesis in Bacteria.

In bacteria, sigma factors play a crucial role in the initiation of RNA synthesis. They are specialized subunits of RNA polymerase that provide promoter recognition, since the core enzyme alone cannot specifically identify where transcription should begin. When a sigma factor binds to the core RNA polymerase, the resulting holoenzyme can recognize conserved promoter sequences, particularly the -35 region and the -10 Pribnow box. By guiding RNA polymerase to these sites, sigma factors ensure that transcription starts at the correct position. They also assist in unwinding the DNA strands to form the transcription bubble, allowing RNA synthesis to begin. Once the first few nucleotides of RNA are added, the sigma factor is often released, leaving the core enzyme to carry out elongation. Different sigma factors exist in bacteria, each recognizing distinct promoter sequences and regulating different sets of genes. For example, σ^{70} controls most housekeeping genes, while alternative sigma factors such as σ^{32} regulate stress responses like the heat shock reaction. This system enables bacteria to rapidly adjust gene expression in response to environmental changes. In short, sigma factors are essential for directing RNA polymerase to the right promoters and ensuring accurate transcription initiation.

Teachers answer:

A sigma factor is one of five subunits within an RNA polymerase involved exclusively in transcription of DNA to RNA. It identifies and binds to a promoter (initiation) site, notably the highly conservative -10 and -35 sites, which creates an RNA polymerase-DNA template complex that facilitates transcription initiation. A sigma factor then dissociates from the template after a short stretch of RNA is synthesized.

2) Explain in bullet points how eukaryotic housekeeping genes are transcribed. Include both TATA dependent and TATA-less promoters.

Housekeeping genes in eukaryotes are transcribed continuously at a basal level because they are essential for basic cellular functions. Their transcription is carried out mainly by RNA polymerase II and requires the action of general transcription factors (TFIID, TFIIA, TFIIB, TFIIF, TFIIIE, and TFIIH). Depending on the promoter structure, housekeeping genes may use either TATA-dependent or TATA-less promoters.

In TATA-dependent promoters, a conserved TATA box sequence is located about 25–30 base pairs upstream of the transcription start site. Transcription begins when the TATA-binding protein (TBP), a component of the TFIID complex, binds directly to the TATA box. This binding is stabilized by TFIIA and TFIIB, which also help position RNA polymerase II at the correct site. RNA polymerase II is brought to the promoter with the help of TFIIF, while TFIIIE and TFIIH complete the

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preinitiation complex. TFIIF then unwinds the DNA and phosphorylates the C-terminal domain (CTD) of RNA polymerase II, enabling transcription initiation.

In TATA-less promoters, the TATA box is absent. Instead, transcription relies on alternative core promoter elements such as the initiator (Inr) sequence at the transcription start site, the downstream promoter element (DPE), or CpG island regions, which keep chromatin open and accessible. In this case, TBP is still required, but it does not bind DNA directly. Instead, TBP-associated factors (TAFs) within TFIID recognize these alternative promoter elements and guide the assembly of the transcription machinery. Once TFIID is bound, the rest of the general transcription factors and RNA polymerase II are recruited in much the same way as in TATA-dependent promoters.

Thus, while TATA-dependent promoters rely on the direct binding of TBP to the TATA box, TATA-less promoters use TAFs to recognize Inr, DPE, or CpG islands. Both mechanisms ultimately recruit RNA polymerase II and general transcription factors, ensuring reliable transcription of housekeeping genes.

Teachers answer:

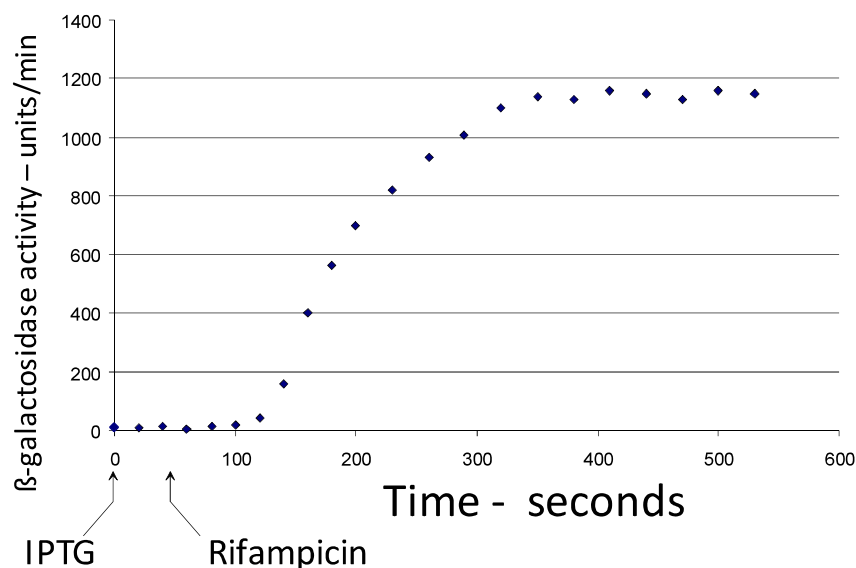
Household genes are transcribed by DNA polIII. The transcription initiates at the promoter, in the following sequence of events:

1. DNA must be accessible to transcription; ie. dense chromatin must be remodelled to the open configuration.
2. Transcription factor TF IID (a complex of TBP and several TAFs (TBP Associated Factors)) binds to the TATA-box – if present – or to the Inr sequence.
3. TF IIB binds to the BRE site (The **BRE site**, or **B recognition element**, is a core promoter element in eukaryotic genes that serves as the binding site for the general transcription factor **TFIIB**. It is usually found immediately adjacent to the TATA box, either just upstream (BREu) or just downstream (BREd). By binding to the BRE, TFIIB stabilizes the interaction of the TATA-binding protein (TBP) with the promoter and helps position RNA polymerase II correctly at the transcription start site. In this way, the BRE plays an important role in ensuring that transcription begins accurately and efficiently.)
4. The RNA polymerase II together with TF IIF is recruited and correctly positioned in the promoter
5. TF IIE and TF IIH binds to the RNA PolIII complex, and the first few nucleotides are transcribed.
6. When entering the elongation mode, the TF IIE and H is dissociated, and the TBP complex is left back on the promoter, potentially facilitating another round of transcription initiation.
7. The RNA polymerase proceeds until termination

Task set C

Problem C1. RNA polymerase transcription rate

Rifampicin is an antibiotic that binds to *E. coli*'s RNA polymerase. Rifampicin inhibits initiation of transcription, but all RNA polymerases that are transcribing will complete ongoing transcriptions. An *E. coli* wild-type strain was grown on a growth medium with glycerol as a carbon source at 30 °C. The lactose analogue IPTG was added at time 0 to induce the synthesis of β -galactosidase encoded by the *lacZ* gene. 45 seconds later, rifampicin was added and at short time intervals samples were taken, and translation inhibited by chloramphenicol addition. After appropriate pretreatment of the samples, the activity of β galactosidase was determined in each sample. The result of the experiment is shown in the figure, where β galactosidase activity is plotted as a function of the time of sampling in chloramphenicol.



1. Describe the curve

Within the first 100 seconds (about the first minute and a half) there is little to no β -galactosidase activity. After this short initiation period with little to no activity we see a big exponential growth in activity until we hit time 350 seconds – about 5,8 minutes. After this rapid growth we see a sort of upper limit, where the activity seems to be constant at a little under 1200 units/minute.

Teachers answer:

β -galactosidase activity begins to rise after 100-120 sec, where the first RNA polymerases reach the end of the β -galactosidase (*lacZ*) gene. Then mRNA begins to break down and after 300-350 seconds there is no more mRNA and therefore no more β -galactosidase accumulate

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2. The β -galactosidase protein is 1023 amino acids long. Calculate the transcription rate (bases added to the mRNA per second) assuming transcription and translation are coupled processes (i.e., the ribosomes follow closely after the RNA polymerase).

The β -galactosidase gene encodes 1023 amino acids, corresponding to 3069 nucleotides. From the graph, transcription takes about 100 seconds, giving a rate of $3069/100 \approx 30$ nucleotides per second, which matches typical *E. coli* transcription rates.

Teachers Answer:

The open reading frame corresponds to $3 \times 1023 = 3069$ bases of mRNA. It takes approximately 100 seconds, so the transcription rate must be $3069/100 = 30$ bases per second.

3. The cell contains several types of RNA. What are they, and how stable are the individual types?

RNA type	Function	Stability	
mRNA	Carries genetic information to ribosomes	Variable (minutes to hours, sometimes days)	
tRNA	Brings amino acids to ribosomes during translation	Very stable (days)	
rRNA	Structural and catalytic core of ribosomes	Very stable (days)	
snRNA	Part of spliceosome, mediates pre-mRNA splicing	Relatively stable	
snoRNA	Guides chemical modification of rRNA	Stable	
miRNA/siRNA	Post-transcriptional gene regulation	Moderately stable when bound to RISC	
antisense RNA	Regulatory; binds complementary mRNA to block or degrade it	Usually unstable to moderately stable	

Teachers Answer:

mRNA, rRNA, tRNA, sRNA, antisenseRNA. The rRNA and tRNA are stable whereas the others are mainly unstable

Explain why the slope of curve is declining and reaches a constant level.

The slope of the curve represents the **rate of β -galactosidase production**. At the beginning, the slope is steep because many new transcripts are being made and translated, so enzyme levels rise quickly. Over time, however, **no new transcription can start** (rifampicin blocks initiation), and only those RNA polymerases that had already begun transcription before the drug addition can finish their transcripts. As these transcripts are completed and translated, the number of active ribosomes decreases. This causes the rate of new enzyme production to gradually decline. Eventually, once all ongoing transcription and translation events are completed, no more β -galactosidase can be produced. At that point, the curve reaches a constant plateau, reflecting the maximum amount of enzyme synthesized from the last round of transcripts.

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In short: The slope declines because transcription initiation is blocked, leaving only the last transcripts to be finished and translated. Once these are exhausted, enzyme synthesis stops, and the curve levels off.

Teachers Answer:

The synthesis rate of β -galactosidase declines as mRNA is degraded. When no more mRNA amount of β -galactosidase becomes constant as the enzyme is stable.

Problem C2. Tissue specific expression

To detect expression of the GLUT1 glucose transporter in different tissues, a Northern blot of RNA isolated from the Atlantic Cod was performed. Different tissues were isolated and the RNA extracted.

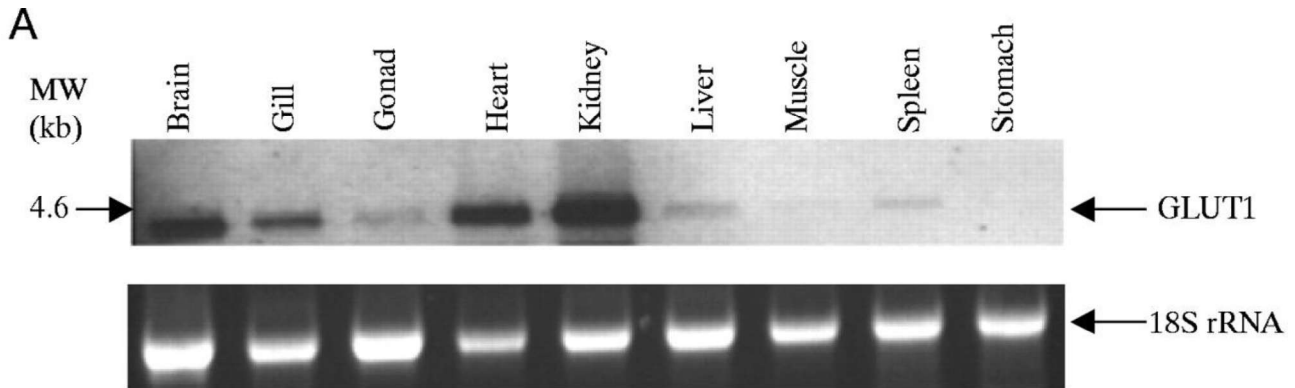


Figure C2.1. Tissue distribution of Atlantic cod GLUT1. (A) Northern blot analysis of total RNA from various cod tissues hybridized with a 664 bp antisense RNA probe specific for Atlantic cod GLUT1. Top panel shows the 4560 bp GLUT1 mRNA transcript, and bottom panel shows the 18S rRNA as detected by staining.

1. Briefly describe the principle of a Northern blot.

In a Northern blot, RNA is first extracted from cells or tissues and separated by size using gel electrophoresis. The separated RNA is then transferred onto a membrane (usually nylon or nitrocellulose), where it can be stably bound. A labelled probe, which is complementary to the RNA sequence of interest, is hybridized to the membrane. The probe binds specifically by base-pairing to its target RNA, allowing detection of the presence and size of the transcript. This method is commonly used to study gene expression across tissues or conditions.

Teachers Answer:

Total RNA is separated on gel bases on size. The RNA is transferred to a filter, and a specific, labelled probe is added. The probe will hybridize by base parring identifying the presence of the specific RNA molecule

2. In which tissue is GLUT1 expressed at the highest level? Rank the expression levels?

From the blot, GLUT1 mRNA is most strongly expressed in the kidney. The next highest expression is seen in the brain, followed by the heart, gills, liver/gonad, spleen, muscle, and finally stomach, which shows little or no signal. This ranking reflects tissue-specific regulation of GLUT1 expression, which makes sense given that tissues like the kidney and brain have high energy demands and require efficient glucose uptake.

Teachers Answer:

In the kidney the expression is the highest, the follows Brain, Hart, Gills, Liver/Gonad, Spleen, Muscle and stomach.

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3. Which RNA polymerase is most likely responsible for expressing GLUT1?

GLUT1 is a **protein-coding gene**, and in eukaryotes, protein-coding genes are transcribed by **RNA polymerase II**. This enzyme synthesizes messenger RNA (mRNA), which is then processed and translated into protein. RNA polymerase I transcribes rRNA (except 5S rRNA), and RNA polymerase III transcribes tRNA and some small RNAs, so they are not involved here.

Teachers Answer:

Most probably RNA polIII, as it transcribe household gene

4. The lower panel shows the amount of 18S rRNA in each sample. Why is that included?

The 18S rRNA serves as a **loading control**. Since rRNA is very abundant and relatively constant across tissues, visualizing the 18S band ensures that approximately equal amounts of total RNA were loaded into each lane of the gel. This allows the GLUT1 signal in the upper panel to be compared reliably across tissues — differences in GLUT1 band intensity can then be attributed to differences in expression rather than unequal sample loading or transfer.

Teachers Answer:

It is shown to verify that approximately the same amount of total RNA is added to the gel.